

Project Data Page:





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DESIGN CALCULATION

In Accordance with ASME Section VIII Division 1

ASME Code Version : 2025

Analysis Performed by : SL User

Job File : ...\\A150NRL13EQ002\\WATER COOLER.pvdb

Date of Analysis : Sep 5, 2026 5:05pm

PV Elite 28, January 2026



Channel side design conditions needs to finalize from PC.
So, channel side nozzles- not finalized here right now.

Vessel Design Summary:

ASME Code, Section VIII Division 1, 2025

Diameter Spec : 1940.000 mm. ID
 Vessel Design Length, Tangent to Tangent 15236.17 mm.
 Specified Datum Line Distance 0.00 mm.
 Shell Side Design Temperature 260 °C
 Channel Side Design Temperature 85 °C
 Shell Side Design Pressure 191.000 bars
 Channel Side Design Pressure 10.000 bars
 Shop Shell Side Test Pressure 249.472 bars
 Shop Channel Side Test Pressure 13.837 bars
 Wind Design Code ASCE/SEI 7-22
 Earthquake Design Code ASCE/SEI 7-22

Materials of Construction:

Component Type	Material	Class	Thickness	UNS #	Material Treatment Options
Shell	SA-516 70	...	...	K02700	
Flange	SA-266 2	...	...	K03506	
Nozzle	SA-106 B	...	...	K03006	
Nozzle	SA-266 2	...	...	K03506	
Re-Pad	SA-516 70	...	...	K02700	
Nozzle Flg	SA-105	...	...	K03504	
Nozzle Flg	SA-350 LF2	1	...	K03011	
Tubes	SA-210 A-1	...	...	K02707	
Tubesheet	SA-266 2	...	...	K03506	
Flg Bolting	SA-193 B7	...	<= 64	G41400	
Hrz Bolting	SA-36	...	...	K02600	
Saddles	SA-285 C	...	...	K02801	

+FG - Fine Grained, +R - rolled, +T - Tempered, +N - Normalized
 +Q - quenched, +CBV - Cb & V added, +IT - Impact tested

Element Pressures and MAWP (bars & mm.):

Element Description or Type	Design Pressure + Stat. head	Ext. Press.	Element M.A.W.P	Total Corrosion Allowance	Str. Flg. Gov.	In Creep Range
CH. COVER-1	10.000	0.00	11.000	3.0000	N/A	No
CH. FLG-1	10.000	0.00	13.000	3.0000	N/A	No
CH. SHELL-1	10.093	0.00	31.000	3.0000	N/A	No
MAIN SHELL	191.112	0.00	192.000	1.5000	N/A	No
CH. SHELL-2	10.190	0.00	31.000	3.0000	N/A	No
CH. FLG-2	10.094	0.00	12.000	3.0000	N/A	No
CH. COVER-2	10.000	0.00	11.000	3.0000	N/A	No

Liquid Level: 1940.00 mm. Dens.: 0.001 kg./cm³ Sp. Gr.: 1.000

Element Types and Properties:

Element Type	"To" Elev mm.	Element Length mm.	Nominal Thickness mm.	Finished Thickness mm.	Reqd Thk Internal mm.	Reqd Thk External mm.	Long Eff	Circ Eff
Body Flg	110.0	105.0	110.0	105.0	101.9	...	1.00	1.00
Body Flg	260.0	150.0	115.0	110.0	107.8	...	1.00	1.00
Cylinder	1860.0	1600.0	25.0	25.0	10.1	...	1.00	1.00

Cylinder	13029.0	10820.0	149.0	149.0	148.2	...	1.00	1.00
Cylinder	14978.0	1600.0	25.0	25.0	10.2	...	1.00	1.00
Body Flg	15131.2	150.0	115.0	110.0	107.8	...	1.00	1.00
Body Flg	15236.2	105.0	110.0	105.0	101.9	...	1.00	1.00

Local Stress Analysis Results:

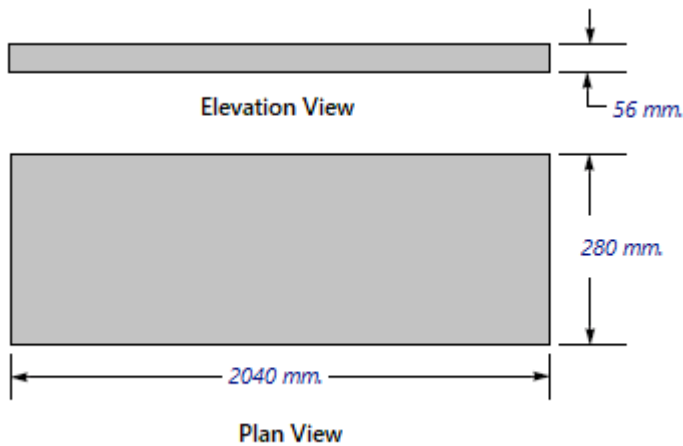
Description	Analysis Type	Max Stress Ratio	High Stress Location	Pass Fail
TS OUT NOZ. T2	WRC-297	0.886	Nozzle	Passed
SH. IN A2	WRC-297	0.986	Shell	Passed
SH. OUT NOZ B2	WRC-297	0.986	Shell	Passed
CH. IN NOZ. A1	WRC-297	0.886	Nozzle	Passed

Foundation/Support Summary

Saddle Parameters:

Saddle Width	240.000 mm.
Saddle Bearing Angle	120.000 deg.
Centerline Dimension	1675.000 mm.
Wear Pad Width	340.000 mm.
Wear Pad Thickness	16.000 mm.
Wear Pad Bearing Angle	132.000 deg.
Distance from Saddle to Tangent	2200.000 mm.
Baseplate Length	2040.000 mm.
Baseplate Thickness	56.000 mm.
Baseplate Width	280.000 mm.
Number of Ribs (including outside ribs)	4
Rib Thickness	16.000 mm.
Web Thickness	16.000 mm.
Height of Center Web	306.000 mm.
Number of Bolts in Baseplate	3

Baseplate Sketch



Summary of Maximum Saddle Loads, Operating Case, Un-Factored:

Maximum Vertical Saddle Load	132693.86	Kgf
Maximum Transverse Saddle Shear Load	9774.60	Kgf
Maximum Longitudinal Saddle Shear Load	19549.20	Kgf

Summary of Maximum Saddle Loads, Pressure Test Case:

Maximum Vertical Saddle Load	109295.68	Kgf
Maximum Transverse Saddle Shear Load	765.07	Kgf
Maximum Longitudinal Saddle Shear Load	332.81	Kgf

Weights:



Fabricated - Bare w/o removable internals	167000.1 kg.
Shop Test - Fabricated + water (full)	214622.4 kg.
Shipping - Fab. + rem. intls.+ shipping app.	167000.1 kg.
Erected - Fab. + rem. intls.+ insul. (etc)	177000.1 kg.
Empty - Fab. + intls. + details + wghts.	177000.1 kg.
Operating - Empty + operating liquid (no ca)	202514.7 kg.

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Nozzle Calculation Summary (bars & mm.):

Description	MAWP	Ext	MAPNC	UG-45	[tr]	Weld Path	Areas or Stresses
TS OUT NOZ. T2	11	...	...	OK	10.08	OK	Passed
DRAIN NOZ. D1	31	...	...	OK	6.22	OK	No Calc[*]
SH. IN A2	192	...	...	OK	49.01	OK	Passed
SH. OUT NOZ B2	192	...	...	OK	49.04	OK	Passed
CH. IN NOZ. A1	11	...	...	OK	10.15	OK	Passed
SAFETY VALVE S1	19	...	...	OK	8.26	OK	Passed
DRAIN NOZ. D2	31	...	...	OK	6.22	OK	No Calc[*]

Nozzle MAWP Summary:

Minimum MAWP Nozzles : 192 Nozzle : SH. IN A2 [Shellside]
 Minimum MAWP Nozzles : 11 Nozzle : CH. IN NOZ. A1 [Tubeside]

[*] - This was a small opening and the areas were not computed.

Note: MAWPs (Internal Case) shown above are at the High Point.

Multiple output lines for the same nozzle indicates required Code calculations in both the longitudinal and circumferential planes of reinforcement where applicable.

Check the spatial relationship between the nozzles:

From Node	Nozzle Description	X Coordinate mm.	Layout Angle deg	Dia. Limit mm.
30	TS OUT NOZ. T2	1060.000	90.000	1485.200
30	DRAIN NOZ. D1	1060.000	-90.000	92.260
40	SH. IN A2	3478.999	90.000	1225.200
40	SH. OUT NOZ B2	11758.998	-90.000	1225.200
50	CH. IN NOZ. A1	14177.998	0.000	1485.200
50	SAFETY VALVE S1	14177.998	90.000	186.650
50	DRAIN NOZ. D2	14127.998	-90.000	92.260

The nozzle spacing is computed by the following:

$$= \sqrt{ll^2 + lc^2} \text{ where}$$

ll - Arc length along the inside vessel surface in the long. direction.
 lc - Arc length along the inside vessel surface in the circ. direction

If any interferences/violations are found, they will be noted below.

No interference violations have been detected!

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Nozzle Schedule:

Description	Nominal or Actual Size	Schd or FVC Type	Flg Type	Nozzle O/Dia in	Wall Thk mm	Reinforcing Diameter	Pad Thk mm	Cut Length mm	Flg Class
DRAIN NOZ. D1	1.500 in	160	LWN	1.900	7.137	...	...	225.30	150
DRAIN NOZ. D2	1.500 in	160	LWN	1.900	7.137	...	...	225.30	150
SAFETY VALVE S1	4.000 in	160	LWN	4.500	13.487	...	...	326.68	150
SH. IN A2	24.000 in Actual		None	28.724	60.000	...	...	758.23	...
SH. OUT NOZ B2	24.000 in Actual		None	28.724	60.000	...	...	758.23	...
TS OUT NOZ. T2	30.000 in	20	LWN	30.000	12.700	1002.00	25.00	452.96	150
CH. IN NOZ. A1	30.000 in	20	LWN	30.000	12.700	1002.00	25.00	452.96	150

General Notes for the above table:

The cut length is the outside projection + inside projection + drop + in-plane shell thickness. This value does not include weld gaps, nor does it account for shrinkage.

In the case of oblique nozzles, the outside diameter must be increased. The re-pad width around the nozzle is calculated as follows:

Width of Pad = (Pad Outside Dia. (per above) - Nozzle Outside Dia.)/2

For hub nozzles, the thickness and diameter shown are those of the smaller and thinner section.

Nozzle Material and Weld Fillet Leg Size Details (mm.):

Description	Material	Shl Grve Weld	Noz Shl/Pad Weld	Pad OD Weld	Pad Grve Weld	Inside Weld
DRAIN NOZ.	SA-106 B	25.000	10.000	...	...	...
DRAIN NOZ.	SA-106 B	25.000	10.000	...	...	...
SAFETY VALV	SA-106 B	8.738	10.000	...	...	...
SH. IN A2	SA-266 2	80.000	10.000	...	...	...
SH. OUT NOZ	SA-266 2	80.000	10.000	...	...	...
TS OUT NOZ.	SA-106 B	...	16.000	20.000	25.000	...
CH. IN NOZ.	SA-106 B	...	16.000	20.000	25.000	...

Note: The Outside projections below do not include the flange thickness.

Nozzle Miscellaneous Data:

Description	Elev/Distance From Datum mm.	Layout Angle deg	Proj Outside mm.	Proj Inside mm.	Installed in Component
DRAIN NOZ. D1	1060.000	-90.0	200.00	0.00	CH. SHELL-1
DRAIN NOZ. D2	14127.997	-90.0	200.00	0.00	CH. SHELL-2
SAFETY VALVE S1	14177.998	90.0	300.00	0.00	CH. SHELL-2
SH. IN A2	3478.999	90.0	455.00	0.00	MAIN SHELL
SH. OUT NOZ B2	11758.998	-90.0	455.00	0.00	MAIN SHELL
TS OUT NOZ. T2	1060.000	90.0	350.00	0.00	CH. SHELL-1
CH. IN NOZ. A1	14177.998	0.0	350.00	0.00	CH. SHELL-2

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Bill of Materials:

QTY	DESCRIPTION	MATERIAL
2	BLIND FLANGE: 110.0mm. THK X 2148.0mm. OD	SA-266 2
2	WELD NECK FLANGE: 115.0mm. THK X 2148.0mm. OD	SA-266 2
2	CYLINDER: 25.0mm. THK X 1940.0mm. ID X 1600.0mm.	SA-516 70
1	CYLINDER: 149.0mm. THK X 1940.0mm. ID X 10820.0mm.	SA-516 70
16	BAFFLES: 1920.0mm DIA X 16.0mm THK	SA-516 70
12	TIE RODS: 19mm DIA X 9500mm	SA-193 B7
2	TUBESHEET: 2240.0mm X 349.0mm THK	SA-266 2
3297	TUBES: 10820.0mm X 19.0mm DIA X 2.8mm THK	SA-210 A-1
2	SADDLE: 240.0mm X 120 DEG	SA-285 C
2	CLASS 150 GR 1.1, 1.5" BLIND FLANGE(S)	SA-105
1	CLASS 150 GR 1.1, 4.0" BLIND FLANGE(S)	SA-105
2	GASKET: 1972.0mm. OD X 1946.0mm. ID	...
96	BODY FLANGE BOLTS: 30.0mm. DIA	SA-193 B7
192	NUTS FOR BODY FLANGE BOLTS: 30.0mm. DIA	...
1	NAMEPLATE	...

Minimum Design Metal Temperature Results Summary (°C):

Description Notes	Curve	Basic MDMT	Reduced MDMT	UG-20(f) MDMT	Thickness ratio	Gov Thk mm.	E*	PWHT reqd
FIXED SADDLE [24]	B	-31						No
SLIDING SADDL [24]	B	-31						No
CH. COVER-1 [11]	B	0	-3		0.944	26.250	1.00	No
CH. FLG-1 [11]	B	-1	-104	-29	0.346	25.000	1.00	No
CH. FLG-2 [11]	B	-1	-104	-29	0.346	25.000	1.00	No
CH. COVER-2 [11]	B	0	-3		0.944	26.250	1.00	No
CH. SHELL-1 [8]	B	-1	-104	-29	0.343	25.000	1.00	No
CH. SHELL-2 [8]	B	-1	-104	-29	0.343	25.000	1.00	No
TS OUT NOZ. T2 [1]	B	-26	-48	-29	0.390	11.113	1.00	No
Nozzle Flg [5]	A	-18	-48					
DRAIN NOZ. D1 [1]	B	-29	-104	-29	0.053	6.245	1.00	No
Nozzle Flg [5]	A	-18	-48					
CH. IN NOZ. A1 [1]	B	-26	-48	-29	0.394	11.113	1.00	No
Nozzle Flg [5]	B	-46	-48					
SAFETY VALVE S [1]	B	-24	-104	-29	0.045	11.801	1.00	No
Nozzle Flg [5]	A	-18	-48					
DRAIN NOZ. D2 [1]	B	-29	-104	-29	0.053	6.245	1.00	No
Nozzle Flg [5]	B	-18	-48					
Tubesheet: CS [14]	B	29	-104		0.160	87.250	1.00	No
Bolting [21]		-48						
Bolting [21]		-48						
Warmest MDMT:		29	-3					

Exchanger Side	Computed MDMT	Required MDMT	Pass/Fail
Channel/Tube	-3.0	0.0	Pass

Notes:

- [!] - This was an impact tested material.
- [1] - Governing Nozzle Weld.
- [4] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-c).
- [5] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-b).
- [6] - MDMT Calculations at the Shell/Head Joint.
- [7] - MDMT Calculations for the Straight Flange.
- [8] - Cylinder/Cone/Flange Junction MDMT.
- [9] - Calculations in the Spherical Portion of the Head.
- [10] - Calculations in the Knuckle Portion of the Head.
- [11] - Calculated (Body Flange) Flange MDMT.
- [12] - Calculated Flat Head MDMT per UCS-66.3
- [13] - Tubesheet MDMT, shell side, if applicable
- [14] - Tubesheet MDMT, tube side, if applicable
- [15] - Nozzle Material
- [16] - Shell or Head Material
- [17] - Impact Testing required
- [18] - Impact Testing not required, see UCS-66(b)(3)
- [20] - Cylinder/Cone Junction MDMT based on Longitudinal Stress considerations
- [21] - Body Flange Bolting Material
- [22] - Nozzle Flange Bolting Material
- [23] - Stiffening Ring to Shell Weld
- [24] - Saddle to Shell Weld

UG-84(b)(2) was not considered.
 UCS-66(g) was not considered.
 UCS-66(i) was not considered.

Notes:

Impact test temps were not entered in and not considered in the analysis.
 UCS-66(i) applies to impact tested materials not by specification and
 UCS-66(g) applies to materials impact tested per UG-84.1 General Note (c).
 The Basic MDMT includes the (30F) PWHT credit if applicable.





Class From To : Basic Element Checks.

Class From To: Check of Additional Element Data

There were no geometry errors or warnings.

PV Elite performs all calculations internally in Imperial Units to remain compliant with the ASME Code and any built in assumptions in the ASME Code formulas. The finalized results are reflected to show the set of selected units for this analysis.

Information Regarding Nozzle Loads

Please note that nozzle loadings, if included, are assumed to be local in nature and will not contribute to or create a net section bending moment. Therefore, the addition of nozzle loads will not affect the support load calculation. If you wish to create a load on the support(s) from nozzle loads, you can enable that feature in tools->configuration->Job Specific Settings.



The level of precision is set to a value of less than 3. At this level you may not obtain proper resolution on some results. This value can be changed in Tools->Set Configuration Parameters->Default Values->Level of Precision.



The ASME Code Section VIII Division 1 does not explicitly require stresses on elements to be computed under test conditions. Please see interpretation VIII-1-10-12 for more information. PV Elite also computes weight due to test fluid and uses the weight in various calculations such as saddle and leg loads. PV Elite also computes static pressure due to operating and test liquids. The static pressure is used in conjunction with the design and test pressures to determine a total pressure in the determination of required thickness, MAWP etc. This ensures compliance with UG-22 requirements.

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Units used in this Analysis (Kgfcmm2mm):

Name	System Unit	Constant	User Unit
Length	Feet	304.8000	mm.
Force	Pounds	0.4536	Kgf
Mass	Pounds	0.4536	kg.
Area	sq. inches	6.4516	cm ²
Moment	ft. lbs.	0.1383	Kg-m.
Stress	lbs./sq.in.	0.0069	N./mm ²
Temperature	Degrees F	0.5556	°C
Pressure	psig	0.0689	bars
Elast. Modulus	lbs./sq.in.	6.8946	KPa.
Pipe Density	lbs./cu.in.	0.0277	kg./cm ³
Ins. Density	lbs./cu.ft.	0.1600E-04	kg./cm ³
Fluid Density	lbs./cu.ft.	0.1600E-04	kg./cm ³
Wind Speed	miles/hr	1.6093	Km/hr
Tray Weight	lbs./sq.ft.	0.0005	Kgs/cm ²
Inertia	in.4	41.6231	cm ⁴
G Load	G's	1.0000	g's
Wind Load	lbs./sq.ft.	4.8825	Kgs/m ²
Elevation	Feet	304.8000	mm.
Volume	in.3	16.3900	cm ³
Diameter	inches	25.4000	mm.
Thickness	inches	25.4000	mm.

PV Elite Vessel Analysis Program: Input Data

Exchanger Design Pressures and Temperatures

Shell Side Design Pressure	191	bars
Channel Side Design Pressure	10	bars
Shell Side Design Temperature	260.0	°C
Channel Side Design Temperature	85.0	°C
Radiography, Shell Side	RT-1	
Radiography, Channel Side	RT-1	
Service Type, Shell Side	Sour	
Service Type, Channel Side	Air/Water/Steam	
MDMT (CET), Shell Side	0.0	°C
MDMT (CET), Tube Side	0.0	°C
User defined MAWP, Shell Side	0	bars
User defined MAWP, Channel Side	0	bars
User defined MAPnc, Shell Side	0	bars
User defined MAPnc, Channel Side	0	bars
User defined Test Pres., Shell Side	0	bars
User defined Test Pres., Channel Side	0	bars

Projection of Nozzle from Vessel Top	0	mm.
Projection of Nozzle from Vessel Bottom	0	mm.
Type of Construction	Welded	
Use Higher Longitudinal Stresses (Flag)	Y	
Select t for Internal Pressure (Flag)	N	
Select t for External Pressure (Flag)	N	
Select t for Axial Stress (Flag)	N	
Select Location for Stiff. Rings (Flag)	N	
Consider Vortex Shedding	N	

Shop Pressure Test:

Type of Pressure Test	UG-99(b)
Pressure Test Position	Horizontal
Test Performed in Corroded Condition	Yes

Load Case 1	NP+EW+WI+FW+BW
Load Case 2	NP+EW+EE+FS+BS
Load Case 3	NP+OW+WI+FW+BW
Load Case 4	NP+OW+EQ+FS+BS

Load Case 5	NP+HW+HI
Load Case 6	NP+HW+HE
Load Case 7	IP+OW+WI+FW+BW
Load Case 8	IP+OW+EQ+FS+BS
Load Case 9	EP+OW+WI+FW+BW
Load Case 10	EP+OW+EQ+FS+BS
Load Case 11	HP+HW+HI
Load Case 12	HP+HW+HE
Load Case 13	IP+WE+EW
Load Case 14	IP+WF+CW
Load Case 15	IP+VO+OW
Load Case 16	IP+VE+EW
Load Case 17	NP+VO+OW
Load Case 18	FS+BS+IP+OW
Load Case 19	FS+BS+EP+OW
Load Case 20	BL+IP+OW

Wind Design Code	ASCE/SEI 7-22	
Wind Load Reduction Scale Factor	1.100	
Basic Wind Speed	[V] 180	Km/hr
Surface Roughness Category	D: Flat, unobstructed	
Type of Surface	0	
Base Elevation	300	mm.
Percent Wind for Hydrotest	33.0	
Using User defined Wind Press. Vs Elev.	N	
Height of Hill or Escarpment	H or Hh 0	mm.
Distance Upwind of Crest	Lh 0	mm.
Distance from Crest to the Vessel	x 0	mm.
Type of Terrain (Hill, Escarpment)	Flat	
Damping Factor (Beta) for Wind (Ope)	0.0100	
Damping Factor (Beta) for Wind (Empty)	0.0000	
Damping Factor (Beta) for Wind (Filled)	0.0000	
Site Elevation above Sea Level	0	mm.

Seismic Design Code	ASCE/SEI 7-22	
Importance Factor	1.500	
Force Modification Factor R	3.000	
Site Class	D	
Component Elevation Ratio	z/h 0.000	
Amplification Factor	Ap 0.000	
Force Factor	0.000	
Consider Vertical Acceleration	Yes	
Minimum Acceleration Multiplier	0.000	
User Value of Sds (used if > 0)	0.000	
User Value of Sd1 (used if > 0)	0.000	
Component resonance duct. fact.	[CAR] 0.000	
Component strength factor	[Rpo] 0.000	
Overstrength factor	[Omega0] 0.000	
Response Acceleration Factor	[SMS] 0.326	
Response Acceleration Factor	[SM1] 0.139	
Building Importance Factor	[Ie] 1.000	
Site Latitude	0.000	deg
Site Longitude	0.000	deg

Design Pressure + Static Head	Y
Consider MAP New and Cold in Noz. Design	N
Consider External Loads for Nozzle Des.	Y
Use ASME VIII-1 Appendix 1-9	N

Perform Blast Load Analysis	No
Material Database Year	Current w/Addenda or Code Year

Configuration Directives:

g0 is taken as min(g0, attached shell thk)	Yes
Do not use Nozzle MDMT Interpretation VIII-1 01-37	No
Use Table G instead of exact equation for "A"	Yes
Shell Head Joints are Tapered	Yes
Compute "K" in corroded condition	Yes
Use Code Case 2286	No



Use the MAWP to compute the MDMT Yes
 For thickness ratios ≤ 0.35 , MDMT will be -155F (-104C) Yes
 For PWHT & P1 Materials the MDMT can be $< -55F (-48C)$ No
 Using Metric Material Databases, ASME II D Yes
 Calculate B31.3 type stress for Nozzles with Loads Yes
 Reduce the MDMT due to lower membrane stress Yes
 Consider Longitudinal Stress in MDMT Calculations Yes

Complete Listing of Vessel Elements and Details:

Element#	1/7
Element From Node	10
Element To Node	20
Element Type	Flange
Description	CH. COVER-1
Distance "FROM" to "TO"	105 mm.
Flange Outside Diameter	2148 mm.
Element Thickness	105 mm.
Internal Corrosion Allowance	3 mm.
Nominal Thickness	110 mm.
External Corrosion Allowance	0 mm.
Design Internal Pressure	10 bars
Design Temperature Internal Pressure	85 °C
Design External Pressure	0 bars
Design Temperature External Pressure	85 °C
Effective Diameter Multiplier	1.2
Material Name	SA-266 2
Allowable Stress, Ambient	138 N./mm ²
Allowable Stress, Operating	138 N./mm ²
Allowable Stress, Hydrotest	225 N./mm ²
Material Density	0.00775 kg./cm ³
P Number Thickness	32.004 mm.
Yield Stress, Operating	229.57 N./mm ²
UCS-66 Chart Curve Designation	B
External Pressure Chart Name	CS-2
UNS Number	K03506
Product Form	Forgings
Perform Flange Stress Calculation (Y/N)	Y
Weld is pre-Heated	No

Element#	2/7
Element From Node	20
Element To Node	30
Element Type	Flange
Description	CH. FLG-1
Distance "FROM" to "TO"	150 mm.
Flange Inside Diameter	1940 mm.
Element Thickness	110 mm.
Internal Corrosion Allowance	3 mm.
Nominal Thickness	115 mm.
External Corrosion Allowance	0 mm.
Design Internal Pressure	10 bars
Design Temperature Internal Pressure	85 °C
Design External Pressure	0 bars
Design Temperature External Pressure	85 °C
Effective Diameter Multiplier	1.2
Material Name	SA-266 2
Perform Flange Stress Calculation (Y/N)	Y
Weld is pre-Heated	No

Element#	3/7
Element From Node	30
Element To Node	40
Element Type	Cylinder
Description	CH. SHELL-1



Distance "FROM" to "TO"	1600	mm.
Inside Diameter	1940	mm.
Element Thickness	25	mm.
Internal Corrosion Allowance	3	mm.
Nominal Thickness	25	mm.
External Corrosion Allowance	0	mm.
Design Internal Pressure	10	bars
Design Temperature Internal Pressure	85	°C
Design External Pressure	0	bars
Design Temperature External Pressure	85	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-516 70	
Allowable Stress, Ambient	138	N./mm ²
Allowable Stress, Operating	138	N./mm ²
Allowable Stress, Hydrotest	234	N./mm ²
Material Density	0.00775	kg./cm ³
P Number Thickness	32.004	mm.
Yield Stress, Operating	242	N./mm ²
UCS-66 Chart Curve Designation	B	
External Pressure Chart Name	CS-2	
UNS Number	K02700	
Product Form	Plate	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	

Element From Node	30	
Detail Type	Liquid	
Detail ID	Liquid: 30	
Dist. from "FROM" Node / Offset dist	0	mm.
Height/Length of Liquid	955	mm.
Liquid Density	0.0009934	kg./cm ³

Element From Node	30	
Detail Type	Nozzle	
Detail ID	TS OUT NOZ. T2	
Dist. from "FROM" Node / Offset dist	800	mm.
Nozzle Diameter	30	in.
Nozzle Schedule	20	
Nozzle Class	150	
Layout Angle	90.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	225.34	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-106 B	

Element From Node	30	
Detail Type	Nozzle	
Detail ID	DRAIN NOZ. D1	
Dist. from "FROM" Node / Offset dist	800	mm.
Nozzle Diameter	1.5	in.
Nozzle Schedule	160	
Nozzle Class	150	
Layout Angle	-90.0	
Blind Flange (Y/N)	Y	
Weight of Nozzle (Used if > 0)	5.3017	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-106 B	

Element#	4/7	
Element From Node	40	
Element To Node	50	
Element Type	Cylinder	
Description	MAIN SHELL	
Distance "FROM" to "TO"	10820	mm.
Inside Diameter	1940	mm.
Element Thickness	149	mm.
Internal Corrosion Allowance	1.5	mm.



Nominal Thickness	149	mm.
External Corrosion Allowance	0	mm.
Design Internal Pressure	191	bars
Design Temperature Internal Pressure	260	°C
Design External Pressure	0	bars
Design Temperature External Pressure	260	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-516 70	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	

Element From Node	40	
Detail Type	Saddle	
Detail ID	FIXED SADDLE	
Dist. from "FROM" Node / Offset dist	2200	mm.
Width of Saddle	240	mm.
Height of Saddle at Bottom	1675	mm.
Saddle Contact Angle	120.0	
Height of Composite Ring Stiffener	0	mm.
Width of Wear Plate	340	mm.
Thickness of Wear Plate	16	mm.
Contact Angle, Wear Plate (degrees)	132.0	
Friction coefficient	0.60000002	
Moment Factor	3.0	
Dimension E at base (optional)	0	mm.
Circumferential Eff. over Saddle	1.0	
Circumferential Eff. at Midspan	1.0	
Tangent to Tangent dist. (optional)	0	mm.

Element From Node	40	
Detail Type	Saddle	
Detail ID	SLIDING SADDLE	
Dist. from "FROM" Node / Offset dist	8620	mm.
Width of Saddle	240	mm.
Height of Saddle at Bottom	1675	mm.
Saddle Contact Angle	120.0	
Height of Composite Ring Stiffener	0	mm.
Width of Wear Plate	340	mm.
Thickness of Wear Plate	16	mm.
Contact Angle, Wear Plate (degrees)	132.0	
Friction coefficient	0.1	
Moment Factor	3.0	
Dimension E at base (optional)	0	mm.
Circumferential Eff. over Saddle	1.0	
Circumferential Eff. at Midspan	1.0	
Tangent to Tangent dist. (optional)	0	mm.

Element From Node	40	
Detail Type	Liquid	
Detail ID	Liquid: 40	
Dist. from "FROM" Node / Offset dist	0	mm.
Height/Length of Liquid	1940	mm.
Liquid Density	0.0005891	kg./cm ³

Element From Node	40	
Detail Type	Nozzle	
Detail ID	SH. IN A2	
Dist. from "FROM" Node / Offset dist	1270	mm.
Nozzle Diameter	24	in.
Nozzle Schedule	None	
Nozzle Class	None	
Layout Angle	90.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	2067	Kgf
Grade of Attached Flange	None	
Nozzle Matl	SA-266 2	

Element From Node	40	
Detail Type	Nozzle	



Detail ID	SH. OUT NOZ B2	
Dist. from "FROM" Node / Offset dist	9550	mm.
Nozzle Diameter	24	in.
Nozzle Schedule	None	
Nozzle Class	None	
Layout Angle	-90.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	2067	Kgf
Grade of Attached Flange	None	
Nozzle Matl	SA-266 2	

Element From Node	40	
Detail Type	Weight	
Detail ID	Margin on weigh	
Dist. from "FROM" Node / Offset dist	5410	mm.
Miscellaneous Weight	10000	Kgf
Offset from Element Centerline	0	mm.

Element From Node	40	
Detail Type	Weight	
Detail ID	PDS weight diff	
Dist. from "FROM" Node / Offset dist	5410	mm.
Miscellaneous Weight	5920	Kgf
Offset from Element Centerline	0	mm.

Element#	5/7	
Element From Node	50	
Element To Node	60	
Element Type	Cylinder	
Description	CH. SHELL-2	
Distance "FROM" to "TO"	1600	mm.
Inside Diameter	1940	mm.
Element Thickness	25	mm.
Internal Corrosion Allowance	3	mm.
Nominal Thickness	25	mm.
External Corrosion Allowance	0	mm.
Design Internal Pressure	10	bars
Design Temperature Internal Pressure	85	°C
Design External Pressure	0	bars
Design Temperature External Pressure	85	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-516 70	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	

Element From Node	50	
Detail Type	Liquid	
Detail ID	Liquid: 50	
Dist. from "FROM" Node / Offset dist	0	mm.
Height/Length of Liquid	1940	mm.
Liquid Density	0.0009934	kg./cm ³

Element From Node	50	
Detail Type	Nozzle	
Detail ID	CH. IN NOZ. A1	
Dist. from "FROM" Node / Offset dist	800	mm.
Nozzle Diameter	30	in.
Nozzle Schedule	20	
Nozzle Class	150	
Layout Angle	0.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	225.34	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-106 B	

Element From Node	50	
Detail Type	Nozzle	



Detail ID	SAFETY VALVE S1	
Dist. from "FROM" Node / Offset dist	800	mm.
Nozzle Diameter	4	in.
Nozzle Schedule	160	
Nozzle Class	150	
Layout Angle	90.0	
Blind Flange (Y/N)	Y	
Weight of Nozzle (Used if > 0)	25.849	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-106 B	

Element From Node	50	
Detail Type	Nozzle	
Detail ID	DRAIN NOZ. D2	
Dist. from "FROM" Node / Offset dist	750	mm.
Nozzle Diameter	1.5	in.
Nozzle Schedule	160	
Nozzle Class	150	
Layout Angle	-90.0	
Blind Flange (Y/N)	Y	
Weight of Nozzle (Used if > 0)	5.3017	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-106 B	

Element#	6/7	
Element From Node	60	
Element To Node	70	
Element Type	Flange	
Description	CH. FLG-2	
Distance "FROM" to "TO"	150	mm.
Flange Inside Diameter	1940	mm.
Element Thickness	110	mm.
Internal Corrosion Allowance	3	mm.
Nominal Thickness	115	mm.
External Corrosion Allowance	0	mm.
Design Internal Pressure	10	bars
Design Temperature Internal Pressure	85	°C
Design External Pressure	0	bars
Design Temperature External Pressure	85	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-266 2	
Allowable Stress, Ambient	138	N./mm ²
Allowable Stress, Operating	138	N./mm ²
Allowable Stress, Hydrotest	225	N./mm ²
Material Density	0.00775	kg./cm ³
P Number Thickness	32.004	mm.
Yield Stress, Operating	229.57	N./mm ²
UCS-66 Chart Curve Designation	B	
External Pressure Chart Name	CS-2	
UNS Number	K03506	
Product Form	Forgings	
Perform Flange Stress Calculation (Y/N)	Y	
Weld is pre-Heated	No	

Element From Node	60	
Detail Type	Liquid	
Detail ID	Liquid: 60	
Dist. from "FROM" Node / Offset dist	0	mm.
Height/Length of Liquid	955	mm.
Liquid Density	0.0009984	kg./cm ³

Element#	7/7	
Element From Node	70	
Element To Node	80	
Element Type	Flange	
Description	CH. COVER-2	



Distance "FROM" to "TO"	105	mm.
Flange Outside Diameter	2148	mm.
Element Thickness	105	mm.
Internal Corrosion Allowance	3	mm.
Nominal Thickness	110	mm.
External Corrosion Allowance	0	mm.
Design Internal Pressure	10	bars
Design Temperature Internal Pressure	85	°C
Design External Pressure	0	bars
Design Temperature External Pressure	85	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-266 2	
Perform Flange Stress Calculation (Y/N)	Y	
Weld is pre-Heated	No	



XY Coordinate Calculations:

From	To	X (Horiz.) mm.	Y (Vert.) mm.	DX (Horiz.) mm.	DY (Vert.) mm.
CH. COVER-1		110	...	105	...
CH. FLG-1		260	...	150	...
CH. SHELL-1		1860	...	1600	...
MAIN SHELL		13029	...	10820	...
CH. SHELL-2		14978	...	1600	...
CH. FLG-2		15131.2	...	150	...
CH. COVER-2		15236.2	...	105	...

The calculated dimensions are based on the given element lengths. Due to variability in manufacturing (weld gaps etc.), the Tangent to Tangent length may not be exact. The length includes gasket thicknesses and raised face heights.

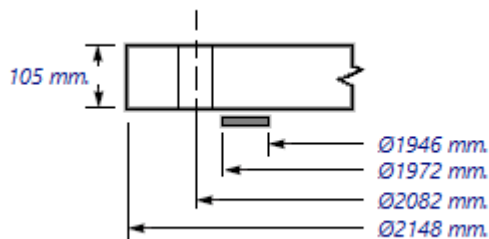
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Flange Input Data Values

Description: CH. COVER-1 :

CH. COVER-1

Flange Type		Blind	
Design Pressure	P	10.00	bars
Design Temperature		85	°C
Internal Corrosion Allowance	ci	3.0000	mm.
External Corrosion Allowance	ce	0.0000	mm.
Use Corrosion Allowance in Thickness Calcs.		Yes	
Flange Outside Diameter	A	2148.000	mm.
Flange Thickness	t	105.0000	mm.
Flange Material		SA-266 2	
Flange Material UNS number		K03506	
Flange Allowable Stress At Temperature	Sfo	138.00	N./mm ²
Flange Allowable Stress At Ambient	Sfa	138.00	N./mm ²
Bolt Material		SA-193 B7	
Bolt Allowable Stress At Temperature	Sb	172.00	N./mm ²
Bolt Allowable Stress At Ambient	Sa	172.00	N./mm ²
Perimeter along the Center of the Bolts	L	6540.796	mm.
Diameter of Bolt Circle	C	2082.000	mm.
Nominal Bolt Diameter	a	30.0000	mm.
Type of Threads		TEMA Metric	
Number of Bolts		48	
Flange Face Outside Diameter	Fod	1978.000	mm.
Flange Face Inside Diameter	Fid	1940.000	mm.
Flange Facing Sketch		Sketch 1a	
Gasket Outside Diameter	Go	1972.000	mm.
Gasket Inside Diameter	Gi	1946.000	mm.
Gasket Factor	m	3.7500	
Gasket Design Seating Stress	y	62.06	N./mm ²
Gasket Thickness	tg	5.0000	mm.
Raised face depth	Fd	0.0000	mm.



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Gasket Contact Width,	$N = (G_o - G_i)/2$	13.000 mm.
Basic Gasket Width,	$bo = N/2$	6.500 mm.
Effective Gasket Width,	$b = C_b \cdot \sqrt{bo}$	6.425 mm.
Gasket Reaction Diameter,	$G = G_o - 2 \cdot b$	1959.151 mm.

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:
 $= \pi/4 \cdot G^2 \cdot P$
 $= \pi/4 \cdot 1959.1509^2 \cdot 10.0$
 $= 307413.031 \text{ Kg}$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P$$

$$= 2 \cdot 6.4246 \cdot \pi \cdot 1959.1509 \cdot 3.75 \cdot 10.0$$

$$= 30242.582 \text{ Kgf}$$

Design Bolt Load for the Operating Condition [Wo]:

$$= \max(Wo,1 + Wo,2 + H'p, 0)$$

$$= \max(307413 + 30243 + 0, 0)$$

$$= 337655.594 \text{ Kgf}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$= y \cdot b \cdot \pi \cdot G + y_{Part} \cdot b_{Part} \cdot l_p$$

$$= 62.06 \cdot 6.4246 \cdot \pi \cdot 1959.151 + 0.0 \cdot 0.0 \cdot 0.0$$

$$= 250212.594 \text{ Kgf}$$

Required Bolt Area [Am]:

$$= \max((Wo + Fa + 4Me/G) / Sb, Wg / Sa)$$

$$= \max((337656 + 0 + 4 \cdot 0 / 1959.15) / 172, 250213 / 172)$$

$$= 192.523 \text{ cm}^2$$

Bolting Information for TEMA Metric Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	192.523	241.423	
Radial Distance between Bolts and Edge:	32.500	33.000	
Circ. Spacing between the Bolts:	69.900	136.169	...

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$= Ab \cdot Sa / (y \cdot \pi (Go + Gi))$$

$$= 241.423 \cdot 172.0 / (62.06 \cdot \pi (1972.0 + 1946.0))$$

$$= 5.436 \text{ mm.}$$

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$= Sa \cdot Ab$$

$$= 172.0 \cdot 241.4232$$

$$= 423419.59 \text{ Kgf}$$

Gasket Load for the Operating Condition [HG]:

$$= Wo$$

$$= 337655.59 \text{ Kgf}$$

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$= (C - G) / 2$$

$$= (2082.0 - 1959.1509) / 2$$

$$= 61.4245 \text{ mm.}$$

Flange Stresses

Tangential Flange Stress, Flat Head (4.6.2.3), Operating [STo]:

$$= 1.9 \cdot Wm1 \cdot hG \cdot Bsc / (t^2 \cdot G) + C \cdot Z \cdot Peq \cdot G^2 / t^2$$

$$= 1.9 \cdot 337656 \cdot 61.4245 \cdot 1.0 / (102.0^2 \cdot 1959.1509) + 0.3 \cdot 1.0 \cdot 10.0 \cdot 1959.1509^2 / 102.0^2$$

$$= 129.64 \text{ N./mm}^2$$

Tangential Flange Stress, Flat Head (4.6.2.4), Seating [STa]:

$$= 1.9 \cdot W \cdot hG \cdot Bsc / (t^2 \cdot G)$$

$$= 1.9 \cdot 423420 \cdot 61.4245 \cdot 1.0 / (102.0^2 \cdot 1959.1509)$$

$$= 23.78 \text{ N./mm}^2$$

Bolt Stress, Operating [BSO]:

$$= Wm1 / Ab$$

$$= 337656 / 241.4232$$

$$= 137.16 \text{ N./mm}^2$$

Bolt Stress, Seating [BSa]:

$$= (Wm2 / Ab)$$

$$= (250213 / 241.4232)$$



= 101.64 N./mm²

Flange Stress Analysis Results: N./mm²

	Actual	Operating Allowed	Gasket Seating Actual Allowed
Tangential Flange	129.64	138.00	23.78 138.00
Bolting	137.16	172.00	101.64 172.00

Reqd. Blind Flange Thickness at Center 101.864 mm.
 Reqd. Blind Flange Thickness at Gasket 45.338 mm.

Estimated M.A.W.P. (Operating) 11 bars
 Estimated Finished Weight of Flange at given Thk. 2949.0 kg.
 Estimated Unfinished Weight of Forging at given Thk 2949.0 kg.

Minimum Design Metal Temperature Results:

Thickness Ratio = 0.944, Temperature Reduction per Fig. UCS 66.1 = 3 °C

Min Metal Temp. w/o impact per UCS-66, Curve B 0 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -3 °C

Note: UCS-66(b)(-c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: N./mm²

	Actual	Operating Allowed	Gasket Seating Actual Allowed
Tangential Flange	122.34	138.00	22.44 138.00
Bolting	137.16	172.00	101.64 172.00

Minimum Req'd Thickness, ambient, no Corrosion: 98.882 mm.

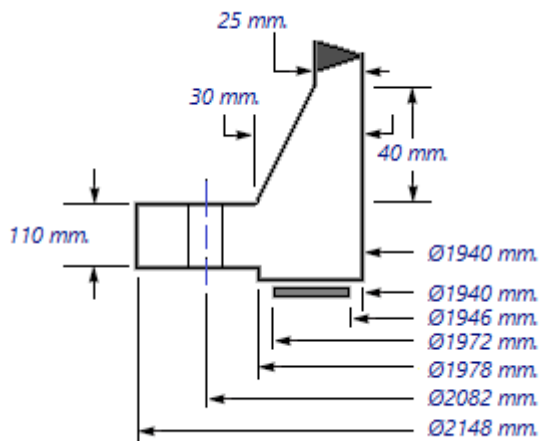
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Flange Input Data Values

Description: CH. FLG-1 :

CH. FLG-1

Flange Type	Integral Weld Neck		
Design Pressure	P	10.00	bars
Design Temperature		85	°C
Internal Corrosion Allowance	ci	3.0000	mm.
External Corrosion Allowance	ce	0.0000	mm.
Use Corrosion Allowance in Thickness Calcs.		Yes	
Flange Inside Diameter	B	1940.000	mm.
Flange Outside Diameter	A	2148.000	mm.
Flange Thickness	t	110.0000	mm.
Thickness of Hub at Small End	go	25.0000	mm.
Thickness of Hub at Large End	gl	30.0000	mm.
Length of Hub	h	40.0000	mm.
Flange Material		SA-266 2	
Flange Material UNS number		K03506	
Flange Allowable Stress At Temperature	Sfo	138.00	N./mm ²
Flange Allowable Stress At Ambient	Sfa	138.00	N./mm ²
Bolt Material		SA-193 B7	
Bolt Allowable Stress At Temperature	Sb	172.00	N./mm ²
Bolt Allowable Stress At Ambient	Sa	172.00	N./mm ²
Diameter of Bolt Circle	C	2082.000	mm.
Nominal Bolt Diameter	a	30.0000	mm.
Type of Threads		TEMA Metric	
Number of Bolts		48	
Flange Face Outside Diameter	Fod	1978.000	mm.
Flange Face Inside Diameter	Fid	1940.000	mm.
Flange Facing Sketch		Sketch 1a	
Gasket Outside Diameter	Go	1972.000	mm.
Gasket Inside Diameter	Gi	1946.000	mm.
Gasket Factor	m	3.7500	
Gasket Design Seating Stress	y	62.06	N./mm ²
Gasket Thickness	tg	3.1750	mm.
Raised face depth	Fd	0.0000	mm.



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Hub Small End Required Thickness due to Internal Pressure:

$$= (P \cdot (D/2 + Ca)) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (10.0 \cdot (1940.0/2 + 3.0)) / (138.0 \cdot 1.0 - 0.6 \cdot 10.0) + Ca$$

= 10.0821 mm.

Small End Hub MAWP:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 22.0) / (973.0 + 0.6 \cdot 22.0)$$

$$= 30.782 \text{ bars}$$

Hub Small End Hub MAPnc:

$$= (S_a \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 25.0) / (970.0 + 0.6 \cdot 25.0)$$

$$= 35.023 \text{ bars}$$

Corroded Flange Thickness, tc = T-ci	107.000 mm.
Corroded Flange ID, Bcor = B+2*ci	1946.000 mm.
Corroded Large Hub, g1Cor = g1-ci	27.000 mm.
Corroded Small Hub, g0Cor = go-ci	22.000 mm.
Code R Dimension, R = ((C-Bcor)/2)-g1cor	41.000 mm.

Gasket Contact Width, N = (Go - Gi)/2	13.000 mm.
Basic Gasket Width, bo = N/2	6.500 mm.
Effective Gasket Width, b = Cb*sqrt(bo)	6.425 mm.
Gasket Reaction Diameter, G = Go - 2 * b	1959.151 mm.

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:

$$= \pi / 4 \cdot G^2 \cdot P$$

$$= \pi / 4 \cdot 1959.1509^2 \cdot 10.0$$

$$= 307413.031 \text{ Kgf}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P$$

$$= 2 \cdot 6.4246 \cdot \pi \cdot 1959.1509 \cdot 3.75 \cdot 10.0$$

$$= 30242.582 \text{ Kgf}$$

Hydrostatic End Load at Flange ID [Hd]:

$$= \pi / 4 \cdot B_{cor}^2 \cdot P$$

$$= \pi / 4 \cdot 1946.0^2 \cdot 10.0$$

$$= 303299.844 \text{ Kgf}$$

Pressure Force on Flange Face [Ht]:

$$= H - H_d$$

$$= 307413 - 303300$$

$$= 4113.188 \text{ Kgf}$$

Design Bolt Load for the Operating Condition [Wo]:

$$= \max(W_{o,1} + W_{o,2} + H'p, 0)$$

$$= \max(307413 + 30243 + 0, 0)$$

$$= 337655.594 \text{ Kgf}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$= y \cdot b \cdot \pi \cdot G + y_{Part} \cdot b_{Part} \cdot l_p$$

$$= 62.06 \cdot 6.4246 \cdot \pi \cdot 1959.151 + 0.0 \cdot 0.0 \cdot 0.0$$

$$= 250212.594 \text{ Kgf}$$

Required Bolt Area [Am]:

$$= \max((W_o + F_a + 4Me/G) / S_b, W_g / S_a)$$

$$= \max((337656 + 0 + 4 \cdot 0 / 1959.15) / 172, 250213 / 172)$$

$$= 192.523 \text{ cm}^2$$

ASME Maximum Circumferential Spacing between Bolts, Table 4.16.11 [Bsmax]:

$$= 2a + 6t / (m + 0.5)$$

$$= 2 \cdot 30.0 + 6 \cdot 107.0 / (3.75 + 0.5)$$

$$= 211.059 \text{ mm.}$$

Actual Circumferential Bolt Spacing [Bs]:

$$= C \cdot \sin(\pi / n)$$

$$= 2082.0 \cdot \sin(\pi / 48)$$

$$= 136.169 \text{ mm.}$$

ASME Moment Multiplier for Bolt Spacing per Table 4.16.11 [Bsc]:

$$= \max(\sqrt{\frac{Bs}{2a + t}}, 1)$$

$$= \max(\sqrt{\frac{136.169}{2 \cdot 30.0 + 107.0}}, 1)$$

$$= 1.0000$$

Bolting Information for TEMA Metric Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	192.523	241.423	
Radial Distance between Hub and Bolts:	40.500	41.000	
Radial Distance between Bolts and Edge:	32.500	33.000	
Circ. Spacing between the Bolts:	69.900	136.169	211.059

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$= Ab \cdot Sa / (y \cdot \pi(Go + Gi))$$

$$= 241.423 \cdot 172.0 / (62.06 \cdot \pi(1972.0 + 1946.0))$$

$$= 5.436 \text{ mm.}$$

Recommended Minimum Gasket Contact Width per Table 4.16.2

Sheet Gaskets including laminated sheet gaskets with or without a metal core: 19 mm
 Preformed comp. gaskets incl. spiral wound, jacketed, & solid flat metal gaskets: 16 mm

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$= Sa \cdot Ab$$

$$= 172.0 \cdot 241.4232$$

$$= 423419.59 \text{ Kgf}$$

Gasket Load for the Operating Condition [HG]:

$$= Wo - H$$

$$= 337656 - 307413$$

$$= 30242.59 \text{ Kgf}$$

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$= (C - G) / 2$$

$$= (2082.0 - 1959.1509) / 2$$

$$= 61.4245 \text{ mm.}$$

Distance to Face Pressure Reaction [ht]:

$$= 0.5((C - B) / 2 + hg)$$

$$= 0.5((2082.0 - 1946.0) / 2 + 61.4245)$$

$$= 64.7123 \text{ mm.}$$

Distance to End Pressure Reaction [hd]:

$$= (C - B - g1) / 2$$

$$= (2082.0 - 1946.0 - 27.0) / 2$$

$$= 54.5000 \text{ mm.}$$

Summary of Moments for Internal Pressure: (Kg-m.)

Loading	Force	Distance	Bolt Corr	Moment
End Pressure, Md	303300.	54.5000	1.0000	16530.
Face Pressure, Mt	4113.	64.7123	1.0000	266.
Gasket Load, Mg	30243.	61.4245	1.0000	1858.
Gasket Seating, Matm	423420.	61.4245	1.0000	26009.

Total Moment for Operation, Mo 18654. Kg-m.
 Total Moment for Gasket seating, Mg 26009. Kg-m.

Effective Hub Length, ho = sqrt(Bcor*goCor) 206.911 mm.
 Hub Ratio, h/h0 = HL / H0 0.193
 Thickness Ratio, g1/g0 = (g1Cor/goCor) 1.227

Flange Factors for Integral Flange:

Factor F per 4.16.5 0.908
 Factor V per 4.16.5 0.468



Factor f per 4.16.5

1.034

Stress factor from Table 4.16.4 [K]:

$$\begin{aligned} &= A / B \\ &= 2148.0 / 1946.0 \\ &= 1.104 \end{aligned}$$

Stress factor from Table 4.16.4 [Y]:

$$\begin{aligned} &= (0.66845 + (5.7169(K^2 \cdot \log_{10}(K))) / (K^2 - 1)) / (K - 1) \\ &= (0.66845 + (5.7169(1.104^2 \cdot \log_{10}(1.104))) / (1.104^2 - 1)) / (1.104 - 1) \\ &= 19.619 \end{aligned}$$

Stress factor from Table 4.16.4 [T]:

$$\begin{aligned} &= (K^2(1 + 8.55246 \cdot \log_{10}(K)) - 1) / ((1.0472 + 1.9448 \cdot K^2) \cdot (K - 1)) \\ &= (1.104^2(1 + 8.55246 \cdot \log_{10}(1.104)) - 1) / ((1.0472 + 1.9448 \cdot 1.104^2) \cdot (1.104 - 1)) \\ &= 1.876 \end{aligned}$$

Stress factor from Table 4.16.4 [U]:

$$\begin{aligned} &= (K^2(1 + 8.55246 \cdot \log_{10}(K)) - 1) / (1.36136(K^2 - 1) \cdot (K - 1)) \\ &= (1.104^2(1 + 8.55246 \cdot \log_{10}(1.104)) - 1) / (1.36136(1.104^2 - 1) \cdot (1.104 - 1)) \\ &= 21.559 \end{aligned}$$

Stress factor from Table 4.16.4 [Z]:

$$\begin{aligned} &= (K^2 + 1) / (K^2 - 1) \\ &= (1.104^2 + 1) / (1.104^2 - 1) \\ &= 10.158 \end{aligned}$$

Stress factor from Table 4.16.4 for integral flanges [d]:

$$\begin{aligned} &= U \cdot g_0^2 \cdot h_0 / V \\ &= 21.559 \cdot 22.0^2 \cdot 206.911 / 0.468 \\ &= 4612139.500 \text{ mm}^3 \end{aligned}$$

Stress factor from Table 4.16.4 for integral flanges [e]:

$$\begin{aligned} &= F / h_0 \\ &= 0.908 / 206.911 \\ &= 0.004 \text{ 1/mm.} \end{aligned}$$

Stress factor from Table 4.16.4 [L]:

$$\begin{aligned} &= (t \cdot e + 1) / T + t^3/d \\ &= (107.0 \cdot 0.004 + 1) / 1.876 + 107.0^3 / 4612140 \\ &= 1.049 \end{aligned}$$

Flange Stresses

Longitudinal Hub Stress, Operating [SHo]:

$$\begin{aligned} &= (f \cdot Mop / Bcor) / (L \cdot g_1^2) \\ &= (1.0339 \cdot 18654 / 1946.0) / (1.0491 \cdot 27.0^2) \\ &= 127.09 \text{ N./mm}^2 \end{aligned}$$

Longitudinal Hub Stress, Seating [SHa]:

$$\begin{aligned} &= (f \cdot Matm / Bcor) / (L \cdot g_1^2) \\ &= (1.0339 \cdot 26009 / 1946.0) / (1.0491 \cdot 27.0^2) \\ &= 177.20 \text{ N./mm}^2 \end{aligned}$$

Radial Flange Stress, Operating [SRo]:

$$\begin{aligned} &= (\beta \cdot Mop / Bcor) / (L \cdot t^2) \\ &= (1.6263 \cdot 18654 / 1946.0) / (1.0491 \cdot 107.0^2) \\ &= 12.73 \text{ N./mm}^2 \end{aligned}$$

Radial Flange Stress, Seating [SRa]:

$$\begin{aligned} &= (\beta \cdot Matm / Bcor) / (L \cdot t^2) \\ &= (1.6263 \cdot 26009 / 1946.0) / (1.0491 \cdot 107.0^2) \\ &= 17.75 \text{ N./mm}^2 \end{aligned}$$

Tangential Flange Stress, Operating [STo]:

$$\begin{aligned} &= (Y \cdot Mo / (t^2 \cdot Bcor)) - Z \cdot SRo \\ &= (19.6189 \cdot 18654 / (107.0^2 \cdot 1946.0)) - 10.1583 \cdot 13 \\ &= 31.79 \text{ N./mm}^2 \end{aligned}$$

Tangential Flange Stress, Seating [STa]:

$$= (y \cdot Matm / (t^2 \cdot Bcor)) - Z \cdot SRA$$

$$= (19.6189 \cdot 26009 / (107.0^2 \cdot 1946.0)) - 10.1583 \cdot 18$$

$$= 44.32 \text{ N./mm}^2$$

Average Flange Stress, Operating [SAo]:

$$= (SHo + \max(SRo, STo)) / 2$$

$$= (127 + \max(13, 32)) / 2$$

$$= 79.44 \text{ N./mm}^2$$

Average Flange Stress, Seating [SAa]:

$$= (SHA + \max(SRA, STA)) / 2$$

$$= (177 + \max(18, 44)) / 2$$

$$= 110.76 \text{ N./mm}^2$$

Bolt Stress, Operating [BSo]:

$$= Wm1 / Ab$$

$$= 337656 / 241.4232$$

$$= 137.16 \text{ N./mm}^2$$

Bolt Stress, Seating [BSa]:

$$= (Wm2 / Ab)$$

$$= (250213 / 241.4232)$$

$$= 101.64 \text{ N./mm}^2$$

Flange Stress Analysis Results: N./mm²

	Actual	Operating Allowed	Gasket Seating Actual	Allowed
Longitudinal Hub	127.09	207.00	177.20	207.00
Radial Flange	12.73	138.00	17.75	138.00
Tangential Flange	31.79	138.00	44.32	138.00
Maximum Average	79.44	138.00	110.76	138.00
Bolting	137.16	172.00	101.64	172.00

Minimum Required Flange Thickness [Rigidity] 107.848 mm.

Estimated M.A.W.P. (Operating) 13 bars

Estimated Finished Weight of Flange at given Thk. 622.1 kg.

Estimated Unfinished Weight of Forging at given Thk 776.4 kg.

Flange rigidity based on required thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:

$$= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 26008.9 / 1.0 \cdot 9806.637 \cdot 0.468 / (1.028 \cdot 201334 \cdot 22.0^2 \cdot 206.911 \cdot 0.3)$$

$$= 1.000 \text{ (should be } \leq 1)$$

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:

$$= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 18654.0 / 1.0 \cdot 9806.637 \cdot 0.468 / (1.028 \cdot 197675 \cdot 22.0^2 \cdot 206.911 \cdot 0.3)$$

$$= 0.731 \text{ (should be } \leq 1)$$

Flange rigidity based on given thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:

$$= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 26008.9 / 1.0 \cdot 9806.637 \cdot 0.468 / (1.049 \cdot 201334 \cdot 22.0^2 \cdot 206.911 \cdot 0.3)$$

$$= 0.981 \text{ (should be } \leq 1)$$

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:

$$= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 18654.0 / 1.0 \cdot 9806.637 \cdot 0.468 / (1.049 \cdot 197675 \cdot 22.0^2 \cdot 206.911 \cdot 0.3)$$

$$= 0.716 \text{ (should be } \leq 1)$$

Minimum Design Metal Temperature Results:

Thickness Ratio = 0.346, Temperature Reduction per Fig. UCS 66.1 = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -1 °C

Min Metal Temp. at Required thickness (UCS 66.1) -104 °C



Min Metal Temp. w/o impact per UG-20(f)

-29 °C

Note: UCS-66(b)(-c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: N./mm²

	Operating		Gasket Seating	
	Actual	Allowed	Actual	Allowed
Longitudinal Hub	109.68	207.00	149.08	207.00
Radial Flange	12.87	138.00	17.50	138.00
Tangential Flange	25.29	138.00	34.38	138.00
Maximum Average	67.49	138.00	91.73	138.00
Bolting	137.16	172.00	101.64	172.00

Minimum Req'd Thickness, ambient, no Corrosion: 74.320 mm.

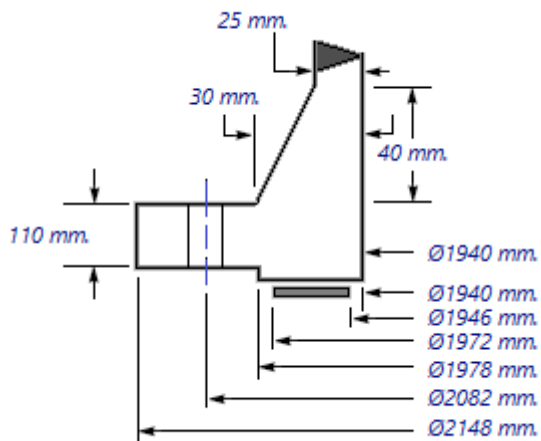
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Flange Input Data Values

Description: CH. FLG-2 :

CH. FLG-2

Flange Type	Integral Weld Neck		
Design Pressure	P	10.09	bars
Design Temperature		85	°C
Internal Corrosion Allowance	ci	3.0000	mm.
External Corrosion Allowance	ce	0.0000	mm.
Use Corrosion Allowance in Thickness Calcs.		Yes	
Flange Inside Diameter	B	1940.000	mm.
Flange Outside Diameter	A	2148.000	mm.
Flange Thickness	t	110.0000	mm.
Thickness of Hub at Small End	go	25.0000	mm.
Thickness of Hub at Large End	gl	30.0000	mm.
Length of Hub	h	40.0000	mm.
Flange Material		SA-266 2	
Flange Material UNS number		K03506	
Flange Allowable Stress At Temperature	Sfo	138.00	N./mm ²
Flange Allowable Stress At Ambient	Sfa	138.00	N./mm ²
Bolt Material		SA-193 B7	
Bolt Allowable Stress At Temperature	Sb	172.00	N./mm ²
Bolt Allowable Stress At Ambient	Sa	172.00	N./mm ²
Diameter of Bolt Circle	C	2082.000	mm.
Nominal Bolt Diameter	a	30.0000	mm.
Type of Threads		TEMA Metric	
Number of Bolts		48	
Flange Face Outside Diameter	Fod	1978.000	mm.
Flange Face Inside Diameter	Fid	1940.000	mm.
Flange Facing Sketch		Sketch 1a	
Gasket Outside Diameter	Go	1972.000	mm.
Gasket Inside Diameter	Gi	1946.000	mm.
Gasket Factor	m	3.7500	
Gasket Design Seating Stress	y	62.06	N./mm ²
Gasket Thickness	tg	3.1750	mm.
Raised face depth	Fd	0.0000	mm.



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Hub Small End Required Thickness due to Internal Pressure:

$$= (P \cdot (D/2 + Ca)) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (10.09 \cdot (1940.0/2 + 3.0)) / (138.0 \cdot 1.0 - 0.6 \cdot 10.09) + Ca$$

= 10.1486 mm.

Small End Hub MAWP:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 22.0) / (973.0 + 0.6 \cdot 22.0)$$

$$= 30.782 \text{ bars}$$

Hub Small End Hub MAPnc:

$$= (S_a \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 25.0) / (970.0 + 0.6 \cdot 25.0)$$

$$= 35.023 \text{ bars}$$

Corroded Flange Thickness, $t_c = T - c_i$	107.000 mm.
Corroded Flange ID, $B_{cor} = B + 2 \cdot c_i$	1946.000 mm.
Corroded Large Hub, $g1_{cor} = g1 - c_i$	27.000 mm.
Corroded Small Hub, $g0_{cor} = g0 - c_i$	22.000 mm.
Code R Dimension, $R = ((C - B_{cor}) / 2) - g1_{cor}$	41.000 mm.

Gasket Contact Width, $N = (G_o - G_i) / 2$	13.000 mm.
Basic Gasket Width, $b_o = N / 2$	6.500 mm.
Effective Gasket Width, $b = C_b \cdot \sqrt{b_o}$	6.425 mm.
Gasket Reaction Diameter, $G = G_o - 2 \cdot b$	1959.151 mm.

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:

$$= \pi / 4 \cdot G^2 \cdot P$$

$$= \pi / 4 \cdot 1959.1509^2 \cdot 10.094$$

$$= 310290.719 \text{ Kgf}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P$$

$$= 2 \cdot 6.4246 \cdot \pi \cdot 1959.1509 \cdot 3.75 \cdot 10.09$$

$$= 30525.682 \text{ Kgf}$$

Hydrostatic End Load at Flange ID [Hd]:

$$= \pi / 4 \cdot B_{cor}^2 \cdot P$$

$$= \pi / 4 \cdot 1946.0^2 \cdot 10.0936$$

$$= 306139.000 \text{ Kgf}$$

Pressure Force on Flange Face [Ht]:

$$= H - H_d$$

$$= 310291 - 306139$$

$$= 4151.716 \text{ Kgf}$$

Design Bolt Load for the Operating Condition [Wo]:

$$= \max(W_o,1 + W_o,2 + H'p, 0)$$

$$= \max(310291 + 30526 + 0, 0)$$

$$= 340816.406 \text{ Kgf}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$= y \cdot b \cdot \pi \cdot G + y_{Part} \cdot b_{Part} \cdot l_p$$

$$= 62.06 \cdot 6.4246 \cdot \pi \cdot 1959.151 + 0.0 \cdot 0.0 \cdot 0.0$$

$$= 250212.594 \text{ Kgf}$$

Required Bolt Area [Am]:

$$= \max((W_o + F_a + 4Me/G) / S_b, W_g / S_a)$$

$$= \max((340816 + 0 + 4 \cdot 0 / 1959.15) / 172, 250213 / 172)$$

$$= 194.325 \text{ cm}^2$$

ASME Maximum Circumferential Spacing between Bolts, Table 4.16.11 [Bsmax]:

$$= 2a + 6t / (m + 0.5)$$

$$= 2 \cdot 30.0 + 6 \cdot 107.0 / (3.75 + 0.5)$$

$$= 211.059 \text{ mm.}$$

Actual Circumferential Bolt Spacing [Bs]:

$$= C \cdot \sin(\pi / n)$$

$$= 2082.0 \cdot \sin(\pi / 48)$$

$$= 136.169 \text{ mm.}$$

ASME Moment Multiplier for Bolt Spacing per Table 4.16.11 [Bsc]:

$$= \max(\sqrt{\frac{Bs}{2a + t}}, 1)$$

$$= \max(\sqrt{\frac{136.169}{2 \cdot 30.0 + 107.0}}, 1)$$

$$= 1.0000$$

Bolting Information for TEMA Metric Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	194.325	241.423	
Radial Distance between Hub and Bolts:	40.500	41.000	
Radial Distance between Bolts and Edge:	32.500	33.000	
Circ. Spacing between the Bolts:	69.900	136.169	211.059

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$= Ab \cdot Sa / (y \cdot \pi(Go + Gi))$$

$$= 241.423 \cdot 172.0 / (62.06 \cdot \pi(1972.0 + 1946.0))$$

$$= 5.436 \text{ mm.}$$

Recommended Minimum Gasket Contact Width per Table 4.16.2

Sheet Gaskets including laminated sheet gaskets with or without a metal core: 19 mm
 Preformed comp. gaskets incl. spiral wound, jacketed, & solid flat metal gaskets: 16 mm

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$= Sa \cdot Ab$$

$$= 172.0 \cdot 241.4232$$

$$= 423419.59 \text{ Kgf}$$

Gasket Load for the Operating Condition [HG]:

$$= Wo - H$$

$$= 340816 - 310291$$

$$= 30525.69 \text{ Kgf}$$

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$= (C - G) / 2$$

$$= (2082.0 - 1959.1509) / 2$$

$$= 61.4245 \text{ mm.}$$

Distance to Face Pressure Reaction [ht]:

$$= 0.5((C - B) / 2 + hg)$$

$$= 0.5((2082.0 - 1946.0) / 2 + 61.4245)$$

$$= 64.7123 \text{ mm.}$$

Distance to End Pressure Reaction [hd]:

$$= (C - B - g1) / 2$$

$$= (2082.0 - 1946.0 - 27.0) / 2$$

$$= 54.5000 \text{ mm.}$$

Summary of Moments for Internal Pressure: (Kg-m.)

Loading	Force	Distance	Bolt Corr	Moment
End Pressure, Md	306139.	54.5000	1.0000	16685.
Face Pressure, Mt	4152.	64.7123	1.0000	269.
Gasket Load, Mg	30526.	61.4245	1.0000	1875.
Gasket Seating, Matm	423420.	61.4245	1.0000	26009.

Total Moment for Operation, Mo 18829. Kg-m.
 Total Moment for Gasket seating, Mg 26009. Kg-m.

Effective Hub Length, ho = sqrt(Bcor*goCor) 206.911 mm.
 Hub Ratio, h/h0 = HL / H0 0.193
 Thickness Ratio, g1/g0 = (g1Cor/goCor) 1.227

Flange Factors for Integral Flange:

Factor F per 4.16.5 0.908
 Factor V per 4.16.5 0.468

Factor f per 4.16.5

1.034

Stress factor from Table 4.16.4 [K]:

$$= A / B$$

$$= 2148.0 / 1946.0$$

$$= 1.104$$

Stress factor from Table 4.16.4 [Y]:

$$= (0.66845 + (5.7169 (K^2 \cdot \log_{10}(K))) / (K^2 - 1)) / (K - 1)$$

$$= (0.66845 + (5.7169 (1.104^2 \cdot \log_{10}(1.104))) / (1.104^2 - 1)) / (1.104 - 1)$$

$$= 19.619$$

Stress factor from Table 4.16.4 [T]:

$$= (K^2(1 + 8.55246 \cdot \log_{10}(K)) - 1) / ((1.0472 + 1.9448 \cdot K^2) \cdot (K - 1))$$

$$= (1.104^2(1 + 8.55246 \cdot \log_{10}(1.104)) - 1) / ((1.0472 + 1.9448 \cdot 1.104^2) \cdot (1.104 - 1))$$

$$= 1.876$$

Stress factor from Table 4.16.4 [U]:

$$= (K^2(1 + 8.55246 \cdot \log_{10}(K)) - 1) / (1.36136(K^2 - 1) \cdot (K - 1))$$

$$= (1.104^2(1 + 8.55246 \cdot \log_{10}(1.104)) - 1) / (1.36136(1.104^2 - 1) \cdot (1.104 - 1))$$

$$= 21.559$$

Stress factor from Table 4.16.4 [Z]:

$$= (K^2 + 1) / (K^2 - 1)$$

$$= (1.104^2 + 1) / (1.104^2 - 1)$$

$$= 10.158$$

Stress factor from Table 4.16.4 for integral flanges [d]:

$$= U \cdot g_0^2 \cdot h_0 / V$$

$$= 21.559 \cdot 22.0^2 \cdot 206.911 / 0.468$$

$$= 4612139.500 \text{ mm}^3$$

Stress factor from Table 4.16.4 for integral flanges [e]:

$$= F / h_0$$

$$= 0.908 / 206.911$$

$$= 0.004 \text{ 1/mm}$$

Stress factor from Table 4.16.4 [L]:

$$= (t \cdot e + 1) / T + t^3/d$$

$$= (107.0 \cdot 0.004 + 1) / 1.876 + 107.0^3 / 4612140$$

$$= 1.049$$

Flange Stresses

Longitudinal Hub Stress, Operating [SHo]:

$$= (f \cdot Mop / Bcor) / (L \cdot g_1^2)$$

$$= (1.0339 \cdot 18829 / 1946.0) / (1.0491 \cdot 27.0^2)$$

$$= 128.28 \text{ N/mm}^2$$

Longitudinal Hub Stress, Seating [SHa]:

$$= (f \cdot Matm / Bcor) / (L \cdot g_1^2)$$

$$= (1.0339 \cdot 26009 / 1946.0) / (1.0491 \cdot 27.0^2)$$

$$= 177.20 \text{ N/mm}^2$$

Radial Flange Stress, Operating [SRo]:

$$= (\beta \cdot Mop / Bcor) / (L \cdot t^2)$$

$$= (1.6263 \cdot 18829 / 1946.0) / (1.0491 \cdot 107.0^2)$$

$$= 12.85 \text{ N/mm}^2$$

Radial Flange Stress, Seating [SRa]:

$$= (\beta \cdot Matm / Bcor) / (L \cdot t^2)$$

$$= (1.6263 \cdot 26009 / 1946.0) / (1.0491 \cdot 107.0^2)$$

$$= 17.75 \text{ N/mm}^2$$

Tangential Flange Stress, Operating [STo]:

$$= (Y \cdot Mo / (t^2 \cdot Bcor)) - Z \cdot SRo$$

$$= (19.6189 \cdot 18829 / (107.0^2 \cdot 1946.0)) - 10.1583 \cdot 13$$

$$= 32.08 \text{ N/mm}^2$$

Tangential Flange Stress, Seating [STa]:

$$= (y \cdot \text{Matm} / (t^2 \cdot \text{Bcor})) - Z \cdot \text{SRa}$$

$$= (19.6189 \cdot 26009 / (107.0^2 \cdot 1946.0)) - 10.1583 \cdot 18$$

$$= 44.32 \text{ N./mm}^2$$

Average Flange Stress, Operating [SAo]:

$$= (S_{Ho} + \max(S_{Ro}, S_{To})) / 2$$

$$= (128 + \max(13, 32)) / 2$$

$$= 80.18 \text{ N./mm}^2$$

Average Flange Stress, Seating [SAa]:

$$= (S_{Ha} + \max(S_{Ra}, S_{Ta})) / 2$$

$$= (177 + \max(18, 44)) / 2$$

$$= 110.76 \text{ N./mm}^2$$

Bolt Stress, Operating [BSo]:

$$= W_{m1} / A_b$$

$$= 340816 / 241.4232$$

$$= 138.44 \text{ N./mm}^2$$

Bolt Stress, Seating [BSa]:

$$= (W_{m2} / A_b)$$

$$= (250213 / 241.4232)$$

$$= 101.64 \text{ N./mm}^2$$

Flange Stress Analysis Results: N./mm²

	Actual	Operating Allowed	Gasket Seating Actual	Allowed
Longitudinal Hub	128.28	207.00	177.20	207.00
Radial Flange	12.85	138.00	17.75	138.00
Tangential Flange	32.08	138.00	44.32	138.00
Maximum Average	80.18	138.00	110.76	138.00
Bolting	138.44	172.00	101.64	172.00

Minimum Required Flange Thickness [Rigidity] 107.848 mm.

Estimated M.A.W.P. (Operating) 13 bars
 Estimated Finished Weight of Flange at given Thk. 622.1 kg.
 Estimated Unfinished Weight of Forging at given Thk 776.4 kg.

Flange rigidity based on required thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:

$$= 52.14 \cdot M_a / B_{sc} \cdot C_{nvfac} \cdot V / (\lambda \cdot E_{amb} \cdot g_o^2 \cdot h_o \cdot K_i)$$

$$= 52.14 \cdot 26008.9 / 1.0 \cdot 9806.637 \cdot 0.468 / (1.028 \cdot 201334 \cdot 22.0^2 \cdot 206.911 \cdot 0.3)$$

$$= 1.000 \text{ (should be } \leq 1)$$

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:

$$= 52.14 \cdot M_o / B_{sc} \cdot C_{nvfac} \cdot V / (\lambda \cdot E_{op} \cdot g_o^2 \cdot h_o \cdot K_i)$$

$$= 52.14 \cdot 18828.6 / 1.0 \cdot 9806.637 \cdot 0.468 / (1.028 \cdot 197675 \cdot 22.0^2 \cdot 206.911 \cdot 0.3)$$

$$= 0.738 \text{ (should be } \leq 1)$$

Flange rigidity based on given thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:

$$= 52.14 \cdot M_a / B_{sc} \cdot C_{nvfac} \cdot V / (\lambda \cdot E_{amb} \cdot g_o^2 \cdot h_o \cdot K_i)$$

$$= 52.14 \cdot 26008.9 / 1.0 \cdot 9806.637 \cdot 0.468 / (1.049 \cdot 201334 \cdot 22.0^2 \cdot 206.911 \cdot 0.3)$$

$$= 0.981 \text{ (should be } \leq 1)$$

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:

$$= 52.14 \cdot M_o / B_{sc} \cdot C_{nvfac} \cdot V / (\lambda \cdot E_{op} \cdot g_o^2 \cdot h_o \cdot K_i)$$

$$= 52.14 \cdot 18828.6 / 1.0 \cdot 9806.637 \cdot 0.468 / (1.049 \cdot 197675 \cdot 22.0^2 \cdot 206.911 \cdot 0.3)$$

$$= 0.723 \text{ (should be } \leq 1)$$

Minimum Design Metal Temperature Results:

Thickness Ratio = 0.346, Temperature Reduction per Fig. UCS 66.1 = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -1 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C



Min Metal Temp. w/o impact per UG-20(f) -29 °C

Note: UCS-66(b)(-c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: N./mm²

	Actual	Operating Allowed	Gasket Seating Actual Allowed	
Longitudinal Hub	110.71	207.00	149.08	207.00
Radial Flange	12.99	138.00	17.50	138.00
Tangential Flange	25.53	138.00	34.38	138.00
Maximum Average	68.12	138.00	91.73	138.00
Bolting	138.44	172.00	101.64	172.00

Minimum Req'd Thickness, ambient, no Corrosion: 74.320 mm.

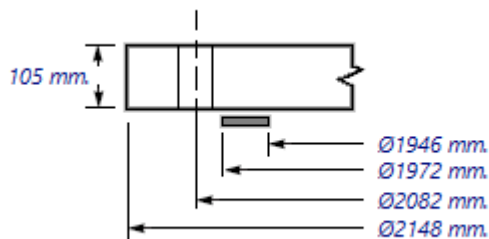
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Flange Input Data Values

Description: New Flange :

CH. COVER-2

Flange Type		Blind	
Design Pressure	P	10.00	bars
Design Temperature		85	°C
Internal Corrosion Allowance	ci	3.0000	mm.
External Corrosion Allowance	ce	0.0000	mm.
Use Corrosion Allowance in Thickness Calcs.		Yes	
Flange Outside Diameter	A	2148.000	mm.
Flange Thickness	t	105.0000	mm.
Flange Material		SA-266 2	
Flange Material UNS number		K03506	
Flange Allowable Stress At Temperature	Sfo	138.00	N./mm ²
Flange Allowable Stress At Ambient	Sfa	138.00	N./mm ²
Bolt Material		SA-193 B7	
Bolt Allowable Stress At Temperature	Sb	172.00	N./mm ²
Bolt Allowable Stress At Ambient	Sa	172.00	N./mm ²
Perimeter along the Center of the Bolts	L	6540.796	mm.
Diameter of Bolt Circle	C	2082.000	mm.
Nominal Bolt Diameter	a	30.0000	mm.
Type of Threads		TEMA Metric	
Number of Bolts		48	
Flange Face Outside Diameter	Fod	1978.000	mm.
Flange Face Inside Diameter	Fid	1940.000	mm.
Flange Facing Sketch		Sketch 1a	
Gasket Outside Diameter	Go	1972.000	mm.
Gasket Inside Diameter	Gi	1946.000	mm.
Gasket Factor	m	3.7500	
Gasket Design Seating Stress	y	62.06	N./mm ²
Gasket Thickness	tg	5.0000	mm.
Raised face depth	Fd	0.0000	mm.



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Gasket Contact Width,	$N = (G_o - G_i)/2$	13.000 mm.
Basic Gasket Width,	$bo = N/2$	6.500 mm.
Effective Gasket Width,	$b = C_b \cdot \sqrt{bo}$	6.425 mm.
Gasket Reaction Diameter,	$G = G_o - 2 \cdot b$	1959.151 mm.

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:
 $= \pi/4 \cdot G^2 \cdot P$
 $= \pi/4 \cdot 1959.1509^2 \cdot 10.0$
 $= 307413.031 \text{ Kg}$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P$$

$$= 2 \cdot 6.4246 \cdot \pi \cdot 1959.1509 \cdot 3.75 \cdot 10.0$$

$$= 30242.582 \text{ Kgf}$$

Design Bolt Load for the Operating Condition [Wo]:

$$= \max(Wo,1 + Wo,2 + H'p, 0)$$

$$= \max(307413 + 30243 + 0, 0)$$

$$= 337655.594 \text{ Kgf}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$= y \cdot b \cdot \pi \cdot G + yPart \cdot bPart \cdot lp$$

$$= 62.06 \cdot 6.4246 \cdot \pi \cdot 1959.151 + 0.0 \cdot 0.0 \cdot 0.0$$

$$= 250212.594 \text{ Kgf}$$

Required Bolt Area [Am]:

$$= \max((Wo + Fa + 4Me/G)/Sb, Wg/Sa)$$

$$= \max((337656 + 0 + 4 \cdot 0/1959.15)/172, 250213/172)$$

$$= 192.523 \text{ cm}^2$$

Bolting Information for TEMA Metric Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	192.523	241.423	
Radial Distance between Bolts and Edge:	32.500	33.000	
Circ. Spacing between the Bolts:	69.900	136.169	...

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$= Ab \cdot Sa / (y \cdot \pi (Go + Gi))$$

$$= 241.423 \cdot 172.0 / (62.06 \cdot \pi (1972.0 + 1946.0))$$

$$= 5.436 \text{ mm.}$$

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$= Sa \cdot Ab$$

$$= 172.0 \cdot 241.4232$$

$$= 423419.59 \text{ Kgf}$$

Gasket Load for the Operating Condition [HG]:

$$= Wo$$

$$= 337655.59 \text{ Kgf}$$

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$= (C - G) / 2$$

$$= (2082.0 - 1959.1509) / 2$$

$$= 61.4245 \text{ mm.}$$

Flange Stresses

Tangential Flange Stress, Flat Head (4.6.2.3), Operating [STo]:

$$= 1.9 \cdot Wm1 \cdot hG \cdot Bsc / (t^2 \cdot G) + C \cdot Z \cdot Peq \cdot G^2 / t^2$$

$$= 1.9 \cdot 337656 \cdot 61.4245 \cdot 1.0 / (102.0^2 \cdot 1959.1509) + 0.3 \cdot 1.0 \cdot 10.0 \cdot 1959.1509^2 / 102.0^2$$

$$= 129.64 \text{ N./mm}^2$$

Tangential Flange Stress, Flat Head (4.6.2.4), Seating [STa]:

$$= 1.9 \cdot W \cdot hG \cdot Bsc / (t^2 \cdot G)$$

$$= 1.9 \cdot 423420 \cdot 61.4245 \cdot 1.0 / (102.0^2 \cdot 1959.1509)$$

$$= 23.78 \text{ N./mm}^2$$

Bolt Stress, Operating [BSO]:

$$= Wm1 / Ab$$

$$= 337656 / 241.4232$$

$$= 137.16 \text{ N./mm}^2$$

Bolt Stress, Seating [BSa]:

$$= (Wm2 / Ab)$$

$$= (250213 / 241.4232)$$



= 101.64 N./mm²

Flange Stress Analysis Results: N./mm²

	Actual	Operating Allowed	Gasket Seating Actual Allowed
Tangential Flange	129.64	138.00	23.78 138.00
Bolting	137.16	172.00	101.64 172.00

Reqd. Blind Flange Thickness at Center 101.864 mm.
 Reqd. Blind Flange Thickness at Gasket 45.338 mm.

Estimated M.A.W.P. (Operating) 11 bars
 Estimated Finished Weight of Flange at given Thk. 2949.0 kg.
 Estimated Unfinished Weight of Forging at given Thk 2949.0 kg.

Minimum Design Metal Temperature Results:

Thickness Ratio = 0.944, Temperature Reduction per Fig. UCS 66.1 = 3 °C

Min Metal Temp. w/o impact per UCS-66, Curve B 0 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -3 °C

Note: UCS-66(b)(-c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: N./mm²

	Actual	Operating Allowed	Gasket Seating Actual Allowed
Tangential Flange	122.34	138.00	22.44 138.00
Bolting	137.16	172.00	101.64 172.00

Minimum Req'd Thickness, ambient, no Corrosion: 98.882 mm.

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Internal Pressure Results Summary:

Element Thickness, Pressure, Diameter and Allowable Stress :

From	To	Int. Press + Liq. Hd bars	Nominal Thickness mm.	Total Corr Allowance mm.	Element Diameter mm.	Allowable Stress(SE) N./mm ²	Element#
CH. COVER-1		10	110	3	2148	138	1
CH. FLG-1		10	115	3	1940	138	2
CH. SHELL-1		10.093	25	3	1940	138	3
MAIN SHELL		191.11	149	1.5	1940	138	4
CH. SHELL-2		10.19	25	3	1940	138	5
CH. FLG-2		10.094	115	3	1940	138	6
CH. COVER-2		10	110	3	2148	138	7

Element Required Thickness and MAWP :

From	To	Design Pressure bars	M.A.W.P. Corroded bars	M.A.P. New & Cold bars	Minimum Thickness mm.	Required Thickness mm.	Element#
CH. COVER-1		10	10.6439	11.2798	105	101.864	1
CH. FLG-1		10	12.54	12.54	110	107.848	2
CH. SHELL-1		10	30.6891	35.0227	25	10.1485	3
MAIN SHELL		191	191.901	194.076	149	148.245	4
CH. SHELL-2		10	30.593	35.0227	25	10.2169	5
CH. FLG-2		10	12.4464	12.54	110	107.848	6
CH. COVER-2		10	10.6439	11.2798	105	101.864	7

Summary of Heat Exchanger Maximum Allowable Working Pressures :

Note:

For Exchanger designs, the following values include MAWPs that consider the tubesheet, tubes, tube/tubesheet joint etc. These values were determined by iteration. Review the tubesheet analysis report for more information.

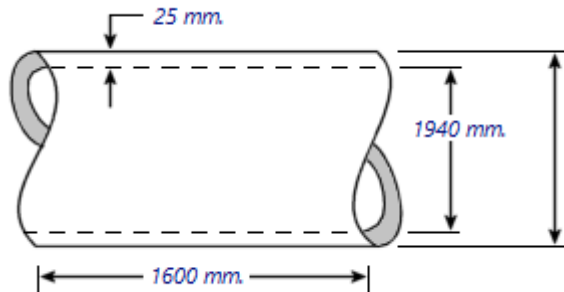
Shell Side MAWP = 192 bars
 Shell Side MAPnc = 194 bars
 Channel Side MAWP = 11 bars
 Channel Side MAPnc = 11 bars

Elements Suitable for Design Internal Pressure.

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Internal Pressure Calculation Results:

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Internal Pressure Results for: CH. SHELL-1

Cylindrical Shell Design Information:					
Design Pressure	P	10.093 bars	Int. Design Temperature		85 °C
Material		SA-516 70	Ext. Design Temperature		85 °C
Length		1600.000 mm.	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	3.0000 mm.	Allowable Stress (ope)	S	138.0 N./mm ²
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	138.0 N./mm ²
Inside Diameter	Di	1940.000 mm.	Actual Stress		45.2 N./mm ²
Spec. Min. Thk t,e, or sa		25.0000 mm.	Spec. Nominal Thk.		25.000 mm.
Surface Area		100028.3 cm ²	SG of Contents		0.995
Element Volume		4730326.5 cm ³	Weight of Contents		2305.6 Kg
Empty Weight		1913.8 Kg	Operating Weight		4219.4 Kg
			Comp. All. Ope. Stress		132.5 N./mm ²

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (10.093 \cdot 973.0) / (138.0 \cdot 1.0 - 0.6 \cdot 10.093)$$

$$= 7.1485 + 3.0000 = 10.1485 \text{ mm.}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.093 bars

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 22.0) / (973.0 + 0.6 \cdot 22.0)$$

$$= 31 - 0.1 = 31 \text{ bars}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 25.0) / (970.0 + 0.6 \cdot 25.0)$$

$$= 35 \text{ bars}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (10.093 \cdot (973.0 + 0.6 \cdot 22.0)) / (1.0 \cdot 22.0)$$

$$= 45.249 \text{ N./mm}^2$$

% Elongation per Table UG-79-1 (50 * tnom / Rf * (1 - Rf / Ro)) 1.272 %

Minimum Design Metal Temperature Results:

Govrn. thk, tg = 25.0, tr = 7.54, c = 3.0 mm., E* = 1.0



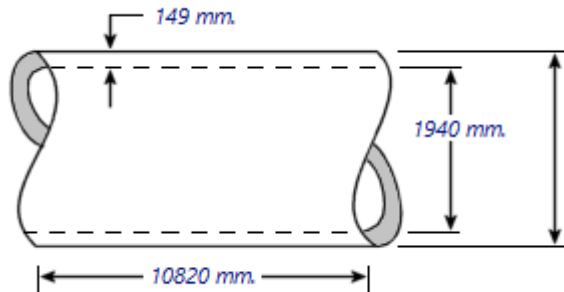
Thickness Ratio = $tr * E^* / (tg - c) = 0.343$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -1 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	10.644 bars
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	0 °C
Governing Thk.	25.000 mm.	Basic MDMT	-1 °C
Stress/Thk. Ratio	0.343	Temperature Difference	103 °C
UCS-66(c) red.	No	Computed Min. Temp.	-104 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E*	100 %

Internal Pressure Calculation Results:

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Internal Pressure Results for: MAIN SHELL

Cylindrical Shell Design Information:					
Design Pressure	P	191.112 bars	Int. Design Temperature	260 °C	
Material		SA-516 70	Ext. Design Temperature	260 °C	
Length		10820.000 mm.	External Pressure Chart	CS-2	
			UNS Number	K02700	
Int. Corr. All.	c	1.5000 mm.	Allowable Stress (ope)	S	138.0 N./mm ²
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	138.0 N./mm ²
Inside Diameter	Di	1940.000 mm.	Actual Stress	137.3 N./mm ²	
Spec. Min. Thk t _e , or s _a		149.0000 mm.	Spec. Nominal Thk.	149.000 mm.	
Surface Area		760741.8 cm ²	SG of Contents	0.590	
Element Volume		21819234.0 cm ³	Weight of Contents	12865.5 Kg	
Empty Weight		82003.4 Kg	Operating Weight	94868.9 Kg	
			Comp. All. Ope. Stress	131.2 N./mm ²	

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (191.112 \cdot 971.5) / (138.0 \cdot 1.0 - 0.6 \cdot 191.112)$$

$$= 146.7448 + 1.5000 = 148.2448 \text{ mm.}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.112 bars

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 147.5) / (971.5 + 0.6 \cdot 147.5)$$

$$= 192 - 0 = 192 \text{ bars}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 149.0) / (970.0 + 0.6 \cdot 149.0)$$

$$= 194 \text{ bars}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (191.112 \cdot (971.5 + 0.6 \cdot 147.5)) / (1.0 \cdot 147.5)$$

$$= 137.350 \text{ N./mm}^2$$

% Elongation per Table UG-79-1 (50 * t_{nom} / R_f * (1 - R_f / R_o)) 7.133 %

Note: Please Check Requirements of UCS-79 as Elongation is > 5%.

Minimum Design Metal Temperature Results:

Warning:

*The minimum metal temperature could not be computed because the governing thickness for this component was greater than 100.000 mm..
This component must be impact tested.*

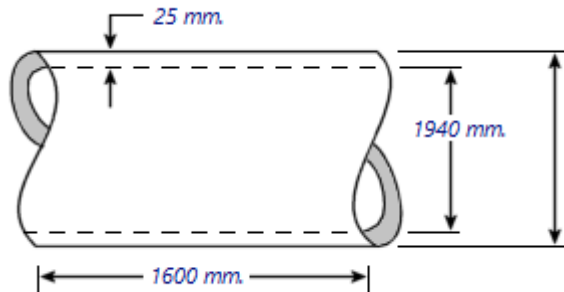
Note:

Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	191.901 bars
UCS-66(b) red.	No	Min. Des. Mtl. Temp.	0 °C
Governing Thk.	149.000 mm.	Basic MDMT	-17 °C
Stress/Thk. Ratio	-1.000	Temperature Difference	0 °C
UCS-66(c) red.	No	Computed Min. Temp.	-17 °C
		Joint Efficiency E*	100 %

Internal Pressure Calculation Results:

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Internal Pressure Results for: CH. SHELL-2

Cylindrical Shell Design Information:					
Design Pressure	P	10.190 bars	Int. Design Temperature		85 °C
Material		SA-516 70	Ext. Design Temperature		85 °C
Length		1600.000 mm.	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	3.0000 mm.	Allowable Stress (ope)	S	138.0 N./mm ²
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	138.0 N./mm ²
Inside Diameter	Di	1940.000 mm.	Actual Stress		45.7 N./mm ²
Spec. Min. Thk t _e or s _a		25.0000 mm.	Spec. Nominal Thk.		25.000 mm.
Surface Area		100028.3 cm ²	SG of Contents		0.995
Element Volume		4730326.5 cm ³	Weight of Contents		4703.8 Kg
Empty Weight		1913.8 Kg	Operating Weight		6617.6 Kg
			Comp. All. Ope. Stress		131.2 N./mm ²

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (10.19 \cdot 973.0) / (138.0 \cdot 1.0 - 0.6 \cdot 10.19)$$

$$= 7.2169 + 3.0000 = 10.2169 \text{ mm.}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.190 bars

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 22.0) / (973.0 + 0.6 \cdot 22.0)$$

$$= 31 - 0 = 31 \text{ bars}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 25.0) / (970.0 + 0.6 \cdot 25.0)$$

$$= 35 \text{ bars}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (10.19 \cdot (973.0 + 0.6 \cdot 22.0)) / (1.0 \cdot 22.0)$$

$$= 45.679 \text{ N./mm}^2$$

% Elongation per Table UG-79-1 (50 * t_{nom} / R_f * (1 - R_f / R_o)) 1.272 %

Minimum Design Metal Temperature Results:

Govern. thk, t_g = 25.0, t_r = 7.54, c = 3.0 mm., E* = 1.0



Thickness Ratio = $tr * E^* / (tg - c) = 0.343$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -1 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	10.644 bars
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	0 °C
Governing Thk.	25.000 mm.	Basic MDMT	-1 °C
Stress/Thk. Ratio	0.343	Temperature Difference	103 °C
UCS-66(c) red.	No	Computed Min. Temp.	-104 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E*	100 %

Hydrostatic Test Pressure Results:

Exchanger Shell Side Hydrostatic Test Pressures:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	249.472 bars
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	248.300 bars
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	252.089 bars
UG100	= 1.10 * M.A.W.P. * Sa/S	211.092 bars
PED	= max(1.43*DP, 1.25*DP*ratio)	273.130 bars
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	273.130 bars
App 27-4	= M.A.W.P.	191.901 bars

Exchanger Channel Side Hydrostatic Test Pressures:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	13.837 bars
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	13.000 bars
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	14.453 bars
UG100	= 1.10 * M.A.W.P. * Sa/S	11.708 bars
PED	= max(1.43*DP, 1.25*DP*ratio)	14.300 bars
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	14.300 bars
App 27-4	= M.A.W.P.	10.644 bars

Horizontal Test performed per: UG-99b

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (N./mm² & bars):

From To	Stress	Allowable	Ratio	Pressure
CH. SHELL-1	62.9	234.0	0.269	14.03
MAIN SHELL	179.4	234.0	0.767	249.66
CH. SHELL-2	62.9	234.0	0.269	14.03

Stress ratios for Nozzle and Pad Materials (N./mm²):

Description	Pad/Nozzle	Ambient	Operating	Ratio
TS OUT NOZ. T2	Nozzle	118.00	118.00	1.000
TS OUT NOZ. T2	Pad	138.00	138.00	1.000
DRAIN NOZ. D1	Nozzle	118.00	118.00	1.000
SH. IN A2	Nozzle	138.00	134.60	1.025
SH. OUT NOZ B2	Nozzle	138.00	134.60	1.025
CH. IN NOZ. A1	Nozzle	118.00	118.00	1.000
CH. IN NOZ. A1	Pad	138.00	138.00	1.000
SAFETY VALVE S1	Nozzle	118.00	118.00	1.000
DRAIN NOZ. D2	Nozzle	118.00	118.00	1.000

Minimum 1.000

Stress ratios for Pressurized Vessel Elements (N./mm²):

Description	Ambient	Operating	Ratio
CH. COVER-1	138.00	138.00	1.000
CH. FLG-1	138.00	138.00	1.000
CH. SHELL-1	138.00	138.00	1.000
MAIN SHELL	138.00	138.00	1.000
CH. SHELL-2	138.00	138.00	1.000
CH. FLG-2	138.00	138.00	1.000
CH. COVER-2	138.00	138.00	1.000

Minimum 1.000

Stress ratios for Exchanger Materials (N./mm²):



Description	Ambient	Operating	Ratio
Tube Material	118.00	118.00	1.000
Tubesheet Material	138.00	134.60	1.025

Minimum 1.000

Hoop Stress in Nozzle Wall during Pressure Test (N./mm²):

Description	Ambient	Operating	Ratio
TS OUT NOZ. T2	54.54	216.00	0.252
DRAIN NOZ. D1	7.62	216.00	0.035
SH. IN A2	145.71	225.00	0.648
SH. OUT NOZ B2	145.71	225.00	0.648
CH. IN NOZ. A1	54.54	216.00	0.252
SAFETY VALVE S1	7.08	216.00	0.033
DRAIN NOZ. D2	7.62	216.00	0.035

Elements Suitable for Test Pressure.

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MDMT Calculations for Stiffening Rings:

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Input Echo, Tubesheet Number 1, Description: FIXED TS

Shell Data:

Main Shell Description: MAIN SHELL

Shell Test Pressure		249.47 bars
Shell Thickness	ts	149.0000 mm.
Shell Internal Corrosion Allowance	cas	0.0000 mm.
Shell External Corrosion Allowance	casext	0.0000 mm.
Inside Diameter of Shell	Ds	1940.000 mm.
Shell Circumferential Joint Efficiency	Esw	1.000
Shell Temperature for Internal Pressure	Ts	21.11 °C
Shell Material		SA-516 70

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Shell Material UNS Number		K02700
Shell Allowable Stress at Temperature	Ss	138.00 N./mm ²
Shell Allowable Stress at Ambient		138.00 N./mm ²

Channel Data:

Channel Description: CH. SHELL-1

Channel Type:		Cylinder
Channel Test Pressure		13.84 bars
Channel Thickness	tc	25.0000 mm.
Channel Internal Corrosion Allowance	cac	0.0000 mm.
Channel External Corrosion Allowance	cacext	0.0000 mm.
Inside Diameter of Channel	Dc	1940.000 mm.
Channel Design Temperature	TEMPC	21.11 °C
Channel Flange Matl		SA-516 70

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Channel Material UNS Number		K02700
Channel Allowable Stress at Temperature	Sc	138.00 N./mm ²
Channel Allowable Stress at Ambient		138.00 N./mm ²

Tube Data:

Number of Tube Holes	Nt	3297
Tube Wall Thickness	et	2.7700 mm.
Tube Outside Diameter	D	19.0500 mm.
Straight Tube Length (bet. inner tubsht faces) L		10820.00 mm.
Design Temperature of the Tubes		21.11 °C
Tube Material		SA-210 A-1
Tube Material UNS Number		K02707
Is this a Welded Tube		No
Tube Material Specification used		Smls. tube
Tube Allowable Stress at Temperature		118.00 N./mm ²
Tube Allowable Stress At Ambient		118.00 N./mm ²
Tube Yield Stress At design Temperature	Syt	255.00 N./mm ²
Tube Pitch (Center to Center Spacing)	P	30.0000 mm.
Tube Layout Pattern		Triangular
Fillet Weld Leg	af	2.7000 mm.
Groove Weld Leg	ag	2.7000 mm.
Tube-Tubesheet Joint Weld Type		Full Strength
Method for Tube-Tubesheet Jt. Allowable		UW-20.3
Is the Tube Inset		No
Radius to Outermost Tube Hole Center	ro	939.980 mm.
Largest Center-to-Center Tube Distance	Ul	0.0000 mm.
Length of Expanded Portion of Tube	ltx	0.0000 mm.
Tube-side pass partition groove depth	hg	0.0000 mm.
Area of Tube Layout	Ap	24988.3 cm ²
Perimeter of Tube Layout	Cp	5991.4 mm.

Tubesheet Data:

Tubesheet TYPE: Fixed Tubesheet Exchanger, Conf A

Tubesheet Design Metal Temperature T 21.11 °C
 Tubesheet Material SA-266 2

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Tubesheet Material UNS Number K03506
 Tubesheet Allowable Stress at Temperature S 138.00 N./mm²
 Tubesheet Allowable Stress at Ambient Tt 138.00 N./mm²
 Thickness of Tubesheet h 349.0000 mm.
 Tubesheet Corr. Allowance (Shell side) Cats 0.0000 mm.
 Tubesheet Corr. Allowance (Channel side) Catc 0.0000 mm.
 Tubesheet Outside Diameter A 2240.000 mm.

Area of the Untubed Lanes AL 0.0 cm²

Additional Data for Fixed/Floating Tubesheet Exchangers:

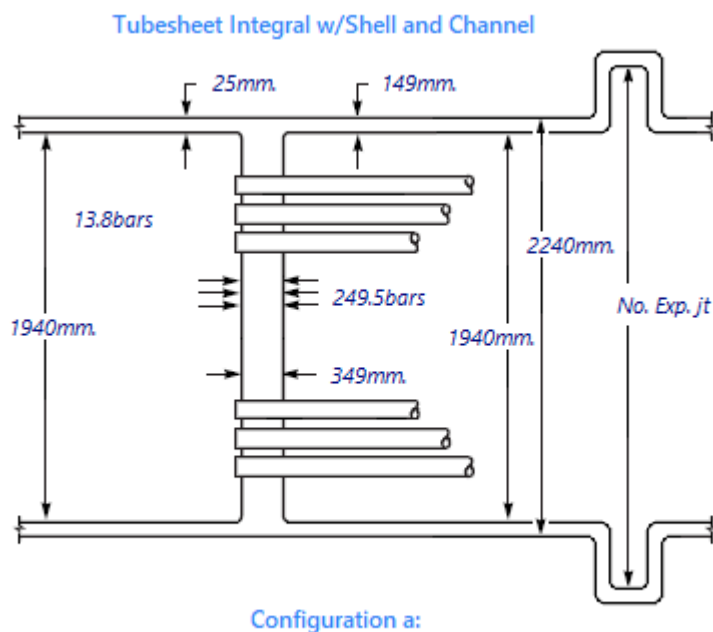
Unsupported Tube Span under consideration l 1510.000 mm.
 Tube End condition corresponding to Span (l) k 0.80

Ignore Radial Thermal Exp. effects (4.18.8/4.18.9) Yes
 Perform Differential Pressure Design No
 Run Multiple Load Cases Yes

Mean Shell Metal Temp. along Shell len. Tsm 21.11 °C
 Mean Tube Metal Temp. along Tube length Ttm 21.11 °C
 Junction Stress Reduction option Perform Plastic Calculation

Tubesheet Integral with Both
 Tubesheet Extended as Flange No

Is Exchanger in Creep range (skip EP, Use 3S for Sps) No



ASME TubeSheet Results per Part UHX/4.18, 2025

Elasticity/Expansion Material Properties:

Shell - TE-1 Carbon & Low Alloy Steels, Group 1
 Shell - TM-1 Carbon Steels with C<= 0.3%

Th. Exp. Coeff. Metal Temp. along Len	21.1 °C	0.0000115191 /°C
Elastic Mod. at Design Temperature	21.1 °C	0.20270E+09 KPa.
Th. Exp. Coeff. Metal Temp. at Tubsht	21.1 °C	0.0000115191 /°C



Elastic Mod. at Metal Temp. along Len	21.1 °C	0.20270E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

Channel - TE-1 Carbon & Low Alloy Steels, Group 1
 Channel - TM-1 Carbon Steels with C<= 0.3%

Th. Exp. Coeff. Metal Temp. at Tubsht	21.1 °C	0.0000115191 /°C
Elastic Mod. at Design Temperature	21.1 °C	0.20270E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

Tubes - TE-1 Carbon & Low Alloy Steels, Group 1
 Tubes - TM-1 Carbon Steels with C<= 0.3%

Th. Exp. Coeff. Metal Temp. along Len	21.1 °C	0.0000115191 /°C
Elastic Mod. at Design Temperature	21.1 °C	0.20270E+09 KPa.
Elastic Mod. at Metal Temp. along Len	21.1 °C	0.20270E+09 KPa.
Elastic Mod. at Tubsht. Design Temp.	21.1 °C	0.20270E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

TubeSheet - TE-1 Carbon & Low Alloy Steels, Group 1
 TubeSheet - TM-1 Carbon Steels with C> 0.3%

Elastic Mod. at Design Temperature	21.1 °C	0.20132E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20132E+09 KPa.

Note:

The Elasticity and Alpha values are taken from Tables in ASME II D.
 Please insure these properties are consistent with the
 type of Material for the tubes, shell, channel etc.

Tube Required Thickness under Internal Pressure (Tubeside pressure):

Thickness Due to Internal Pressure:
 $= (P \cdot (D/2 - CAE)) / (S \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= (13.84 \cdot (19.05/2 - 0)) / (118.0 \cdot 1.0 + 0.4 \cdot 13.84)$
 $= 0.1112 + 0.0000 = 0.1112 \text{ mm.}$

Tube Required Thickness under External Pressure (Shellside pressure) :

External Pressure Chart CS-2 at 21.11 °C
 Elastic Modulus for Material 199943392.00 KPa.

Results for Max. Allowable External Pressure (Emawp):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
2.7700	19.05	10820.00	6.88	50.0000	0.0232575	122.73

EMAWP = $(2.167 / (D/T) - 0.0833) \cdot B = 284.4698 \text{ bars}$

Results for Req'd Thickness for Ext. Pressure (Tca):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
2.5195	19.05	10820.00	7.56	50.0000	0.0192417	122.73

EMAWP = $(2.167 / (D/T) - 0.0833) \cdot B = 249.5038 \text{ bars}$

Summary of Tube Required Thickness Results:

Total Required Thickness including Corrosion all.	2.5195 mm.
Allowable Internal Pressure at Corroded thickness	388.30 bars
Required Internal Design Pressure	13.84 bars
Allowable External Pressure at Corroded thickness	284.47 bars
Required External Design Pressure	249.47 bars
Required Thickness due to Shell Side pressure	2.5195 mm.

Detailed Results for load Case D3 un-corr. (Psd,max + Ptd,max)

Results for ASME Fixed Tubesheet Calculations for Configuration a:

Results for Tubesheet Calculations Original Thickness:

ASME Section VIII-2 Part 4.18.8.4 Step 1:



Tube expansion depth ratio [ρ]:

$$\begin{aligned} &= l_{tx} / h \text{ (modified for corrosion if present)} \\ &= 0/349.0 \\ &= 0.0000 \text{ (must be } 0 \leq \rho \leq 1 \text{)} \end{aligned}$$

Effective tube hole diameter [d^*]:

$$\begin{aligned} &= \max(dt - 2 \cdot tt(EtT/E) \cdot (StT/S) \cdot (\rho); dt - 2 \cdot tt) \\ &= \max(19.05 - 2 \cdot 2.77 (0.20270E+09/.20132E+09) \cdot \\ &\quad (117/137) \cdot (0); 19.05 - 2 \cdot 2.77) \\ &= 19.0500 \text{ mm.} \end{aligned}$$

Equivalent outer tube limit circle diameter [Do]:

$$\begin{aligned} &= 2 \cdot ro + dt \\ &= 2 \cdot 939.98 + 19.05 \\ &= 1899.010 \text{ mm.} \end{aligned}$$

Determine the basic ligament efficiency for shear [μ]:

$$\begin{aligned} &= (p - dt) / p \\ &= (30.0 - 19.05) / 30.0 \\ &= 0.365 \end{aligned}$$

Equivalent outer tube limit radius [ao]:

$$\begin{aligned} &= Do / 2 \\ &= 1899.01/2 \\ &= 949.5050 \text{ mm.} \end{aligned}$$

Effective tube pitch [p^*]:

$$\begin{aligned} &= p / \sqrt{ (1 - 4 \cdot \min(AL \cdot CNVfactor; 4 \cdot Do \cdot p) / (\pi \cdot Do^2)) } \\ &= 30.0 / \sqrt{ (1 - 4 \cdot \min(0 \cdot 100.0 ; 4 \cdot 1899.01 \cdot 30.0) / (\pi \cdot 1899.01^2)) } \\ &= 30.0000 \text{ mm.} \end{aligned}$$

Effective ligament efficiency for bending [μ^*]:

$$\begin{aligned} &= (p^* - d^*) / p^* \\ &= (30.0 - 19.05) / 30.0 \\ &= 0.3650 \end{aligned}$$

Ratio [ps]:

$$\begin{aligned} &= as / ao \\ &= 970.0/949.505 \\ &= 1.0216 \end{aligned}$$

Ratio [pc]:

$$\begin{aligned} &= ac / ao \\ &= 970.0/949.505 \\ &= 1.0216 \end{aligned}$$

Compute parameter [xt]:

$$\begin{aligned} &= 1 - Nt((dt - 2 \cdot tt) / (2 \cdot ao))^2 \\ &= 1 - 3297 ((19.05 - 2 \cdot 2.77) / (2 \cdot 949.505))^2 \\ &= 0.8331 \end{aligned}$$

Determine parameter [xs]:

$$\begin{aligned} &= 1 - Nt(dt / (2 \cdot ao))^2 \\ &= 1 - 3297 (19.05 / (2 \cdot 949.505))^2 \\ &= 0.6682 \end{aligned}$$

Determine the value [$h'g$]:

$$\begin{aligned} &= \max((hg - CATC); 0) \text{ (For pressure only cases)} \\ &= \max((0 - 0); 0) \\ &= 0.000 \text{ mm.} \end{aligned}$$

ASME Section VIII-2 Part 4.18.8.4 Step 2:

Axial Shell Stiffness [Ks]:

$$\begin{aligned} &= \pi \cdot ts(Ds + ts) Es / L \\ &= \pi \cdot 149.0 (1940.0 + 149.0) .20270E+09 / 10820.0 \\ &= 1868100.3750 \text{ Kg/mm.} \end{aligned}$$

Axial Tube Stiffness [Kt]:



$$= \pi \cdot t_t (D_t - t_t) E_t / L$$

$$= \pi \cdot 2.77 (19.05 - 2.77) \cdot 20270E+09 / 10820.0$$

$$= 270.6511 \text{ Kg/mm.}$$

Stiffness Factor [Ks,t]:

$$= K_s / (N_t \cdot K_t)$$

$$= 1868100 / (3297 \cdot 270.651)$$

$$= 2.09349$$

Rigidity Ratio [J]:

$$= 1 / (1 + K_s / K_j)$$

$$= 1 / (1 + 1868100 / 0)$$

$$= 1.0000 (= 1 \text{ if No Exp. Jt.})$$

Shell coefficient [β_s]:

$$= ((12(1 - \nu_s^2))^{1/4}) / ((D_s + t_s) t_s)$$

$$= ((12(1 - 0.3^2))^{1/4}) / ((1940.0 + 149.0) 149.0)$$

$$= 0.0033 \text{ 1/mm.}$$

Shell coefficient [ks]:

$$= \beta_s \cdot E_s \cdot t_s^3 / (6(1 - \nu_s^2))$$

$$= 0.003 \cdot 0.20270E+09 \cdot 149.0^3 / (6(1 - 0.3^2))$$

$$= 4001435648.0000 \text{ bars} \cdot \text{mm.}^2$$

Shell coefficient [λ_s]:

$$= 6(D_s + t_s) \cdot k_s / h^3 \cdot (1 + h \cdot \beta_s + (h \cdot \beta_s)^2 / 2)$$

$$= 6(1940.0 + 149.0) \cdot 0.40014E+1 / 349.0^3 \cdot (1 + 349.0 \cdot 0.003 + (349.0 \cdot 0.003)^2 / 2)$$

$$= 3284379.2500 \text{ bars}$$

Shell coefficient [δ_s]:

$$= D_s^2 / (4 \cdot E_s \cdot t_s) \cdot (1 - D_s \cdot \nu_s / (2(D_s + t_s)))$$

$$= 1940.0^2 / (4 \cdot 0.20270E+09 \cdot 149.0) \cdot (1 - 1940.0 \cdot 0.3 / (2(1940.0 + 149.0)))$$

$$= 0.26812E-01 \text{ mm./N./mm}^2$$

Channel coefficient [β_c]:

$$= ((12(1 - \nu_c^2))^{1/4}) / ((D_c + t_c) t_c)$$

$$= ((12(1 - 0.3^2))^{1/4}) / ((1940.0 + 25.0) 25.0)$$

$$= 0.0082 \text{ 1/mm.}$$

Channel coefficient [kc]:

$$= \beta_c \cdot E_c \cdot t_c^3 / (6(1 - \nu_c^2))$$

$$= 0.008 \cdot 0.20270E+09 \cdot 25.0^3 / (6(1 - 0.3^2))$$

$$= 47576020.0000 \text{ bars} \cdot \text{mm.}^2$$

Channel coefficients [CC, CqDelta, CqTheta, CmTheta, CmDelta]:

$$CqDelta = CqTheta = CmTheta = CmDelta = 1$$

Channel coefficient [λ_c]:

$$= 6(D_c + t_c / \cos(\theta)) \cdot k_c / h^3 \cdot (CmTheta + h \cdot \beta_c \cdot (Cm\Delta + CqTheta) / 2 + (h \cdot \beta_c)^2 / 2 \cdot Cq\Delta)$$

$$= 6(1940.0 + 25.0 / \cos(0)) \cdot 47576020 / 349.0^3 \cdot (1.0 + 349.0 \cdot 0.008 \cdot (1.0 + 1.0) / 2 + (349.0 \cdot 0.008)^2 / 2 \cdot 1.0)$$

$$= 105023.6 \text{ bars}$$

Channel coefficient [δ_c]:

$$= D_c^2 / (4 \cdot E_c \cdot t_c) \cdot (1 - D_c \cdot \nu_c / (2(D_c + t_c)))$$

$$= 1940.0^2 / (4 \cdot 0.20270E+09 \cdot 25.0) \cdot (1 - 1940.0 \cdot 0.3 / (2(1940.0 + 25.0)))$$

$$= 0.15817E+00 \text{ mm./N./mm}^2$$

Channel coefficient [zetac]:

$$zetac = 0$$

ASME Section VIII-2 Part 4.18.8.4 Step 3:

$$h/p = 11.633333 ; \mu^* = 0.365000 ; E^* = 73897896 \text{ KPa.}$$

$$E^*/E = 0.367063 ; \nu^* = 0.320752$$

Note: As h/p (11.633) is > 2 , data values for $h/p = 2$ were used.

Tube Bundle Stiffness Factor [X_a]:



$$= \left(\frac{24(1 - v^2)Nt \cdot Et \cdot tt(dt - tt)ao^2}{E \cdot L \cdot H^3} \right)^{1/4}$$

$$= \left(\frac{24(1 - 0.321^2)3297 \cdot 0.20270E+09 \cdot 2.77}{(19.05 - 2.77)949.505^2} \right)^{1/4}$$

$$= 2.0368$$

Values from Table 4.18.3

$$Zd = 0.164501 ; Zv = 0.161008 ; Zm = 0.671985$$

$$Za = 0.826311E+00 ; Zw = 0.161008$$

ASME Section VIII-2 Part 4.18.8.4 Step 4:

Diameter ratio [K]:

$$= A / Do$$

$$= 2240.0/1899.01$$

$$= 1.1796$$

Compute coefficient [F]:

$$= (1 - v) / E \cdot (\lambda_s + \lambda_c + E \cdot \ln(K))$$

$$= (1 - 0.32) / 73897896 \cdot (3284379 + 105023.62 + 0.20132E+09 \cdot \ln(1.18))$$

$$= 3.4210$$

Compute parameter [φ]:

$$= (1 + v)F$$

$$= (1 + 0.3208)3.421$$

$$= 4.5183$$

Compute parameter [Q1]:

$$= (ps - 1 - \phi \cdot Zv) / (1 + \phi \cdot Zm)$$

$$= (1.0216 - 1 - 4.5183 \cdot 0.161) / (1 + 4.5183 \cdot 0.672)$$

$$= -0.174891$$

Compute Parameter [Qz1]:

$$= (Zd + Q1 \cdot Zw) / 2 \cdot Xa^4$$

$$= (0.1645 + -0.17489 \cdot 0.16101) / 2 \cdot 2.03684^4$$

$$= 1.1734$$

Compute parameter [Qz2]:

$$= (Zv + Q1 \cdot Zm) / 2 \cdot Xa^4$$

$$= (0.16101 + -0.17489 \cdot 0.67199) / 2 \cdot 2.03684^4$$

$$= 0.3742$$

Compute parameter [U]:

$$= (Zw + (ps - 1)Zm)Xa^4 / (1 + \phi \cdot Zm)$$

$$= (0.161 + (1.0216 - 1)0.672)2.03684^4 / (1 + 4.5183 \cdot 0.672)$$

$$= 0.7484$$

ASME Section VIII-2 Part 4.18.8.4 Step 5:

Determine factor [yb]:

$$= 0 \text{ (config a)}$$

Compute Parameter [γ]:

$$= 0.000 \text{ mm. (For Pressure only cases)}$$

Calculate parameter [ws]:

$$= ps \cdot ks \cdot \beta_s \cdot \delta_s(1 + h \cdot \beta_s)$$

$$= 1.0216 \cdot 0.40014E+1 \cdot 0.0033 \cdot 0.026812(1 + 349.0 \cdot 0.0033)$$

$$= 76325.7578 \text{ mm.}^2$$

Calculate parameter [w*s]:

$$= ao^2(ps^2 - 1) \cdot (ps - 1) / 4 - ws$$

$$= 949.505^2(1.022^2 - 1) \cdot (1.022 - 1) / 4 - 76325.758$$

$$= -76113.4688 \text{ mm.}^2$$

Parameter [wc]:

$$= pc \cdot kc[\delta C(CmTheta \cdot \beta C + Cq\Delta \cdot h \cdot \beta C^2) - \zeta c(CmTheta + h \cdot CqTheta \cdot h \cdot \beta C / 2)]$$

$$= 1.022 \cdot 47576020 [0.158167(1.0 + 0.008 \cdot 1.0 \cdot 349.0) - 0(1.0/349.0 + 1.0 \cdot 0.008/2)]$$



$$= 24353.6348 \text{ mm.}^2$$

Parameter [$\omega \cdot c$]:

$$= ao^2 [(pc^2 + 1) \cdot (pc - 1)/4 - (ps - 1)/2] - \omega c$$

$$= 949.505^2 [(1.02158^2 + 1) \cdot (1.02158 - 1)/4 - (1.02158 - 1)/2] - 24353.63477$$

$$= -24141.3457 \text{ mm.}^2$$

Pressure [$P \cdot S$]:

$$= 0 \text{ For Pressure only cases or Configurations d,e,f,A,B,C,D}$$

Pressure [$P \cdot C$]:

$$= 0 \text{ For Pressure only cases or Configurations b,c,d,B,C,D}$$

ASME Section VIII-2 Part 4.18.8.4 Step 6:

Pressure [$P's$]:

$$= Ps \{ xs + 2(1 - xs)vt + [2/Kst \cdot (Ds/Do)^2] vs - [(ps^2 - 1)/(J \cdot Kst)] [(1 - J)/(2J \cdot Kst)] [(Dj^2 - (Ds)^2)/Do^2] \}$$

$$= 249.472 \{ 0.668 + 2(1 - 0.668)0.3 + [2/2.093 \cdot (1940.0/1899.01)^2] 0.3 - [(1.022^2 - 1)/(1.0 \cdot 2.093)] - [(1 - 1.0)/(2 \cdot 1.0 \cdot 2.093)] [(0^2 - (1940.0)^2)/1899.01^2] \}$$

$$= 285.7829 \text{ bars}$$

Pressure [$P't$]:

$$= [xt + 2(1 - xt)vt + 1/(J \cdot Kst)] Pt$$

$$= [0.833 + 2(1 - 0.833)0.3 + 1/(1. \cdot 2.093)] 13.837$$

$$= 19.5230 \text{ bars}$$

Pressure [P_y]:

$$= Nt \cdot Kt \cdot \gamma / (\pi \cdot ao^2)$$

$$= 3297 \cdot 270.651 \cdot 0 / (3.142 \cdot 949.505^2)$$

$$= 0.0000 \text{ bars}$$

Pressure [P_w]:

$$= -\gamma b \cdot U \cdot W \cdot / (2 \cdot \pi \cdot ao^2)$$

$$= -0 \cdot 0.748 \cdot 0 / (2 \cdot \pi \cdot 949.505^2)$$

$$= -0.0000 \text{ bars}$$

Calculate the pressure [Prim]:

$$= -(U/ao^2) \cdot (\omega \cdot s \cdot Ps - \omega \cdot c \cdot Pt)$$

$$= -(0.748/949.505^2) \cdot (-76113.469 \cdot 249.472 - -24141.346 \cdot 13.837)$$

$$= 15.4860 \text{ bars}$$

Calculate the pressure [P_w]:

$$= (U/ao^2) \cdot (\omega s \cdot P's - \omega c \cdot P \cdot c)$$

$$= (0.748/949.505^2) \cdot (76325.7578 \cdot 0 - 24353.6348 \cdot 0)$$

$$= 0.0000 \text{ bars}$$

Determine the effective pressure [Pe]:

$$= J \cdot Kst / (1 + J \cdot Kst(Qz1 + (ps - 1)Qz2)) \cdot (P's - P't + P_y + P_w + Pw + Prim)$$

$$= 0.1000E+01 \cdot 2.093 / (1 + 1.0 \cdot 2.093 (1.173 + (1.022 - 1)0.374)) \cdot (285.783 - 19.523 + 0 + 0 + 0 + 15.486)$$

$$= 169.8185 \text{ bars}$$

ASME Section VIII-2 Part 4.18.8.4 Step 7:

Determine factor [$Q2$]:

$$= [((\omega \cdot s \cdot Ps - \omega \cdot c \cdot Pt) - (\omega s \cdot P's - \omega c \cdot P \cdot c)) + W \cdot \gamma b / (2 \cdot \pi)] / (1 + \phi \cdot Zm)$$

$$= [((-76113.469 \cdot 249.472 - -24141.346 \cdot 13.837) - (76325.758 \cdot 0 - 24353.635 \cdot 0)) \cdot 0.01 + 0 \cdot 0 / (2 \cdot \pi)] / (1 + 4.51834 \cdot 0.67199)$$

$$= -47129.4336 \text{ Kg}$$

Calculate factor [$Q3$]:

$$= Q1 + 2 \cdot Q2 / (Pe \cdot ao^2)$$

$$= -0.175 + 2 \cdot -47129.434 / (169.818 \cdot 949.505^2)$$

$$= -0.23526$$



Fm Value from Table 4.18.3 = 0.117632

Tubesheet Bending Stress - original thickness [σ]:

$$\begin{aligned} &= (1.5 \cdot F_m / \mu) \cdot (2 \cdot a_o / (H - h'g))^2 \cdot P_e \\ &= (1.5 \cdot 0.1176 / 0.365) \cdot (2 \cdot 949.505 / (349.0 - 0))^2 \cdot 169.82 \\ &= 243.0739 \text{ N./mm}^2 \end{aligned}$$

Allowable Tubesheet Bending Stress [σ allowed]:

$$= 1.5 \cdot \beta_T \cdot S_y = 1.5 \cdot 0.95 \cdot 250.0 = 356.24 \text{ N./mm}^2, 4.18.233$$

Tubesheet Bending Stress - final thickness [σ_f]:

$$\begin{aligned} &= (1.5 \cdot F_m / \mu) \cdot (2 \cdot a_o / (h - h'g))^2 \cdot P_e \\ &= (1.5 \cdot 0.113 / 0.365) \cdot (2 \cdot 949.505 / (275.735 - 0))^2 \cdot 161.68 \\ &= 356.2411 \text{ N./mm}^2 \end{aligned}$$

Reqd Tubesheet Thickness, for Bending Stress (including CA) [HReqB]:

$$\begin{aligned} &= h + C_{ats} + C_{atc} \\ &= 275.7352 + 0 + 0 \\ &= 275.7352 \text{ mm.} \end{aligned}$$

ASME Section VIII-2 Part 4.18.8.4 Step 8:

Shear Stress check [τ_{limit}]:

$$\begin{aligned} &= (2 \cdot \mu \cdot h / a_o) \cdot (0.8 \cdot \beta_T \cdot S_y) \\ &= (2 \cdot 0.365 \cdot 349.0 / 949.505) \cdot 189.997 \\ &= 50.9796 \text{ N./mm}^2 \end{aligned}$$

Shear Stress is not required to be computed; [P_e] <= τ_{limit}

Tubesheet Shear Stress - final thickness [τ_f]:

$$\begin{aligned} &= (A_p \cdot P_e) / (\mu \cdot h \cdot C_p) \\ &= (24988.3 \cdot 60.16) / (0.365 \cdot 36.414 \cdot 5991.4) \\ &= 188.7957 \text{ N./mm}^2 \end{aligned}$$

Reqd Tubesheet Thickness, for Shear Stress (including CA) [HreqS]:

$$\begin{aligned} &= H + C_{ats} + C_{atc} \\ &= 36.4141 + 0 + 0 \\ &= 36.4141 \text{ mm.} \end{aligned}$$

Reqd Tubesheet Thickness for Given Loadings (including CA) [Hreqd] :

$$\begin{aligned} &= \max(H_{reqB}; H_{reqS}) \\ &= \max(275.7352; 36.4141) \\ &= 275.7352 \text{ mm.} \end{aligned}$$

ASME Section VIII-2 Part 4.18.8.4 Step 9:

The F_{tmin} and F_{tmax} Coefficients from Table 4.18.4:

$$F_{tmin} = 0.8296, F_{tmax} = 1.0897$$

First Extreme Tube Axial Stress from among all the tubes [σ_{t1}]:

$$\begin{aligned} &= ((P_s \cdot x_s - P_t \cdot x_t) - P_e \cdot F_{tmin}) / (X_t - X_s) \\ &= ((249.47 \cdot 0.6682 - 13.84 \cdot 0.8331) - (169.818) \cdot 0.83) / \\ &\quad (0.8331 - 0.6682) \\ &= 8.6709 \text{ N./mm}^2 \end{aligned}$$

Second Extreme value of Tube Axial Stress from among all the tubes [σ_{t2}]:

$$\begin{aligned} &= ((P_s \cdot x_s - P_t \cdot x_t) - P_e \cdot F_{tmax}) / (X_t - X_s) \\ &= ((249.47 \cdot 0.6682 - 13.84 \cdot 0.8331) - (169.818) \cdot 1.09) / \\ &\quad (0.8331 - 0.6682) \\ &= -18.1180 \text{ N./mm}^2 \end{aligned}$$

Maximum Tube Axial Stress [$\sigma_{t,max}$]:

$$= \max(|\sigma_{t,1}|; |\sigma_{t,2}|) = 18.118 \text{ N./mm}^2$$

Allowable Tube Stress [$\sigma_{t,A}$]:

$$= \beta_T \cdot S_{y,t} = 242.2456 \text{ N./mm}^2$$

Check for Buckling as some of the tubes are in compression.

Determine the Factor of Safety [Fs]:



$$\begin{aligned} &= \max((3.25 - 0.25(Z_d + Q_3 \cdot Z_w)X_a^4); 1.25) \\ &= \max((3.25 - 0.25(0.165 + -0.235 \cdot 0.161)2.037^4); 1.25) \\ &= 2.0000 \text{ (Should be } \leq 2 \text{)} \end{aligned}$$

Determine the factor [rt]:

$$\begin{aligned} &= (\sqrt{ dt^2 + (dt - 2 \cdot tt)^2 }) / 4 \\ &= (\sqrt{ 19.05^2 + (19.05 - 2 \cdot 2.77)^2 }) / 4 \\ &= 5.8386 \text{ mm.} \end{aligned}$$

Determine the factor [Ct]:

$$\begin{aligned} &= \sqrt{ (2 \cdot \pi^2 \cdot Et / S_{yt}) } \\ &= \sqrt{ (2 \cdot \pi^2 \cdot 0.20270E+09 / 254) } \\ &= 133.6543 \end{aligned}$$

Determine the factor [Ft]:

$$\begin{aligned} &= k \cdot L / r \\ &= 0.8 \cdot 1510.0 / 5.839 \\ &= 206.8999 \end{aligned}$$

Buckling Allowable Stress [Stb]:

$$\begin{aligned} &= (1 / F_s) \cdot (\pi^2 \cdot Et / Ft^2) (Ct \leq Ft) \\ &= (1 / 2.0) \cdot (\pi^2 \cdot 0.20270E+09 / 206.8999^2) \\ &= 242.246 \text{ N./mm}^2 \text{ (Never greater than } S_{ot} \text{)} \end{aligned}$$

Note: The Axial Compressive stress in tubes is within limits.

Largest tube-to-tubesheet Joint Load [Wt]:

$$\begin{aligned} &= \sigma_{t,max} \cdot \text{Tube Area} \\ &= 18.12 \cdot 1.4167 \\ &= 261.74 \text{ Kgf} \end{aligned}$$

Tube Metal Area [At]:

$$\begin{aligned} &= \pi(d_o - t)t \\ &= \pi(19.05 - 2.77)2.77 \\ &= 1.417 \text{ cm}^2 \end{aligned}$$

Tube Weld Size Results per UW-20.3:

Tube Strength [Ft]:

$$\begin{aligned} &= \pi \cdot t(d_o - t)S \\ &= \pi \cdot 2.77 (19.05 - 2.77)242.246 \\ &= 3499.543 \text{ Kgf} \end{aligned}$$

Groove Weld Strength [Fg]:

$$\begin{aligned} &= 0.85 \cdot \pi \cdot a_g(d_o + 0.67 \cdot a_g)S_w \text{ (but not } > Ft \text{)} \\ &= 0.85 \cdot \pi \cdot 2.7 (19.05 + 0.67 \cdot 2.7)237.5 \\ &= 3499.5427 \text{ Kgf} \end{aligned}$$

Weld Strength Factor [fw]:

$$\begin{aligned} &= S / S_w \text{ where } S_w = \min(S \text{ and } S_t) \\ &= 242.246 / 237.496 \\ &= 1.020 \end{aligned}$$

Design Strength Ratio [fd]:

$$= 1 \text{ for full strength welds}$$

Fillet Weld Strength [Ff]:

$$\begin{aligned} &= 0.55 \cdot \pi \cdot a_f(d_o + 0.67 \cdot a_f)S_w \text{ (but not } > Ft \text{)} \\ &= 0.55 \cdot \pi \cdot 2.7 (19.05 + 0.67 \cdot 2.7)237.5 \\ &= 2356.6602 \text{ Kgf} \end{aligned}$$

Minimum Length of Weld [ar]:

$$\begin{aligned} &= 2(\sqrt{ (0.75 \cdot d_o)^2 + 1.07 \cdot t \cdot (d_o - t)fw \cdot fd } - 0.75 \cdot d_o) \\ &= 2(\sqrt{ (0.75 \cdot 19.05)^2 + 1.07 \cdot 2.77 \cdot (19.05 - 2.77)1.02 \cdot 1.0 } - 0.75 \cdot 19.05) \\ &= 3.259 \text{ mm.} \end{aligned}$$

Minimum Required Fillet Weld Leg [afr]: 1.6620 mm.
Minimum Required Groove Weld Leg [agr]: 1.6620 mm.



Tube-Tubesheet Jt allowable, 3499.54 is >= tube strength 3499.54 Kgf

Note: This tube-tubesheet joint is a Full Strength joint

ASME Section VIII-2 Part 4.18.8.4 Step 10:

Shell Axial Membrane Allowable Stress:
 $= \beta T \cdot S_{y,s} = 0.95 \cdot 260.0 = 370.49 \text{ N./mm}^2$

Axial Membrane Stress in Shell $[\sigma_s, m]$:
 $= a o^2 / ((D_s + t_s) t_s) \cdot [P_e + (p s^2 - 1)(P_s - P_t)] + a s^2 \cdot P_t / ((D_s + t_s) t_s)$
 $= 949.505^2 / ((1940.0 + 149.0) 149.0) \cdot [169.82 + (1.022^2 - 1)$
 $(249.47 - 13.84)] + 970.0^2 \cdot 13.84 / ((1940.0 + 149.0) 149.0)$
 $= 56.3517 \text{ N./mm}^2$

ASME Section VIII-2 Part 4.18.8.4 Step 11:

Note:
 For a given Shell thickness of 149.0 mm., the minimum Shell length adjacent to the tubesheet should be 967.757 mm.

Stress/Force Summary for Loadcase D3 un-corr. (Psd,max + Ptd,max):

Stress Description	Actual	Allowable	Pass/Fail
Tubesheet Bend. Stress	243.1 <=	356.2 N./mm ²	Pass
Tubesheet Shear Stress	55.6 <=	190.0 N./mm ²	Pass
Maximum Tube Stress	18.1 <=	242.2 N./mm ²	Pass
Minimum Tube Stress (Buckling)	-18.1 <=	-242.2 N./mm ²	Pass
Maximum Force on any one Tube	261.7 <=	3499.5 Kgf	Pass
Axial Membrane Stress in Shell	56.4 <=	370.5 N./mm ²	Pass

Thickness Results for Loadcase D3 un-corr. (Psd,max + Ptd,max):

Thickness (mm.)	Required	Actual	P/F
Tubesheet Thickness :	275.735	349.000	Ok
Tube-Tubesheet Fillet Weld Leg :	1.662	2.700	Ok
Tube-Tubesheet Groove Weld Leg :	1.662	2.700	Ok

Fixed Tubesheet results per ASME Div. 2 Part 4.18.8 2025

Results for 3 Load Cases:

Case#	--Reqd. Thk. + CA Tbsht Extnsn	----	Tubesheet Stresses Bend Allwd Shear Allwd	Case Type	Pass/Fail
D1uc	9.597 ...	14	356 4 190	Ps+Pt-Th D1	Ok
D2uc	286.116 ...	257	356 60 190	Ps+Pt-Th D2	Ok
D3uc	275.735 ...	243	356 56 190	Ps+Pt-Th D3	Ok
Max:	286.1160 ... mm.	0.721	0.313 (Str. Ratio)		

Load Case Definitions:

[Ps & Pt]:
 Shell-side and Tube-side Design or Operating Pressures
 derived from Psd,min Ptd,max, Psox,min, Ptox,max etc. per the Load Case Tables

[(+/-)Th]:
 With or Without Thermal Expansion, Tt,mx & Ts,mx

[c]:
 With or Without Corrosion Allowance

[D1, D2, D3]:
 Design Load Cases using the Maximum and Minimum Design Pressures



[D4]:
 Design Load Case using the Minimum (Vacuum) Pressures (if specified)

[O1, O2, O3, O4]:
 Operating Load Cases using the Maximum and Minimum Operating Pressures and
 Operating Temperatures

Shell Axial Membrane Stress Summary:

---- Shell Stresses ----					:	---- Shell Band Stress ----					:	Pass
Case#	Ten	Allwd	Cmp	Allwd	:	Ten	Allwd	Cmp	Allwd	:	Fail	
D1uc	1	370	...	...	:	...	...	...	...	:		Ok
D2uc	56	370	...	...	:	...	...	...	...	:		Ok
D3uc	56	370	...	...	:	...	...	...	...	:		Ok
Max RATIO	0.152		...		:	...		...		:		

Tube, Shell and Channel Stress Summary:

---- Tube Stresses ----					Tube Loads		Shell Stress		Channel Stress		Pass
Case#	Ten	Allwd	Cmp	Allwd	Ld	Allwd	Stress	Allwd	Stress	Allwd	Fail
D1uc	1	242	-1	-242	20	3499	...	370	...	370	Ok
D2uc	19	242	-19	-242	281	3499	...	370	...	370	Ok
D3uc	18	242	-18	-242	262	3499	...	370	...	370	Ok
Max RATIO	0.080		0.080		0.080		...		...		

Summary of Thickness Comparisons for 3 Load Cases:

Thickness (mm.)	Required	Actual	P/F
Tubesheet thickness :	286.116	349.000	Ok
Tube thickness :	2.520	2.770	Ok
Tube-tubesheet fillet weld leg :	1.662	2.700	Ok
Tube-tubesheet groove weld leg :	1.662	2.700	Ok

Min Shell length of thk, (149.000) adj. to tubesheet: 967.757 mm.

Note: This is a full strength Tube to Tubesheet Joint.

Summary of Axial Differential Expansion between Shell and Tubes :

Due to Thermal Expansion Shell Extends by : +.000 mm.
 Due to Pressure Shell Compresses by : -1.305 mm.
 Due to Pressure + Thermal Shell Compresses by : -1.305 mm.

Summary of Exchanger Pressure and Thermal Displacements: (mm.)

Load Case#	Due to Pressure		Due to Thermal Exp.		Computed Delta Ms	
	UnCorroded	Corroded	UnCorroded	Corroded	UnCorroded	Corroded
1	0.0096	...	...	...	...	...
2	-1.3054	...	...	...	...	...
3	-1.2958	...	...	...	...	...
4	...	...	...	...	...	...
5	...	...	...	...	...	...
6	...	...	...	...	...	...
7	...	...	...	...	...	...
8	...	...	...	...	...	...

External Pressure Calculation Results Summary:

ASME Code, Section VIII Division 1, 2025

External Pressure Calculations:

From	To	Section Length mm.	Outside Diameter mm.	Corroded Thickness mm.	Factor A	Factor B N./mm ²
10	20	No Calc	...	102	No Calc	No Calc
20	30	No Calc	...	107	No Calc	No Calc
30	Tsht	1599.97	1990	22	0.0019946	103.696
Tsht	40	No Calc	...	22	No Calc	No Calc
40	Tsht	10820	2238	147.5	0.0052998	93.2527
Tsht	50	No Calc	...	147.5	No Calc	No Calc
50	60	1600.03	1990	22	0.0019946	103.695
60	70	No Calc	...	107	No Calc	No Calc
70	80	No Calc	...	102	No Calc	No Calc

External Pressure Calculations:

From	To	External Actual T. mm.	External Required T. mm.	External Design Pressure bars	External M.A.W.P. bars
10	20	105	45.3378	...	No Calc
20	30	110	107.848	...	No Calc
30	Tsht	25	No Calc	...	15.2843
Tsht	40	25	No Calc	...	No Calc
40	Tsht	149	No Calc	...	81.942
Tsht	50	149	No Calc	...	No Calc
50	60	25	No Calc	...	15.2841
60	70	110	107.848	...	No Calc
70	80	105	45.3378	...	No Calc

Minimum

15

External Pressure Calculations:

From	To	Actual Length Bet. Stiffeners mm.	Allowable Length Bet. Stiffeners mm.	Ring Inertia Required cm ⁴	Ring Inertia Available cm ⁴
10	20	No Calc	No Calc	No Calc	No Calc
20	30	No Calc	No Calc	No Calc	No Calc
30	Tsht	1599.97	No Calc	No Calc	No Calc
Tsht	40	No Calc	No Calc	No Calc	No Calc
40	Tsht	10820	No Calc	No Calc	No Calc
Tsht	50	No Calc	No Calc	No Calc	No Calc
50	60	1600.03	No Calc	No Calc	No Calc
60	70	No Calc	No Calc	No Calc	No Calc
70	80	No Calc	No Calc	No Calc	No Calc

Elements Suitable for External Pressure.

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Element and Detail Weights:

From	To	Element Metal Wgt. kg.	Element ID Volume cm ³	Corroded Metal Wgt. kg.	Corroded ID Volume cm ³	Extra due Misc % kg.
10	20	3089.45	...	3089.45	...	...
20	30	647.937	460357	625.938	461089	...
30	40	1913.83	4730326	1686.74	4759631	...
40	50	82003.4	21819234	81236.1	21918246	...
50	60	1913.83	4730326	1686.74	4759631	...
60	70	647.937	460357	625.938	461089	...
70	80	3089.45	...	3089.45	...	...
Total		93305	32200598.00	92040	32359688.00	0



For elements specified as shell side elements, the volume(s) shown above for those elements, reflects the displacement of the tubes.

Weight of Details:

From	Type	Weight of Detail kg.	X Offset, Dtl. Cent. mm.	Y Offset, Dtl. Cent. mm.	Z Offset Dtl. Cent. mm.	Description
30	Liqd	2305.6	800	492.5	...	Liquid: 30
30	Nozl	225.335	800	1338.3	...	TS OUT NOZ. T2
30	Nozl	5.30174	800	-986.993	...	DRAIN NOZ. D1
40	Sadl	679.839	2200	1240	...	FIXED SADDLE
40	Sadl	679.839	8620	1240	...	SLIDING SADDLE
40	Liqd	12865.5	5410	...	...	Liquid: 40
40	Nozl	2067.02	1270	1274.8	...	SH. IN A2
40	Nozl	2067.02	9550	-1274.8	...	SH. OUT NOZ B2
40	Wght	10000	5410	...	...	Margin on weigh
40	Wght	5920	5410	...	...	PDS weight diff
50	Liqd	4703.8	800	...	...	Liquid: 50
50	Nozl	225.335	800	...	1338.3	CH. IN NOZ. A1
50	Nozl	25.849	800	1013.66	...	SAFETY VALVE S1
50	Nozl	5.30174	750	-986.993	...	DRAIN NOZ. D2
60	Liqd	225.509	75	492.5	...	Liquid: 60
30	FTsh	8117.74	1774.5	...	...	FIXED TS
30	Tube	41697.5	7359	...	...	Tubes
30	RTsh	8117.74	12943.5	...	...	Tubesheet
40	Bafl	241.24	910	...	...	Baffles
40	Bafl	241.24	1510	...	...	Baffles
40	Bafl	241.24	2110	...	...	Baffles
40	Bafl	241.24	2710	...	...	Baffles
40	Bafl	241.24	3310	...	...	Baffles
40	Bafl	241.24	3910	...	...	Baffles
40	Bafl	241.24	4510	...	...	Baffles
40	Bafl	241.24	5110	...	...	Baffles
40	Bafl	241.24	5710	...	...	Baffles
40	Bafl	241.24	6310	...	...	Baffles
40	Bafl	241.24	6910	...	...	Baffles
40	Bafl	241.24	7510	...	...	Baffles
40	Bafl	241.24	8110	...	...	Baffles
40	Bafl	241.24	8710	...	...	Baffles
40	Bafl	241.24	9310	...	...	Baffles
40	Bafl	241.877	9910	...	...	Baffles

Total Weight of Each Detail Type:

Saddles	1359.7
Liquid	20100.4
Nozzles	4621.2
Weights	15920.0



Exchanger Components 61793.5
 Liquid in Tubes 5414.2
 Sum of the Detail Weights 109208.9 kg.

Weight Summation Results: (kg.)

	Fabricated	Shop Test	Shipping	Erected	Empty	Operating
Main Elements	93305.8	93305.8	93305.8	93305.8	93305.8	93305.8
Saddles	1359.7	1359.7	1359.7	1359.7	1359.7	1359.7
Nozzles	4621.2	4621.2	4621.2	4621.2	4621.2	4621.2
Wld Weights	5920.0	5920.0	5920.0	5920.0	5920.0	5920.0
Exchanger	61793.5	61793.5	61793.5	61793.5	61793.5	61793.5
Empty Weights	...	...	...	10000.0	10000.0	...
Test Weights	...	10000.0	...	...	...	...
Ope Weights	...	...	...	...	...	10000.0
Ope. Liquid	...	...	...	...	...	20100.4
Tube Ope Lqd	...	...	...	...	...	5414.2
Test Liquid	...	32180.9	...	...	...	...
Tube Tst Lqd	...	5441.4	...	...	...	...
Totals	167000.1	214622.4	167000.1	177000.1	177000.1	202514.7

Weight Summary:

Fabricated Wt. - Bare weight without removable internals 167000.1 kg.
 Shop Test Wt. - Fabricated weight + water (full) 214622.4 kg.
 Shipping Wt. - Fab. weight + removable intls.+ shipping app. 167000.1 kg.
 Erected Wt. - Fab. wt + or - loose items (trays,platforms etc.) 177000.1 kg.
 Ope. Wt. no Liq - Fab. weight + internals. + details + weights 177000.1 kg.
 Operating Wt. - Empty weight + operating liq. uncorroded 202514.7 kg.
 Oper. Wt. + CA - Corr wt. + operating liquid 201249.2 kg.

Exchanger Tube Data

Volume of Exchanger tubes : 5444706.0 cm³
 Weight of Ope Liq in tubes : 5414.2 kg.
 Weight of Water in tubes : 5441.4 kg.

The Corroded Weight and thickness are used in the Horizontal Vessel Analysis (Ope Case) and Earthquake Load Calculations.

Field Test weight is computed in the corroded condition.

Surface Areas of Elements:

From	To	Outside Surface Area cm ²	Inside Surface Area cm ²
10	20	43660.5	...
20	30	16951.9	9446.77
30	40	100028	97515
40	50	760742	659445
50	60	100028	97515
60	70	16951.9	9446.77
70	80	43660.5	...

Total 1082023.1 873369.0 cm²
 Total 1164.7 940.1 ft²



Nozzle Flange MAWP Results: (bars & °C)

Nozzle Description	Flange Rating		Des. Temp	Class	Grade/Group	Equiv. Press	Max Pressure		
	Ope.	Ambient					UG-44	50%	DNV
TS OUT NOZ. T2	18.12	19.60	85	150	GR 1.1	...	...	...	...
DRAIN NOZ. D1	18.12	19.60	85	150	GR 1.1	...	...	...	...
CH. IN NOZ. A1	18.12	19.60	85	150	GR 1.1	...	...	...	...
SAFETY VALVE S	18.12	19.60	85	150	GR 1.1	...	...	...	...
DRAIN NOZ. D2	18.12	19.60	85	150	GR 1.1	...	...	...	...

Shellside Flange Rating

Channelside Flange Rating

Lowest Flange Pressure Rating was (Ope)[TubeSide]: 18.120 bars
 Lowest Flange Pressure Rating was (Amb)[TubeSide]: 19.600 bars

Flange Pressure Ratings are per ASME B16.5 2025 Metric Edition

Large Diameter Flange Pressure Ratings are per ASME B16.47 2025 Edition

Warning:

There are nozzles in this model, but no flange MAWPs were de-rated. Be sure this is what you intended. There is a check box in the nozzle dialog that instructs PV Elite to perform the de-rating for each nozzle flange. See ASME VIII-1, UG-44(b) for more information.

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Wind Load Calculation Results:

Input Values:

Wind Design Code	ASCE/SEI 7-22
Wind Load Reduction Scale Factor	1.100
Basic Wind Speed	[V] 180 Km/hr
Surface Roughness Category	D: Flat, unobstructed
Type of Surface	0
Base Elevation	300 mm.
Percent Wind for Hydrotest	33.0
Using User defined Wind Press. Vs Elev.	N
Height of Hill or Escarpment H or Hh	0 mm.
Distance Upwind of Crest Lh	0 mm.
Distance from Crest to the Vessel x	0 mm.
Type of Terrain (Hill, Escarpment)	Flat
Damping Factor (Beta) for Wind (Ope)	0.0100
Damping Factor (Beta) for Wind (Empty)	0.0000
Damping Factor (Beta) for Wind (Filled)	0.0000
Site Elevation above Sea Level	0 mm.

Wind Analysis Results

The importance factor is not used in the calculation of wind pressure vs. elevation for the selected wind code.

Static Gust-Effect Factor, Operating Case [G]:
 $= \min(0.85, 0.925((1 + 1.7 \cdot gQ \cdot I_{zbar} \cdot Q) / (1 + 1.7 \cdot gV \cdot I_{zbar})))$
 $= \min(0.85, 0.925((1 + 1.7 \cdot 3.4 \cdot 0.194 \cdot 0.98) / (1 + 1.7 \cdot 3.4 \cdot 0.194)))$
 $= \min(0.85, 0.915)$
 $= 0.850$

Natural Frequency of Vessel (Operating)	33.000 Hz
Natural Frequency of Vessel (Empty)	33.000 Hz
Natural Frequency of Vessel (Test)	33.000 Hz

User Entered Importance Factor is	1.000
Force Coefficient	[Cf] 0.808
Structure Height to Diameter ratio	7.342

This is classified as a rigid structure. Static analysis performed.

Sample Calculation for the First Element:

The ASCE code performs calculations in Imperial Units. The wind pressure is therefore computed in these units.

Value of [α] and [Zg]:
 Exposure Category: D from Table 26.11-1
 $\alpha = 11.5$: $Z_g = 589788$ mm.

Effective Height [z]:
 $= \text{Centroid Height} + \text{Vessel Base Elevation}$
 $= 1675.0 + 300.0 = 1975.0$ mm.
 $= 6.48$ ft. Imperial Units

Velocity Pressure coefficient evaluated at height z [Kz]:
 Because z (6.48 ft.) < 15 ft.
 $= 2.41(15 / Z_g)^{2/\alpha}$
 $= 2.41(15 / 1935.0)^{2/11.5}$
 $= 1.035$

Type of Hill: No Hill

Wind Directionality Factor [Kd]:
 $= 1$, per Table 26.6-1

As there is No Hill Present: [Kzt]:
 $K_1 = 0$, $K_2 = 0$, $K_3 = 0$

Topographical Factor [Kzt]:

$$= (1 + K1 \cdot K2 \cdot K3)^2$$

$$= (1 + 0.0 \cdot 0.0 \cdot 0.0)^2$$

$$= 1.0$$

Velocity Pressure evaluated at height z, Imperial Units [qz]:

$$= \max(16, 0.00256 \cdot Kz \cdot Kzt \cdot Kd \cdot Ke \cdot V(\text{mph})^2)$$

$$= \max(16, 0.00256 \cdot 1.035 \cdot 1.0 \cdot 1.0 \cdot 1.0 \cdot 111.85^2)$$

$$= 33.1 \text{ psf } [161.849 \text{ Kgs/m}^2]$$

Force on the first element [F]:

$$= qz \cdot G \cdot Cf \cdot \text{WindArea}$$

$$= 33.149 \cdot 0.85 \cdot 0.808 \cdot 2.913$$

$$= 66.3 \text{ lbs. } [30.1 \text{ Kgf}]$$

Element	Hgt (z) mm.	K1	K2	K3	Kz	Kzt	qz Kgs/m ²
CH. COVER-1	1975.0	0.000	0.000	0.000	1.035	1.000	161.849
CH. FLG-1	1975.0	0.000	0.000	0.000	1.035	1.000	161.849
CH. SHELL-1	1975.0	0.000	0.000	0.000	1.035	1.000	161.849
MAIN SHELL	1975.0	0.000	0.000	0.000	1.035	1.000	161.849
CH. SHELL-2	1975.0	0.000	0.000	0.000	1.035	1.000	161.849
CH. FLG-2	1975.0	0.000	0.000	0.000	1.035	1.000	161.849
CH. COVER-2	1975.0	0.000	0.000	0.000	1.035	1.000	161.849

Wind Loads on Masses/Equipment/Piping

ID	Wind Area cm ²	Elevation mm.	Pressure Kgs/m ²	Force Kgf
Margin on wei	0.00	1975.00	161.85	0.00
PDS weight di	0.00	1975.00	161.85	0.00

Wind Load Calculation:

From	To	Wind Height mm.	Wind Diameter mm.	Wind Area cm ²	Wind Pressure Kgs/m ²	Element Wind Load Kgf
10	20	1975	2577.6	2706.48	161.849	33.0773
20	30	1975	2328	3492	161.849	42.6776
30	40	1975	2388	38208	161.849	466.96
40	50	1975	2685.6	290582	161.849	3551.36
50	60	1975	2388	38208	161.849	466.96
60	70	1975	2328	3492	161.849	42.6776
70	80	1975	2577.6	2706.48	161.849	33.0773

The Wind Loads calculated and printed in the Wind Load calculation report have been factored by the input scalar/load reduction factor of: 1.100.

Be sure the wind speed is in accordance with the specified wind design code.

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Earthquake Load Calculation Results:

Earthquake Load Calculation:

Input Values:

Seismic Design Code		ASCE/SEI 7-22	
Importance Factor		1.500	
Force Modification Factor R		3.000	
Site Class		D	
Component Elevation Ratio	z/h	0.000	
Amplification Factor	Ap	0.000	
Force Factor		0.000	
Consider Vertical Acceleration		Yes	
Minimum Acceleration Multiplier		0.000	
User Value of Sds (used if > 0)		0.000	
User Value of Sd1 (used if > 0)		0.000	
Component resonance duct. fact.	[CAR]	0.000	
Component strength factor	[Rpo]	0.000	
Overstrength factor	[Omega0]	0.000	
Response Acceleration Factor	[SMS]	0.326	
Response Acceleration Factor	[SM1]	0.139	
Building Importance Factor	[Ie]	1.000	
Site Latitude		0.000	deg
Site Longitude		0.000	deg

Seismic Analysis Results:

$$SDS = 2/3 * Sms = 2/3 * 0.326 = 0.217$$

$$SD1 = 2/3 * Sm1 = 2/3 * 0.139 = 0.093$$

Check Approximate Fundamental Period from 12.8-7 [Ta]:
 = $C_t \cdot h_{nx}$ where $C_t = 0.020$, $x = 0.75$ and h_n = Structural Height (ft.)
 = $0.020(9.0190.75)$
 = 0.104 seconds

The Coefficient C_u from Table 12.8-1 is : 1.700

Fundamental Period (1/Frequency) [T]:
 = $(1/\text{Natural Frequency}) = (1/33.0)$
 = 0.030

Check the Value of T which is the smaller of $C_u \cdot T_a$ and T:
 = Minimum Value of $(1.7 \cdot 0.104, 0.03)$ per 12.8.2
 = 0.030

As the time period is < 0.06 second, use section 15.4.2.

Base Shear per equation 15.4-5, [V]:
 = $0.3 \cdot Sds \cdot W \cdot I$
 = $0.3 \cdot 0.217 \cdot 199890 \cdot 1.5$
 = 19549.203 Kgf

Vertical load per 12.4-4, [Ev]:
 = $0.2 \cdot Sds \cdot W$
 = $0.2 \cdot 0.217 \cdot 199890$
 = 8688.5 Kgf

Earthquake Load Calculation:

From	To	Earthquake Height mm.	Earthquake Weight Kgf	Element Ope Load Kgf
10	20	1074	22210	2349.05
20	30	970	22210	2121.58
30	40	970	22210	2121.58
40	Sad1	970	22210	2121.58
Sad1	50	970	22210	2121.58
40	50	970	22210	2121.58



50	60	970	22210	2121.58
60	70	970	22210	2121.58
70	80	1074	22210	2349.05

The Earthquake Weight does not include the weight of the saddles.

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Shop/Field Installation Options :

Note : The CG is computed from the first Element From Node

Center of Gravity of the Saddles	7618.999 mm.
Center of Gravity of the Liquid	8485.000 mm.
Center of Gravity of the Nozzles	7655.629 mm.
Center of Gravity of the Added Weights (Operating)	7618.999 mm.
Center of Gravity of the Added Weights (Empty)	7618.999 mm.
Center of Gravity of the Tubesheet(s)	7619.000 mm.
Center of Gravity of the Tubes	7619.000 mm.
Center of Gravity of the Baffles	7619.739 mm.
Center of Gravity of Bare Shell New and Cold	7619.125 mm.
Center of Gravity of Bare Shell Corroded	7619.126 mm.
Vessel CG in the Operating Condition	7706.407 mm.
Vessel CG in the Fabricated (Shop/Empty) Condition	7620.039 mm.
Vessel CG in the Test Condition	7619.862 mm.

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ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : Operating Case

Note:

Wear Pad Width (340.00) is less than $1.56 \cdot \sqrt{r_m \cdot t}$
 and less than $2a$. The wear plate will be ignored.

Minimum Wear Plate Width to be considered in analysis [b1]:

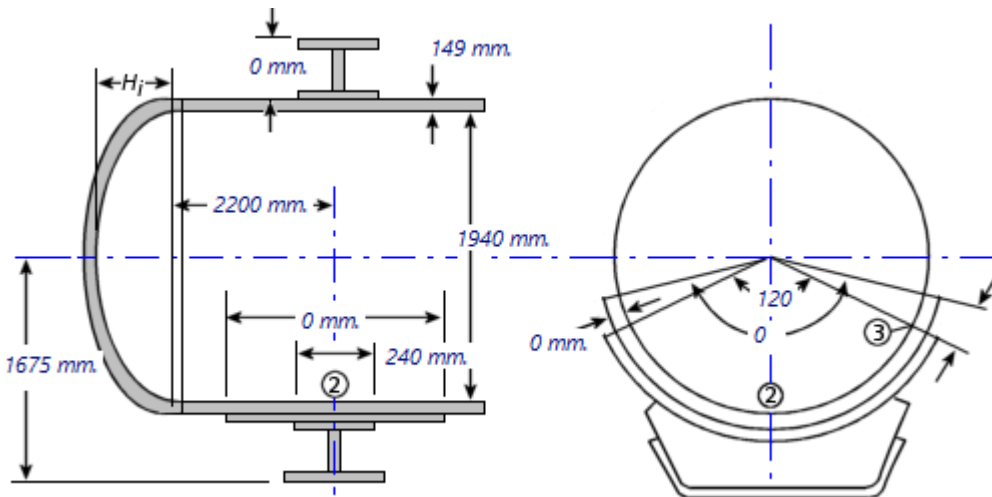
$$= \min(b + 1.56 \cdot \sqrt{R_m \cdot t}, 2a)$$

$$= \min(240.0 + 1.56 \cdot \sqrt{1045.25 \cdot 147.5}, 2 \cdot 2200.0)$$

$$= 852.5347 \text{ mm.}$$

Input and Calculated Values:

Vessel Mean Radius	Rm	1045.25	mm.
Shell Thickness used in this Case	t	149.000	mm.
Stiffened Vessel Length per 4.15.6	L	10820.00	mm.
Distance from Saddle to Vessel tangent	a1 or a	2200.00	mm.
Saddle Width	b1 or b	240.00	mm.
Saddle Bearing Angle	delta or theta	120.00	degrees
Shell Allowable Stress used in Calculation		138.00	N./mm ²
Head Allowable Stress used in Calculation		138.00	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1675.00	mm.
Coefficient of Friction	mu	0.60	



Results for Horizontal Vessel Analysis:

Saddle Force Q, Operating Case		126534.00	Kgf
Pressure used in Analysis	P	191.056	bars

Horizontal Vessel Analysis Results:	Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan	66.34	138.00
Long. Stress at Bottom of Midspan	69.06	138.00
Long. Stress at Top of Saddles	76.82	138.00
Long. Stress at Bottom of Saddles	62.64	138.00
Tangential Shear in Shell	5.59	110.40
Circ. Stress at Horn of Saddle	6.99	172.50
Circ. Compressive Stress in Shell	0.75	138.00

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:
 $= Ftr(Ft/Num\ of\ Saddles + Z\ Force\ Load) \cdot B / E$
 $= 3.0(4636.8/2 + 0) \cdot 1675.0/1965.8777$
 $= 5926.1\ Kg$

Longitudinal Saddle Reaction Force [Fwl]:
 $= \max(Fl,wind; Sum\ of\ X\ Forces) \cdot B / Ls$
 $= \max(1008.5, 0) \cdot 1675.0/6420.001$
 $= 263.1\ Kg$

Saddle Reaction Force due to Earthquake Fl [Fsl]:
 $= \max(Fl,eq; Sum\ of\ X\ Forces) \cdot B / Ls$
 $= \max(19549.2, 0) \cdot 1675.0/6420.001$
 $= 5100.5\ Kg$

Saddle Reaction Force due to Earthquake Ft [Fst]:
 $= Ftr(Ft/Num\ of\ Saddles + Z\ Force\ Load) \cdot B / E$
 $= 3.0(19549/2 + 0) \cdot 1675.0/1965.8777$
 $= 24985.0\ Kg$

Load Combination Results for Q + Wind or Seismic [Q]:
 $= Saddle\ Load + \max(Fwl, Fwt, Fsl, Fst)$
 $= 101549 + \max(263, 5926, 5100, 24985)$
 $= 126534.0\ Kg$

Longitudinal Wind Force [Fl]:
 $= WindScalar \cdot WindPress(Platform\ Area + (\pi/4(OD \cdot WindDiaMult)^2))$
 $= 1.1 \cdot 1587.169(0.0 + \pi/4(2.238 \cdot 1.2)^2)$
 $= 9889.890\ N$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	127213.84	Kgf
Transverse Shear Load	9774.60	Kgf
Longitudinal Shear Load	19549.20	Kgf

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:
 $= \min(2(ShellID/2 + t + WearPadThickness) \sin(\theta/2), 2 \cdot Rm)$
 $= \min(2(1940.0/2 + 149.0 + 16.0) \sin(120.0/2), 2 \cdot 1045.25)$
 $= 1965.878\ mm$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	

Note: Dimension a is greater than or equal to $Rm/2$.

Moment per Equation 4.15.1 [M1]:
 $= -Q \cdot a [1 - (1 - a/L + (Rm^2 - hm^2)/(2a \cdot L)) / (1 + (4hm)/3L)]$
 $= -126534 \cdot 2200.0 [1 - (1 - 2200.0/10820.0 + (1045.25^2 - 0.0^2) / (2 \cdot 2200.0 \cdot 10820.0)) / (1 + (4 \cdot 0.0) / (3 \cdot 10820.0))]$
 $= -50213.8\ Kg \cdot m$

Moment per Equation 4.15.2 [M2]:
 $= Q \cdot L / 4 ((1 + 2(Rm^2 - hm^2)/(L^2)) / (1 + (4hm^2)/(3L)) - 4a/L)$
 $= 126534 \cdot 10820 / 4 ((1 + 2(1045^2 - 0^2)/(10820^2)) / (1 + (4 \cdot 0) / (3 \cdot 10820)) - 4 \cdot 2200 / 10820)$
 $= 70289.4\ Kg \cdot m$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:
 $= P \cdot Rm / (2t) - M2 / (\pi \cdot Rm^2 t)$
 $= 191.056 \cdot 1045.25 / (2 \cdot 147.5) - 70289.4 / (\pi \cdot 1045.2^2 \cdot 147.5)$
 $= 66.34\ N/mm^2$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot Rm / (2t) + M2 / (\pi \cdot Rm^2 \cdot t)$$

$$= 191.056 \cdot 1045.25 / (2 \cdot 147.5) + 70289.4 / (\pi \cdot 1045.2^2 \cdot 147.5)$$

$$= 69.06 \text{ N./mm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ^*3]:

$$= P \cdot Rm / (2t) - M1 / (K1 \cdot \pi \cdot Rm^2 \cdot t)$$

$$= 191.056 \cdot 1045.25 / (2 \cdot 147.5) - 50213.8 / (0.1066 \cdot \pi \cdot 1045.2^2 \cdot 147.5)$$

$$= 76.82 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ^*4]:

$$= P \cdot Rm / (2t) + M1 / (K1 \cdot \pi \cdot Rm^2 \cdot t)$$

$$= 191.056 \cdot 1045.25 / (2 \cdot 147.5) + 50213.8 / (0.1923 \cdot \pi \cdot 1045.2^2 \cdot 147.5)$$

$$= 62.64 \text{ N./mm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L-2a) / (L + (4 \cdot hm / 3))$$

$$= 126534(10820.0 - 2 \cdot 2200.0) / (10820.0 + (4 \cdot 0.0 / 3))$$

$$= 75078.4 \text{ Kg}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [$\tau2$]:

$$= K2 \cdot T / (Rm \cdot t)$$

$$= 1.1707 \cdot 75078.4 / (1045.25 \cdot 147.5)$$

$$= 5.59 \text{ N./mm}^2$$

Decay Length (4.15.20) [$x1, x2$]:

$$= 0.78 \cdot \sqrt{(Rm \cdot t)}$$

$$= 0.78 \cdot \sqrt{(1045.25 \cdot 147.5)}$$

$$= 306.267 \text{ mm.}$$

Circumferential Stress in shell, no rings (4.15.21) [$\sigma6$]:

$$= -K5 \cdot Q \cdot k / (t(b + X1 + X2))$$

$$= -0.7603 \cdot 126534 \cdot 0.1 / (147.5(240.0 + 306.27 + 306.27))$$

$$= -0.75 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8Rm$ (4.15.22) [$\sigma7$]:

$$= -Q / (4 \cdot t(b + X1 + X2)) - 3 \cdot K7 \cdot Q / (2 \cdot t^2)$$

$$= -126534 / (4 \cdot 147.5(240.0 + 306.267 + 306.267)) - 3 \cdot 0.0529 \cdot 126534 / (2 \cdot 147.5^2)$$

$$= -6.99 \text{ N./mm}^2$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(Rm \cdot t)}, 2a)$$

$$= \min(240.0 + 1.56 \cdot \sqrt{(1045.25 \cdot 147.5)}, 2 \cdot 2200.0)$$

$$= 852.53 \text{ mm.}$$

Distance between Saddle Supports [Ls]:

$$= 6420.0 \text{ mm.}$$

Free Un-Restrained Thermal Expansion between the Saddles [Exp]:

$$= \alpha \cdot Ls(\text{Design Temperature} - \text{Ambient Temperature})$$

$$= 0.000013 \cdot 6420.001(260.0 - 21.1)$$

$$= 20.150 \text{ mm.}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	2040.0001	mm.
Baseplate Thickness	Bpthk	56.0000	mm.
Baseplate Width	Bpwid	280.0000	mm.
Number of Ribs (inc. outside ribs)	Nribs	4	
Rib Thickness	Ribtk	16.0000	mm.
Web Thickness	Webtk	16.0000	mm.
Web Location	Webloc	Side	
Saddle Design Temperature		21.1	°C
Saddle Yield Stress	Sy	168.4	N./mm ²
Height of Web at Center	Hw,c	306.0	mm.
Friction Coefficient	mu	0.600	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})t^3) / 12$$

$$= ((1.56 \cdot \sqrt{(971.5 \cdot 147.5)} + 340.0)147.5^3)/12$$

$$= 24884.260 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C1 - Y)^2$$

$$= 24884.26 + 1372.532(148.692 - 73.75)^2$$

$$= 101970.734 \text{ cm}^4$$

	B mm.	D mm.	Y mm.	A cm ²	AY mm. ³	I + AD ⁽²⁾ cm ⁴
Shell	930.5	147.5	73.8	1372.5	10122421.0	0.102E+06
Wearplate	340.0	16.0	155.5	54.4	845919.9	36.8
Web	16.0	484.0	405.5	77.4	3140192.2	0.662E+05
BasePlate	280.0	56.0	675.5	156.8	10591841.0	0.436E+06
Totals	...	...	...	1661.2	24700374.0	0.604E+06

Distance to Centroid [C1]:

$$= AY / A$$

$$= 24700374/1661.172$$

$$= 148.692 \text{ mm.}$$

Angle [β]:

$$= 180 - \text{Saddle Angle}/2$$

$$= 180 - 120.0/2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta)\cos(\beta))$$

$$= (1 + \cos(120.0) - 0.5 \cdot \sin(120.0)^2) / (\pi - 2.094 + \sin(120.0)\cos(120.0))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.204 \cdot 126534.0$$

$$= 25752.4121 \text{ Kg}$$

$$\text{Tension Stress, } St = (Fh/As) = 8.7496 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 101.0403 \text{ N./mm}^2$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$

$$= 1675.0 - 971.5 \cdot \sin(120.0/2) / 1.0472$$

$$= 871.576 \text{ mm.}$$

$$\text{Bending Moment, } M = Fh \cdot d = 22445.6289 \text{ Kg-m.}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 5.4209 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 2/3 \cdot \text{Yield Str} = 112.2670 \text{ N./mm}^2$$

Minimum Thickness of Baseplate per Moss:

The maximum of 2/3*Yield and the Saddle Allowable Stress value was used to determine the baseplate thickness.

$$= \sqrt{3(Q + \text{SaddleWt})\text{BasePlateWidth} / (2 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt{3(126534 + 680)280.0 / (2 \cdot 2040.0 \cdot 137.997)}$$

$$= 43.143 \text{ mm.}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120.0/2) (970.0 + 149.0 + 16.0)$$

$$= 1965.878 \text{ mm.}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nrifs} - 1)$$

$$= 1965.8777 / (4 - 1)$$

$$= 655.293 \text{ mm.}$$



Baseplate Pressure Area [Ap]:

$$= e \cdot Bpwid / 2$$

$$= 655.2925 \cdot 280.0 / 2$$

$$= 917.410 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + SaddleWt) / (BasePlateLength \cdot BasePlateWidth)$$

$$= 127213.836 / (2040.0 \cdot 280.0)$$

$$= 22.271 \text{ Kg/cm}^2$$

Axial Load [P]:

$$= Ap \cdot Bp$$

$$= 917.4 \cdot 22.27$$

$$= 20431.930 \text{ Kg}$$

Area of the Rib and Web [Ar]:

$$= Rib \text{ Area} + Web \text{ Area}$$

$$= 35.84 + 52.423$$

$$= 88.263 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P / Ar$$

$$= 20431.9 / 88.2634$$

$$= 22.702 \text{ N./mm}^2$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib	16.0	224.0	128.0	35.8	458752.0	0.332E+04
Web	327.6	16.0	8.0	52.4	41938.7	0.126E+04
Totals	...	...	...	88.3	500690.7	0.458E+04

Rib dimension [D]:

$$= Saddle \text{ Width} - Web \text{ Thickness}$$

$$= 240.0 - 16.0$$

$$= 224.000 \text{ mm.}$$

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 500691 / 88.263$$

$$= 56.727 \text{ mm.}$$

Distance to Centroid [C1]:

$$= \max(ytot, Saddle \text{ Width} - ytot)$$

$$= \max(56.727, 240.0 - 56.727)$$

$$= 183.273 \text{ mm.}$$

Radius of Gyration [r]:

$$= \sqrt{ (Total \text{ Inertia} / Total \text{ Area}) }$$

$$= \sqrt{ (4575.1 / 88.263) }$$

$$= 71.996 \text{ mm.}$$

Length of Outer Rib [L]:

$$= Saddle \text{ Height} - \cos(\theta / 2) (radius + shlthk + wpdthk) - bpthk$$

$$= 1675.0 - \cos(120.0 / 2) (970.0 + 149.0 + 16.0) - 56.0$$

$$= 1051.500 \text{ mm.}$$

Intermediate Term [Cc]:

$$= \sqrt{ (2 \cdot \pi^2 \cdot Elastic \text{ Modulus} / Yield \text{ Stress}) }$$

$$= \sqrt{ (2 \cdot \pi^2 \cdot 0.18822E+09 / 168.4) }$$

$$= 148.540$$

Slenderness ratio [KL/r]:

$$= KL / r$$

$$= 1 \cdot 1051.5 / 71.996$$

$$= 14.605$$

Bending Moment [Rm]:

$$= FI / (2 \cdot B_{pln}) \cdot e \cdot L / 2$$

$$= 19549.2 / (2 \cdot 2040.0) \cdot 655.293 \cdot 1051.5 / 2$$

$$= 1650.791 \text{ Kg-m.}$$

Compressive Allowable, $KL/r < C_c$ (14.6049 < 148.54) per AISC E2-1 [Sca]:

$$= (1 - (Klr)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (Klr) / (8 \cdot C_c) - (Klr^3) / (8 \cdot C_c^3))$$

$$= (1 - (14.6)^2 / (2 \cdot 148.54^2)) 168 /$$

$$(5/3 + 3 \cdot (14.6) / (8 \cdot 148.54) - (14.6^3) / (8 \cdot 148.54^3))$$

$$= 98.38 \text{ N./mm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (R_m \cdot C_1 / I) / S_{ba}$$

$$= 22.7 / 98.38 + (1650.79 \cdot 183.273 / 45750880) / 112.27$$

$$= 0.808$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	16.0	224.0	128.0	35.8	458752.0	0.436E+04
Web	655.3	16.0	8.0	104.8	83877.4	0.100E+04
Totals	...	...	...	140.7	542629.4	0.537E+04

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 542629 / 140.687$$

$$= 38.570 \text{ mm.}$$

Distance to Centroid [C1]:

$$= \max(y_{tot}, \text{abs}(\text{Saddle Width} - y_{tot}))$$

$$= \max(38.57, \text{abs}(240.0 - 38.57))$$

$$= 201.430 \text{ mm.}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \sqrt{ (Ro + W_{pdthk})^2 - (Pitch/2)^2 } - B_{pthk}$$

$$= 1675.0 - \sqrt{ (1135.0 + 16.0)^2 - (655.293 / 2)^2 } - 56.0$$

$$= 532.320 \text{ mm.}$$

Radius of Gyration [r]:

$$= \sqrt{ (\text{Total Inertia} / \text{Total Area}) }$$

$$= \sqrt{ (5367.2 / 140.687) }$$

$$= 61.765 \text{ mm.}$$

Slenderness ratio [KL/r]:

$$= KL / r$$

$$= 1 \cdot 532.32 / 61.765$$

$$= 8.618$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 19549.203 / (2 \cdot 2040.0)$$

$$= 4.791 \text{ Kg/mm.}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 4.791 \cdot 655.293 \cdot 532.32$$

$$= 1671.420 \text{ Kg-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 1671.42 / 266452.938$$

$$= 61.516 \text{ N./mm}^2$$

Compressive Allowable, $KL/r < C_c$ (8.6184 < 148.54) per AISC E2-1 [Sca]:

$$= (1 - (Klr)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (Klr) / (8 \cdot C_c) - (Klr^3) / (8 \cdot C_c^3))$$

$$= (1 - (8.62)^2 / (2 \cdot 148.54^2)) 168 /$$

$$(5/3 + 3 \cdot (8.62) / (8 \cdot 148.54) - (8.62^3) / (8 \cdot 148.54^3))$$



$$= 99.57 \text{ N./mm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= Sc/Sca + (Mbase,c \cdot C1/I)/Sba$$

$$= 28.02/99.57 + (1671.42 \cdot 201.43/5367.156)/112.27$$

$$= 0.829$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	3
Total Number of Bolts in Tension/Baseplate	Nbt	1
Bolt Design Temperature		21.1 °C
Bolt Material Specification		SA-36
Bolt Allowable Stress	Stba	149.62 N./mm ²
Bolt Corrosion Allowance	Bca	1.5 mm.
Distance from Bolts to Edge	Edgedis	85.0 mm.
Nominal Bolt Diameter	Bnd	42.0000 mm.
Thread Series	Series	TEMA Metric
BasePlate Allowable Stress	S	107.80 N./mm ²
Area Available in a Single Bolt	BltArea	8.5576 cm ²
Saddle Load QO (Weight)	QO	102228.9 Kg
Saddle Load QL (Wind/Seismic contribution)	QL	5100.5 Kg
Maximum Transverse Force	Ft	9774.6 Kg
Maximum Longitudinal Force	Fl	19549.2 Kg
Saddle Bolted to Steel Foundation		No

Shear Stress in a Single Bolt, Longitudinal Direction [tb,l]:

$$= Fl / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 19549.2 / (8.558 \cdot 3)$$

$$= 74.7 \text{ N./mm}^2. \text{ Must be less than } 99.8 \text{ N./mm}^2.$$

Shear Stress in a Single Bolt, Transverse Direction [tb,t]:

$$= Ft / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 9774.6 / (8.558 \cdot 3)$$

$$= 37.3 \text{ N./mm}^2. \text{ Must be less than } 99.8 \text{ N./mm}^2.$$

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:

$$= 0.0 \text{ (QO > QL --> No Uplift in Longitudinal direction)}$$

Bolt Area due to Shear Load [Bltarears]:

$$= Fl / (\text{BoltShearAllowable} \cdot \text{Nbolts})$$

$$= 19549.2 / (99.75 \cdot 3.0)$$

$$= 6.4065 \text{ cm}^2$$

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:

$$= B \cdot Ft + \text{Sum of X Moments}$$

$$= 1675.0 \cdot 9774.6 + 0.0$$

$$= 16372.78 \text{ Kg-m.}$$

Eccentricity (e):

$$= Rmom / QO$$

$$= 16372.78/102228.88$$

$$= 160.15 \text{ mm.} < Bplen/6 \text{ --> No Uplift in Transverse direction}$$

Bolt Area due to Transverse Load [Bltareart]:

$$= 0 \text{ (No Uplift)}$$

Required Area of a Single Bolt [Bltarear]:

$$= \max[Bltarearl, Bltarears, Bltareart]$$

$$= \max[0.0, 6.4065, 0.0]$$

$$= 6.4065 \text{ cm}^2$$

MDMT Calculations for: FIXED SADDLE, Saddle to Wearplate Junction, Curve: B

$$\text{Govrn. thk, tg} = 16.0, c = 1.5 \text{ mm.}, E* = 1.0$$

Thickness Ratio = $tr * E / (tg - c) = -1.0$, Temp. Reduction = 0 °C
 Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B -14 °C
 Min Metal Temp per UCS-66 and UCS-68(c), PWHT credit -31 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Note:

Wear Pad Width (340.00) is less than $1.56 * \sqrt{rm * t}$
 and less than 2a. The wear plate will be ignored.

Minimum Wear Plate Width to be considered in analysis [b1]:
 $= \min(b + 1.56 * \sqrt{ Rm * t }, 2a)$
 $= \min(240.0 + 1.56 * \sqrt{ 1045.25 * 147.5 }, 2 * 2200.0)$
 $= 852.5347 \text{ mm.}$

Input and Calculated Values:

Vessel Mean Radius	Rm	1045.25	mm.
Shell Thickness used in this Case	t	149.000	mm.
Stiffened Vessel Length per 4.15.6	L	10820.00	mm.
Distance from Saddle to Vessel tangent	a1 or a	2200.00	mm.
Saddle Width	b1 or b	240.00	mm.
Saddle Bearing Angle	delta or theta	120.00	degrees
Shell Allowable Stress used in Calculation		138.00	N./mm ²
Head Allowable Stress used in Calculation		138.00	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1675.00	mm.
Coefficient of Friction	mu	0.10	

Results for Horizontal Vessel Analysis:

Saddle Force Q, Operating Case		132014.02	Kgf
Pressure used in Analysis	P	191.056	bars

Horizontal Vessel Analysis Results:	Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan	66.28	138.00
Long. Stress at Bottom of Midspan	69.12	138.00
Long. Stress at Top of Saddles	77.22	138.00
Long. Stress at Bottom of Saddles	62.42	138.00
Tangential Shear in Shell	5.83	110.40
Circ. Stress at Horn of Saddle	7.29	172.50
Circ. Compressive Stress in Shell	0.78	138.00

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:
 $= Ftr(Ft/Num \text{ of Saddles} + Z \text{ Force Load}) * B / E$
 $= 3.0(4636.8/2 + 0) * 1675.0/1965.8777$
 $= 5926.1 \text{ Kgf}$

Longitudinal Saddle Reaction Force [Fwl]:
 $= \max(Fl, wind; \text{Sum of X Forces}) * B / Ls$
 $= \max(1008.5, 0) * 1675.0/6420.001$
 $= 263.1 \text{ Kgf}$

Saddle Reaction Force due to Earthquake Fl [Fsl]:
 $= \max(Fl, eq; \text{Sum of X Forces}) * B / Ls$
 $= \max(19549.2, 0) * 1675.0/6420.001$
 $= 5100.5 \text{ Kgf}$

Saddle Reaction Force due to Earthquake Ft [Fst]:
 $= Ftr(Ft/Num\ of\ Saddles + Z\ Force\ Load) \cdot B / E$
 $= 3.0(19549/2 + 0) \cdot 1675.0/1965.8777$
 $= 24985.0\ Kg$

Load Combination Results for Q + Wind or Seismic [Q]:
 $= Saddle\ Load + \max(Fwl, Fwt, Fsl, Fst)$
 $= 107029 + \max(263, 5926, 5100, 24985)$
 $= 132014.0\ Kg$

Longitudinal Wind Force [FI]:
 $= WindScalar \cdot WindPress(Platform\ Area + (\pi/4(OD \cdot WindDiaMult)^2))$
 $= 1.1 \cdot 1587.169(0.0 + \pi/4(2.238 \cdot 1.2)^2)$
 $= 9889.890\ N$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	132693.84	Kgf
Transverse Shear Load	9774.60	Kgf
Longitudinal Shear Load	19549.20	Kgf

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:
 $= \min(2(ShellID/2 + t + WearPadThickness) \sin(\theta/2), 2 \cdot Rm)$
 $= \min(2(1940.0/2 + 149.0 + 16.0) \sin(120.0/2), 2 \cdot 1045.25)$
 $= 1965.878\ mm$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	

Note: Dimension a is greater than or equal to $Rm/2$.

Moment per Equation 4.15.1 [M1]:
 $= -Q \cdot a [1 - (1 - a/L + (Rm^2 - hm^2)/(2a \cdot L)) / (1 + (4hm)/3L)]$
 $= -132014 \cdot 2200.0 [1 - (1 - 2200.0/10820.0 + (1045.25^2 - 0.0^2) / (2 \cdot 2200.0 \cdot 10820.0)) / (1 + (4 \cdot 0.0) / (3 \cdot 10820.0))]$
 $= -52388.5\ Kg \cdot m$

Moment per Equation 4.15.2 [M2]:
 $= Q \cdot L/4 [(1 + 2(Rm^2 - hm^2)/(L^2)) / (1 + (4hm^2)/(3L)) - 4a/L]$
 $= 132014 \cdot 10820/4 [(1 + 2(1045^2 - 0^2)/(10820^2)) / (1 + (4 \cdot 0) / (3 \cdot 10820)) - 4 \cdot 2200/10820]$
 $= 73333.6\ Kg \cdot m$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:
 $= P \cdot Rm/(2t) - M2/(\pi \cdot Rm^2 t)$
 $= 191.056 \cdot 1045.25/(2 \cdot 147.5) - 73333.6/(\pi \cdot 1045.2^2 \cdot 147.5)$
 $= 66.28\ N/mm^2$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:
 $= P \cdot Rm/(2t) + M2/(\pi \cdot Rm^2 \cdot t)$
 $= 191.056 \cdot 1045.25/(2 \cdot 147.5) + 73333.6/(\pi \cdot 1045.2^2 \cdot 147.5)$
 $= 69.12\ N/mm^2$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_3]:
 $= P \cdot Rm/(2t) - M1/(K1 \cdot \pi \cdot Rm^2 t)$
 $= 191.056 \cdot 1045.25/(2 \cdot 147.5) - 52388.5/(0.1066 \cdot \pi \cdot 1045.2^2 \cdot 147.5)$
 $= 77.22\ N/mm^2$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ_4]:
 $= P \cdot Rm/(2t) + M1/(K1* \cdot \pi \cdot Rm^2 \cdot t)$
 $= 191.056 \cdot 1045.25/(2 \cdot 147.5) + 52388.5/(0.1923 \cdot \pi \cdot 1045.2^2 \cdot 147.5)$
 $= 62.42\ N/mm^2$

Maximum Shear Force in the Saddle (4.15.3) [T]:
 $= Q(L - 2a)/(L + (4 \cdot hm/3))$
 $= 132014(10820.0 - 2 \cdot 2200.0) / (10820.0 + (4 \cdot 0.0/3))$

= 78329.9 Kgf

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K_2 \cdot T / (R_m \cdot t)$$

$$= 1.1707 \cdot 78329.94 / (1045.25 \cdot 147.5)$$

$$= 5.83 \text{ N./mm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(1045.25 \cdot 147.5)}$$

$$= 306.267 \text{ mm.}$$

Circumferential Stress in shell, no rings (4.15.21) [σ_6]:

$$= -K_5 \cdot Q \cdot k / (t(b + X_1 + X_2))$$

$$= -0.7603 \cdot 132014 \cdot 0.1 / (147.5(240.0 + 306.27 + 306.27))$$

$$= -0.78 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8R_m$ (4.15.22) [σ_7]:

$$= -Q / (4 \cdot t(b + X_1 + X_2)) - 3 \cdot K_7 \cdot Q / (2 \cdot t^2)$$

$$= -132014 / (4 \cdot 147.5(240.0 + 306.267 + 306.267)) -$$

$$3 \cdot 0.0529 \cdot 132014 / (2 \cdot 147.5^2)$$

$$= -7.29 \text{ N./mm}^2$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(240.0 + 1.56 \cdot \sqrt{(1045.25 \cdot 147.5)}, 2 \cdot 2200.0)$$

$$= 852.53 \text{ mm.}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	2040.0001	mm.
Baseplate Thickness	Bpthk	56.0000	mm.
Baseplate Width	Bpwid	280.0000	mm.
Number of Ribs (inc. outside ribs)	Nribs	4	
Rib Thickness	Ribtk	16.0000	mm.
Web Thickness	Webtk	16.0000	mm.
Web Location	Webloc	Side	
Saddle Design Temperature		21.1	°C
Saddle Yield Stress	Sy	168.4	N./mm ²
Height of Web at Center	Hw,c	306.0	mm.
Friction Coefficient	mu	0.100	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})t^3)/12$$

$$= ((1.56 \cdot \sqrt{(971.5 \cdot 147.5)} + 340.0)147.5^3)/12$$

$$= 24884.260 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C_1 - Y)^2$$

$$= 24884.26 + 1372.532(148.692 - 73.75)^2$$

$$= 101970.734 \text{ cm}^4$$

	B mm.	D mm.	Y mm.	A cm ²	AY mm. ³	I + AD ⁽²⁾ cm ⁴
Shell	930.5	147.5	73.8	1372.5	10122421.0	0.102E+06
Wearplate	340.0	16.0	155.5	54.4	845919.9	36.8
Web	16.0	484.0	405.5	77.4	3140192.2	0.662E+05
BasePlate	280.0	56.0	675.5	156.8	10591841.0	0.436E+06
Totals	...	...	...	1661.2	24700374.0	0.604E+06

Distance to Centroid [C1]:

$$= AY / A$$

$$= 24700374/1661.172$$

$$= 148.692 \text{ mm.}$$

Angle [β]:

$$= 180 - \text{Saddle Angle}/2$$



$$= 180 - 120.0/2$$
$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$
$$= (1 + \cos(120.0) - 0.5 \cdot \sin(120.0)^2) / (\pi - 2.094 + \sin(120.0) \cos(120.0))$$
$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$
$$= 0.204 \cdot 132014.016$$
$$= 26867.7148 \text{ Kgf}$$

$$\text{Tension Stress, } St = (Fh / As) = 9.1286 \text{ N./mm}^2$$
$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 101.0403 \text{ N./mm}^2$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$
$$= 1675.0 - 971.5 \cdot \sin(120.0/2) / 1.0472$$
$$= 871.576 \text{ mm.}$$

$$\text{Bending Moment, } M = Fh \cdot d = 23417.7188 \text{ Kg-m.}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 5.6557 \text{ N./mm}^2$$
$$\text{Allowed Stress, } Sa = 2/3 \cdot \text{Yield Str} = 112.2670 \text{ N./mm}^2$$

Minimum Thickness of Baseplate per Moss:

The maximum of $2/3 \cdot \text{Yield}$ and the Saddle Allowable Stress value was used to determine the baseplate thickness.

$$= \sqrt{3(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth} / (2 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$
$$= \sqrt{3(132014 + 680) \cdot 280.0 / (2 \cdot 2040.0 \cdot 137.997)}$$
$$= 44.062 \text{ mm.}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$
$$= 2 \cdot \cos(90 - 120.0/2) (970.0 + 149.0 + 16.0)$$
$$= 1965.878 \text{ mm.}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nribs} - 1)$$
$$= 1965.8777 / (4 - 1)$$
$$= 655.293 \text{ mm.}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$
$$= 655.2925 \cdot 280.0/2$$
$$= 917.410 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$
$$= 132693.844 / (2040.0 \cdot 280.0)$$
$$= 23.231 \text{ Kgf/cm}^2$$

Axial Load [P]:

$$= Ap \cdot Bp$$
$$= 917.4 \cdot 23.23$$
$$= 21312.078 \text{ Kgf}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$
$$= 35.84 + 52.423$$
$$= 88.263 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P / Ar$$
$$= 21312.1 / 88.2634$$

= 23.680 N./mm²

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib	16.0	224.0	128.0	35.8	458752.0	0.332E+04
Web	327.6	16.0	8.0	52.4	41938.7	0.126E+04
Totals	...	...	...	88.3	500690.7	0.458E+04

Rib dimension [D]:

= Saddle Width - Web Thickness
 = 240.0 - 16.0
 = 224.000 mm.

Distance to Centroid from Datum [ytot]:

= AY / A
 = 500691/88.263
 = 56.727 mm.

Distance to Centroid [C1]:

= max(ytot, Saddle Width - ytot)
 = max(56.727, 240.0 - 56.727)
 = 183.273 mm.

Radius of Gyration [r]:

= $\sqrt{\text{Total Inertia} / \text{Total Area}}$
 = $\sqrt{4575.1/88.263}$
 = 71.996 mm.

Length of Outer Rib [L]:

= Saddle Height - cos(theta/2)(radius + shlthk + wpdthk) - bpthk
 = 1675.0 - cos(120.0/2)(970.0 + 149.0 + 16.0) - 56.0
 = 1051.500 mm.

Intermediate Term [Cc]:

= $\sqrt{2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress}}$
 = $\sqrt{2 \cdot \pi^2 \cdot 0.18822\text{E}+09/168.4}$
 = 148.540

Slenderness ratio [KL/r]:

= KL/r
 = 1 • 1051.5/71.996
 = 14.605

Bending Moment [Rm]:

= FI / (2 • Bplen) • e • L / 2
 = 19549.2 / (2 • 2040.0) • 655.293 • 1051.5/2
 = 1650.791 Kg-m.

Compressive Allowable, KL/r < Cc (14.6049 < 148.54) per AISC E2-1 [Sca]:

= (1 - (KL/r)² / (2 • Cc²)) Fy / (5/3 + 3 • (KL/r) / (8 • Cc) - (KL/r)³ / (8 • Cc³))
 = (1 - (14.6)² / (2 • 148.54²)) 168 /
 (5/3 + 3 • (14.6) / (8 • 148.54) - (14.6³) / (8 • 148.54³))
 = 98.38 N./mm²

AISC Unity Check of Outside Ribs (must be ≤ 1)

= Sc/Sca + (Rm • C1 / I) / Sba
 = 23.68/98.38 + (1650.79 • 183.273/45750880) / 112.27
 = 0.818

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	16.0	224.0	128.0	35.8	458752.0	0.436E+04
Web	655.3	16.0	8.0	104.8	83877.4	0.100E+04



Totals	...	...	...	140.7	542629.4	0.537E+04
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Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 542629/140.687$$

$$= 38.570 \text{ mm.}$$

Distance to Centroid [C1]:

$$= \max(ytot, \text{abs}(\text{Saddle Width} - ytot))$$

$$= \max(38.57, \text{abs}(240.0 - 38.57))$$

$$= 201.430 \text{ mm.}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \sqrt{ (Ro + Wpdthk)^2 - (Pitch/2)^2 } - Bpthk$$

$$= 1675.0 - \sqrt{ (1135.0 + 16.0)^2 - (655.293/2)^2 } - 56.0$$

$$= 532.320 \text{ mm.}$$

Radius of Gyration [r]:

$$= \sqrt{ \text{Total Inertia} / \text{Total Area} }$$

$$= \sqrt{ 5367.2/140.687 }$$

$$= 61.765 \text{ mm.}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 532.32/61.765$$

$$= 8.618$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 19549.203/(2 \cdot 2040.0)$$

$$= 4.791 \text{ Kg/mm.}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 4.791 \cdot 655.293 \cdot 532.32$$

$$= 1671.420 \text{ Kg-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 1671.42/266452.938$$

$$= 61.516 \text{ N./mm}^2$$

Compressive Allowable, $KL/r < C_c$ (8.6184 < 148.54) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (8.62)^2 / (2 \cdot 148.54^2)) 168 /$$

$$(5/3 + 3 \cdot (8.62) / (8 \cdot 148.54) - (8.62^3) / (8 \cdot 148.54^3))$$

$$= 99.57 \text{ N./mm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= Sc/Sca + (M_{base,c} \cdot C1/I) / S_{ba}$$

$$= 29.23/99.57 + (1671.42 \cdot 201.43/5367.156) / 112.27$$

$$= 0.841$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	3
Total Number of Bolts in Tension/Baseplate	Nbt	1
Bolt Design Temperature		21.1 °C
Bolt Material Specification		SA-36
Bolt Allowable Stress	Stba	149.62 N./mm ²
Bolt Corrosion Allowance	Bca	1.5 mm.
Distance from Bolts to Edge	Edgedis	85.0 mm.
Nominal Bolt Diameter	Bnd	42.0000 mm.
Thread Series	Series	TEMA Metric
BasePlate Allowable Stress	S	107.80 N./mm ²
Area Available in a Single Bolt	BlArea	8.5576 cm ²
Saddle Load Q0 (Weight)	Q0	107708.9 Kg
Saddle Load QL (Wind/Seismic contribution)	QL	5100.5 Kg
Maximum Transverse Force	Ft	9774.6 Kg
Maximum Longitudinal Force	Fl	19549.2 Kg

Saddle Bolted to Steel Foundation

No

Shear Stress in a Single Bolt, Longitudinal Direction [tb,l]:
= $F_l / (\text{Bolt Area} \cdot \text{Number of Bolts})$
= $19549.2 / (8.558 \cdot 3)$
= 74.7 N./mm^2 . Must be less than 99.8 N./mm^2 .

Shear Stress in a Single Bolt, Transverse Direction [tb,t]:
= $F_t / (\text{Bolt Area} \cdot \text{Number of Bolts})$
= $9774.6 / (8.558 \cdot 3)$
= 37.3 N./mm^2 . Must be less than 99.8 N./mm^2 .

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:
= 0.0 (QO > QL --> No Uplift in Longitudinal direction)

Bolt Area due to Shear Load [Bltarears]:
= $F_l / (\text{BoltShearAllowable} \cdot \text{Nbols})$
= $19549.2 / (99.75 \cdot 3.0)$
= 6.4065 cm^2

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:
= $B \cdot F_t + \text{Sum of X Moments}$
= $1675.0 \cdot 9774.6 + 0.0$
= 16372.78 Kg-m .

Eccentricity (e):
= $Rmom / QO$
= $16372.78 / 107708.9$
= 152.01 mm . < Bplen/6 --> No Uplift in Transverse direction

Bolt Area due to Transverse Load [Bltareart]:
= 0 (No Uplift)

Required Area of a Single Bolt [Bltarear]:
= $\max[\text{Bltarearl}, \text{Bltarears}, \text{Bltareart}]$
= $\max[0.0, 6.4065, 0.0]$
= 6.4065 cm^2

MDMT Calculations for: SLIDING SADDLE, Saddle to Wearplate Junction, Curve: B

Govrn. thk, tg = 16.0, c = 1.5 mm., $E^* = 1.0$
Thickness Ratio = $tr \cdot E^* / (tg - c) = -1.0$, Temp. Reduction = 0°C
Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B	-14 °C
Min Metal Temp per UCS-66 and UCS-68(c),PWHT credit	-31 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Review notes about nozzle loadings and supports in the Warnings report.

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ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : Test Case

Note:

Wear Pad Width (340.00) is less than $1.56 \cdot \sqrt{r_m \cdot t}$
 and less than $2a$. The wear plate will be ignored.

Minimum Wear Plate Width to be considered in analysis [b1]:

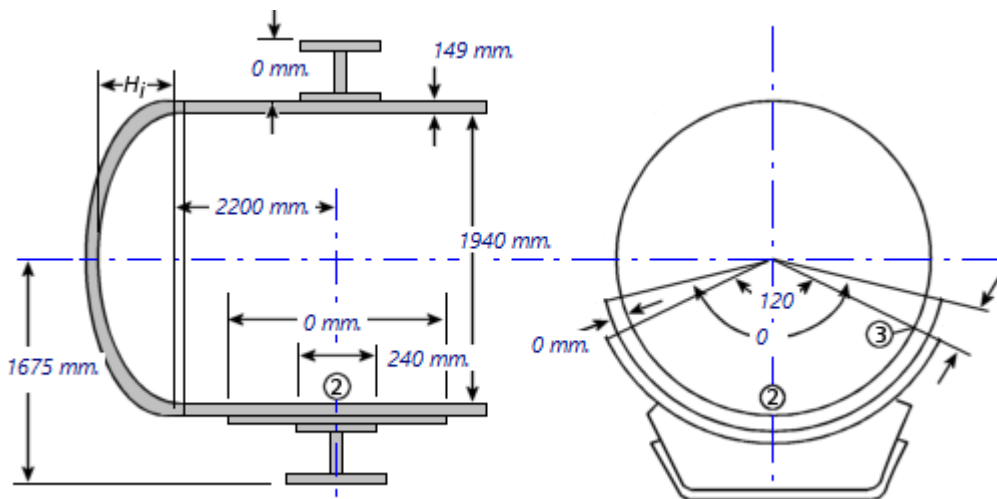
$$= \min(b + 1.56 \cdot \sqrt{R_m \cdot t}, 2a)$$

$$= \min(240.0 + 1.56 \cdot \sqrt{1045.25 \cdot 147.5}, 2 \cdot 2200.0)$$

$$= 852.5347 \text{ mm.}$$

Input and Calculated Values:

Vessel Mean Radius	R_m	1045.25	mm.
Shell Thickness used in this Case	t	149.000	mm.
Stiffened Vessel Length per 4.15.6	L	10820.00	mm.
Distance from Saddle to Vessel tangent	a_1 or a	2200.00	mm.
Saddle Width	b_1 or b	240.00	mm.
Saddle Bearing Angle	δ or θ	120.00	degrees
Shell Allowable Stress used in Calculation		234.00	N./mm ²
Head Allowable Stress used in Calculation		234.00	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1675.00	mm.
Coefficient of Friction	μ	0.60	



Results for Horizontal Vessel Analysis:

Saddle Force Q, Test Case, no Ext. Forces		108558.13	Kgf
Pressure used in Analysis	P	249.574	bars

Horizontal Vessel Analysis Results:	Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan	87.27	234.00
Long. Stress at Bottom of Midspan	89.60	234.00
Long. Stress at Top of Saddles	96.26	234.00
Long. Stress at Bottom of Saddles	84.10	234.00
Tangential Shear in Shell	4.80	138.58
Circ. Stress at Horn of Saddle	6.00	350.99
Circ. Compressive Stress in Shell	0.64	234.00

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:
 $= F_{tr} (F_t / \text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$
 $= 3.0 (1530.1/2 + 0) \cdot 1675.0/1965.8777$
 $= 1955.6 \text{ Kgf}$

Longitudinal Saddle Reaction Force [Fwl]:
 $= \max(F_{l, \text{wind}}, \text{Sum of X Forces}) \cdot B / L_s$
 $= \max(332.81, 0) \cdot 1675.0/6420.001$
 $= 86.8 \text{ Kgf}$

Load Combination Results for Q + Wind or Seismic [Q]:
 $= \text{Saddle Load} + \max(F_{wl}, F_{wt}, F_{sl}, F_{st})$
 $= 106603 + \max(87, 1956, 0, 0)$
 $= 108558.1 \text{ Kgf}$

Longitudinal Wind Force [Fl]:
 $= \text{WindScalar} \cdot \text{WindPress} (\text{Platform Area} + (\pi/4 (OD \cdot \text{WindDiaMult})^2))$
 $= 1.1 \cdot 1587.169 (0.0 + \pi/4 (2.238 \cdot 1.2)^2)$
 $= 9889.890 \text{ N}$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	109237.98	Kgf
Transverse Shear Load	765.07	Kgf
Longitudinal Shear Load	332.81	Kgf

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:
 $= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness}) \sin(\theta/2), 2 \cdot R_m)$
 $= \min(2(1940.0/2 + 149.0 + 16.0) \sin(120.0/2), 2 \cdot 1045.25)$
 $= 1965.878 \text{ mm}$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	

Note: Dimension a is greater than or equal to $R_m/2$.

Moment per Equation 4.15.1 [M1]:
 $= -Q \cdot a [1 - (1 - a/L + (R_m^2 - h_m^2)/(2a \cdot L)) / (1 + (4h_m)/3L)]$
 $= -108558 \cdot 2200.0 [1 - (1 - 2200.0/10820.0 + (1045.25^2 - 0.0^2) / (2 \cdot 2200.0 \cdot 10820.0)) / (1 + (4 \cdot 0.0) / (3 \cdot 10820.0))]$
 $= -43080.2 \text{ Kg-m}$

Moment per Equation 4.15.2 [M2]:
 $= Q \cdot L/4 ((1 + 2(R_m^2 - h_m^2)/(L^2)) / (1 + (4h_m^2)/(3L)) - a/L)$
 $= 108558 \cdot 10820.0 / 4 ((1 + 2(1045^2 - 0^2)/(10820^2)) / (1 + (4 \cdot 0)/(3 \cdot 10820)) - 4 \cdot 2200/10820)$
 $= 60303.9 \text{ Kg-m}$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:
 $= P \cdot R_m / (2t) - M_2 / (\pi \cdot R_m^2 \cdot t)$
 $= 249.574 \cdot 1045.25 / (2 \cdot 147.5) - 60303.9 / (\pi \cdot 1045.2^2 \cdot 147.5)$
 $= 87.27 \text{ N./mm}^2$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:
 $= P \cdot R_m / (2t) + M_2 / (\pi \cdot R_m^2 \cdot t)$
 $= 249.574 \cdot 1045.25 / (2 \cdot 147.5) + 60303.9 / (\pi \cdot 1045.2^2 \cdot 147.5)$
 $= 89.60 \text{ N./mm}^2$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ^*3]:
 $= P \cdot R_m / (2t) - M_1 / (K_1 \cdot \pi \cdot R_m^2 \cdot t)$
 $= 249.574 \cdot 1045.25 / (2 \cdot 147.5) - 43080.2 / (0.1066 \cdot \pi \cdot 1045.2^2 \cdot 147.5)$
 $= 96.26 \text{ N./mm}^2$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ^*4]:

$$= P \cdot R_m / (2t) + M1 / (K1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 249.574 \cdot 1045.25 / (2 \cdot 147.5) + 43080.2 / (0.1923 \cdot \pi \cdot 1045.2^2 \cdot 147.5)$$

$$= 84.10 \text{ N./mm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L-2a) / (L + (4 \cdot h_m / 3))$$

$$= 108558(10820.0 - 2 \cdot 2200.0) / (10820.0 + (4 \cdot 0.0 / 3))$$

$$= 64412.5 \text{ Kgf}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K2 \cdot T / (R_m \cdot t)$$

$$= 1.1707 \cdot 64412.5 / (1045.25 \cdot 147.5)$$

$$= 4.80 \text{ N./mm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(1045.25 \cdot 147.5)}$$

$$= 306.267 \text{ mm.}$$

Circumferential Stress in shell, no rings (4.15.21) [σ_6]:

$$= -K5 \cdot Q \cdot k / (t(b + X1 + X2))$$

$$= -0.7603 \cdot 108558 \cdot 0.1 / (147.5(240.0 + 306.27 + 306.27))$$

$$= -0.64 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8R_m$ (4.15.22) [σ_7]:

$$= -Q / (4 \cdot t(b + X1 + X2)) - 3 \cdot K7 \cdot Q / (2 \cdot t^2)$$

$$= -108558 / (4 \cdot 147.5(240.0 + 306.267 + 306.267)) - 3 \cdot 0.0529 \cdot 108558 / (2 \cdot 147.5^2)$$

$$= -6.00 \text{ N./mm}^2$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(240.0 + 1.56 \cdot \sqrt{(1045.25 \cdot 147.5)}, 2 \cdot 2200.0)$$

$$= 852.53 \text{ mm.}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	2040.0001	mm.
Baseplate Thickness	Bpthk	56.0000	mm.
Baseplate Width	Bpwid	280.0000	mm.
Number of Ribs (inc. outside ribs)	Nribs	4	
Rib Thickness	Ribtk	16.0000	mm.
Web Thickness	Webtk	16.0000	mm.
Web Location	Webloc	Side	
Saddle Yield Stress	Sy	205.0	N./mm ²
Height of Web at Center	Hw,c	306.0	mm.
Friction Coefficient	mu	0.600	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})^3) / 12$$

$$= ((1.56 \cdot \sqrt{(971.5 \cdot 147.5)} + 340.0)^3) / 12$$

$$= 24884.260 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C1 - Y)^2$$

$$= 24884.26 + 1372.532(148.692 - 73.75)^2$$

$$= 101970.734 \text{ cm}^4$$

	B mm.	D mm.	Y mm.	A cm ²	AY mm. ³	I + AD ⁽²⁾ cm ⁴
Shell	930.5	147.5	73.8	1372.5	10122421.0	0.102E+06
Wearplate	340.0	16.0	155.5	54.4	845919.9	36.8
Web	16.0	484.0	405.5	77.4	3140192.2	0.662E+05
BasePlate	280.0	56.0	675.5	156.8	10591841.0	0.436E+06
Totals	...	...	...	1661.2	24700374.0	0.604E+06

Distance to Centroid [C1]:

$$= AY / A$$

$$= 24700374/1661.172$$
$$= 148.692 \text{ mm.}$$

Angle [β]:

$$= 180 - \text{Saddle Angle}/2$$
$$= 180 - 120.0/2$$
$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$
$$= (1 + \cos(120.0) - 0.5 \cdot \sin(120.0)^2) / (\pi - 2.094 + \sin(120.0) \cos(120.0))$$
$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$
$$= 0.204 \cdot 108558.133$$
$$= 22093.9336 \text{ Kgf}$$

$$\text{Tension Stress, } St = (Fh/As) = 7.5066 \text{ N./mm}^2$$
$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 122.9978 \text{ N./mm}^2$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$
$$= 1675.0 - 971.5 \cdot \sin(120.0/2) / 1.0472$$
$$= 871.576 \text{ mm.}$$

$$\text{Bending Moment, } M = Fh \cdot d = 19256.9258 \text{ Kg-m.}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 4.6508 \text{ N./mm}^2$$
$$\text{Allowed Stress, } Sa = 2/3 \cdot \text{Yield Str} = 136.6642 \text{ N./mm}^2$$

Minimum Thickness of Baseplate per Moss:

The maximum of $2/3 \cdot \text{Yield}$ and the Saddle Allowable Stress value was used to determine the baseplate thickness.

$$= \sqrt{3(Q + \text{SaddleWt}) \text{BasePlateWidth} / (2 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$
$$= \sqrt{3(108558 + 680)280.0 / (2 \cdot 2040.0 \cdot 137.997)}$$
$$= 39.978 \text{ mm.}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$
$$= 2 \cdot \cos(90 - 120.0/2) (970.0 + 149.0 + 16.0)$$
$$= 1965.878 \text{ mm.}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nr ribs} - 1)$$
$$= 1965.8777 / (4 - 1)$$
$$= 655.293 \text{ mm.}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$
$$= 655.2925 \cdot 280.0/2$$
$$= 917.410 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$
$$= 109237.977 / (2040.0 \cdot 280.0)$$
$$= 19.124 \text{ Kgf/cm}^2$$

Axial Load [P]:

$$= Ap \cdot Bp$$
$$= 917.4 \cdot 19.12$$
$$= 17544.812 \text{ Kgf}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$
$$= 35.84 + 52.423$$

$$= 88.263 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P/A_r$$

$$= 17544.8/88.2634$$

$$= 19.494 \text{ N./mm}^2$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib	16.0	224.0	128.0	35.8	458752.0	0.332E+04
Web	327.6	16.0	8.0	52.4	41938.7	0.126E+04
Totals	...	...	...	88.3	500690.7	0.458E+04

Rib dimension [D]:

$$= \text{Saddle Width} - \text{Web Thickness}$$

$$= 240.0 - 16.0$$

$$= 224.000 \text{ mm.}$$

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 500691/88.263$$

$$= 56.727 \text{ mm.}$$

Distance to Centroid [C1]:

$$= \max(y_{tot}, \text{Saddle Width} - y_{tot})$$

$$= \max(56.727, 240.0 - 56.727)$$

$$= 183.273 \text{ mm.}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(4575.1/88.263)}$$

$$= 71.996 \text{ mm.}$$

Length of Outer Rib [L]:

$$= \text{Saddle Height} - \cos(\theta/2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 1675.0 - \cos(120.0/2) (970.0 + 149.0 + 16.0) - 56.0$$

$$= 1051.500 \text{ mm.}$$

Intermediate Term [Cc]:

$$= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$$

$$= \sqrt{(2 \cdot \pi^2 \cdot 0.18822\text{E}+09/205.0)}$$

$$= 134.630$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 1051.5/71.996$$

$$= 14.605$$

Bending Moment [Rm]:

$$= FI / (2 \cdot B_{plen}) \cdot e \cdot L / 2$$

$$= 332.8 / (2 \cdot 2040.0) \cdot 655.293 \cdot 1051.5/2$$

$$= 28.103 \text{ Kg-m.}$$

Compressive Allowable, $KL/r < C_c$ ($14.6049 < 134.6301$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (14.6)^2 / (2 \cdot 134.63^2)) 205 /$$

$$(5/3 + 3 \cdot (14.6) / (8 \cdot 134.63) - (14.6^3) / (8 \cdot 134.63^3))$$

$$= 119.4 \text{ N./mm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= Sc/Sca + (R_m \cdot C_1 / I) / S_{ba}$$

$$= 19.49/119.37 + (28.1 \cdot 183.273/45750880) / 136.66$$

$$= 0.171$$

Check of Inside Ribs:



Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	16.0	224.0	128.0	35.8	458752.0	0.436E+04
Web	655.3	16.0	8.0	104.8	83877.4	0.100E+04
Totals	...	...	...	140.7	542629.4	0.537E+04

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 542629/140.687$$

$$= 38.570 \text{ mm.}$$

Distance to Centroid [C1]:

$$= \max(ytot, \text{abs}(\text{Saddle Width} - ytot))$$

$$= \max(38.57, \text{abs}(240.0 - 38.57))$$

$$= 201.430 \text{ mm.}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \sqrt{ (Ro + Wpdthk)^2 - (Pitch/2)^2 } - Bpthk$$

$$= 1675.0 - \sqrt{ (1135.0 + 16.0)^2 - (655.293/2)^2 } - 56.0$$

$$= 532.320 \text{ mm.}$$

Radius of Gyration [r]:

$$= \sqrt{ (\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{ (5367.2/140.687)}$$

$$= 61.765 \text{ mm.}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 532.32/61.765$$

$$= 8.618$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 332.806/(2 \cdot 2040.0)$$

$$= 0.082 \text{ Kg/mm.}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 0.082 \cdot 655.293 \cdot 532.32$$

$$= 28.454 \text{ Kg-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 28.454/266452.938$$

$$= 1.047 \text{ N./mm}^2$$

Compressive Allowable, $KL/r < C_c$ (8.6184 < 134.6301) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (8.62)^2 / (2 \cdot 134.63^2)) 205 / (5/3 + 3 \cdot (8.62) / (8 \cdot 134.63) - (8.62^3) / (8 \cdot 134.63^3))$$

$$= 121 \text{ N./mm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= Sc/Sca + (Mbase,c \cdot C1/I) / Sba$$

$$= 24.06/121.01 + (28.45 \cdot 201.43/5367.156) / 136.66$$

$$= 0.207$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	3
Total Number of Bolts in Tension/Baseplate	Nbt	1
Bolt Material Specification	SA-36	
Bolt Allowable Stress	Stba	114.00 N./mm ²
Bolt Corrosion Allowance	Bca	1.5 mm.
Distance from Bolts to Edge	Edgedis	85.0 mm.
Nominal Bolt Diameter	Bnd	42.0000 mm.
Thread Series	Series	TEMA Metric
BasePlate Allowable Stress	S	108.00 N./mm ²
Area Available in a Single Bolt	BltArea	8.5576 cm ²



Saddle Load Q0 (Weight)	Q0	107282.4 Kgf
Saddle Load QL (Wind/Seismic contribution)	QL	86.8 Kgf
Maximum Transverse Force	Ft	765.1 Kgf
Maximum Longitudinal Force	F1	332.8 Kgf
Saddle Bolted to Steel Foundation		No

Shear Stress in a Single Bolt, Longitudinal Direction [rb,l]:
= $F1 / (\text{Bolt Area} \cdot \text{Number of Bolts})$
= $332.8 / (8.558 \cdot 3)$
= 1.3 N./mm². Must be less than 68.4 N./mm².

Shear Stress in a Single Bolt, Transverse Direction [rb,t]:
= $Ft / (\text{Bolt Area} \cdot \text{Number of Bolts})$
= $765.1 / (8.558 \cdot 3)$
= 2.9 N./mm². Must be less than 68.4 N./mm².

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:
= 0.0 (Q0 > QL --> No Uplift in Longitudinal direction)

Bolt Area due to Shear Load [Bltarears]:
= $F1 / (\text{BoltShearAllowable} \cdot \text{Nbols})$
= $332.81 / (68.4 \cdot 3.0)$
= 0.1591 cm²

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:
= $B \cdot Ft + \text{Sum of X Moments}$
= $1675.0 \cdot 765.07 + 0.0$
= 1281.52 Kg-m.

Eccentricity (e):
= $Rmom / Q0$
= $1281.52 / 107282.38$
= 11.95 mm. < Bplen/6 --> No Uplift in Transverse direction

Bolt Area due to Transverse Load [Bltareart]:
= 0 (No Uplift)

Required Area of a Single Bolt [Bltarear]:
= $\max[\text{Bltarearl}, \text{Bltarears}, \text{Bltareart}]$
= $\max[0.0, 0.1591, 0.0]$
= 0.1591 cm²

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Note:

Wear Pad Width (340.00) is less than $1.56 \cdot \sqrt{rm \cdot t}$
and less than 2a. The wear plate will be ignored.

Minimum Wear Plate Width to be considered in analysis [b1]:
= $\min(b + 1.56 \cdot \sqrt{ Rm \cdot t }, 2a)$
= $\min(240.0 + 1.56 \cdot \sqrt{ 1045.25 \cdot 147.5 }, 2 \cdot 2200.0)$
= 852.5347 mm.

Input and Calculated Values:

Vessel Mean Radius	Rm	1045.25	mm.
Shell Thickness used in this Case	t	149.000	mm.
Stiffened Vessel Length per 4.15.6	L	10820.00	mm.
Distance from Saddle to Vessel tangent	a1 or a	2200.00	mm.
Saddle Width	b1 or b	240.00	mm.
Saddle Bearing Angle	delta or theta	120.00	degrees
Shell Allowable Stress used in Calculation		234.00	N./mm ²
Head Allowable Stress used in Calculation		234.00	N./mm ²

Circumferential Efficiency in Plane of Saddle 1.00
 Circumferential Efficiency at Mid-Span 1.00
 Distance from Saddle Base to Centerline B 1675.00 mm.
 Coefficient of Friction mu 0.10

Results for Horizontal Vessel Analysis:

Saddle Force Q, Test Case, no Ext. Forces 108615.84 Kgf
 Pressure used in Analysis P 249.574 bars

Horizontal Vessel Analysis Results:	Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan	87.27	234.00
Long. Stress at Bottom of Midspan	89.60	234.00
Long. Stress at Top of Saddles	96.27	234.00
Long. Stress at Bottom of Saddles	84.09	234.00
Tangential Shear in Shell	4.80	138.58
Circ. Stress at Horn of Saddle	6.00	350.99
Circ. Compressive Stress in Shell	0.64	234.00

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:
 $= Ftr(Ft/Num\ of\ Saddles + Z\ Force\ Load) \cdot B / E$
 $= 3.0(1530.1/2 + 0) \cdot 1675.0/1965.8777$
 $= 1955.6\ Kgf$

Longitudinal Saddle Reaction Force [Fwl]:
 $= \max(Fl,wind; Sum\ of\ X\ Forces) \cdot B / Ls$
 $= \max(332.81, 0) \cdot 1675.0/6420.001$
 $= 86.8\ Kgf$

Load Combination Results for Q + Wind or Seismic [Q]:
 $= Saddle\ Load + \max(Fwl, Fwt, Fsl, Fst)$
 $= 106660 + \max(87, 1956, 0, 0)$
 $= 108615.8\ Kgf$

Longitudinal Wind Force [Fl]:
 $= WindScalar \cdot WindPress(Platform\ Area + (\pi/4(OD \cdot WindDiaMult)^2))$
 $= 1.1 \cdot 1587.169(0.0 + \pi/4(2.238 \cdot 1.2)^2)$
 $= 9889.890\ N$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight) 109295.68 Kgf
 Transverse Shear Load 765.07 Kgf
 Longitudinal Shear Load 332.81 Kgf

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:
 $= \min(2(ShellID/2 + t + WearPadThickness)\sin(\theta/2), 2 \cdot Rm)$
 $= \min(2(1940.0/2 + 149.0 + 16.0)\sin(120.0/2), 2 \cdot 1045.25)$
 $= 1965.878\ mm.$

The Computed K values from Table 4.15.1:

K1 = 0.1066 K2 = 1.1707 K3 = 0.8799 K4 = 0.4011
 K5 = 0.7603 K6 = 0.0529 K7 = 0.0529 K8 = 0.3405
 K9 = 0.2711 K10 = 0.0581 K1* = 0.1923

Note: Dimension a is greater than or equal to Rm/2.

Moment per Equation 4.15.1 [M1]:
 $= -Q \cdot a [1 - (1 - a/L + (Rm^2 - hm^2) / (2a \cdot L)) / (1 + (4hm) / 3L)]$
 $= -108616 \cdot 2200.0 [1 - (1 - 2200.0/10820.0 + (1045.25^2 - 0.0^2) / (2 \cdot 2200.0 \cdot 10820.0)) / (1 + (4 \cdot 0.0) / (3 \cdot 10820.0))]$
 $= -43103.1\ Kg \cdot m.$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L / 4 \cdot ((1 + 2(Rm^2 - hm^2) / (L^2)) / (1 + (4hm^2) / (3L)) - 4a / L)$$

$$= 108616 \cdot 10820 / 4 \cdot ((1 + 2(1045^2 - 0^2) / (10820^2)) / (1 + (4 \cdot 0) / (3 \cdot 10820)) - 4 \cdot 2200 / 10820)$$

$$= 60335.9 \text{ Kg-m.}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot Rm / (2t) - M2 / (\pi \cdot Rm^2 \cdot t)$$

$$= 249.574 \cdot 1045.25 / (2 \cdot 147.5) - 60335.9 / (\pi \cdot 1045.2^2 \cdot 147.5)$$

$$= 87.27 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot Rm / (2t) + M2 / (\pi \cdot Rm^2 \cdot t)$$

$$= 249.574 \cdot 1045.25 / (2 \cdot 147.5) + 60335.9 / (\pi \cdot 1045.2^2 \cdot 147.5)$$

$$= 89.60 \text{ N./mm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_3]:

$$= P \cdot Rm / (2t) - M1 / (K1 \cdot \pi \cdot Rm^2 \cdot t)$$

$$= 249.574 \cdot 1045.25 / (2 \cdot 147.5) - 43103.1 / (0.1066 \cdot \pi \cdot 1045.2^2 \cdot 147.5)$$

$$= 96.27 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ_4]:

$$= P \cdot Rm / (2t) + M1 / (K1 \cdot \pi \cdot Rm^2 \cdot t)$$

$$= 249.574 \cdot 1045.25 / (2 \cdot 147.5) + 43103.1 / (0.1923 \cdot \pi \cdot 1045.2^2 \cdot 147.5)$$

$$= 84.09 \text{ N./mm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L - 2a) / (L + (4 \cdot hm / 3))$$

$$= 108616(10820.0 - 2 \cdot 2200.0) / (10820.0 + (4 \cdot 0.0 / 3))$$

$$= 64446.7 \text{ Kgf}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K2 \cdot T / (\pi \cdot Rm \cdot t)$$

$$= 1.1707 \cdot 64446.73 / (\pi \cdot 1045.25 \cdot 147.5)$$

$$= 4.80 \text{ N./mm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{Rm \cdot t}$$

$$= 0.78 \cdot \sqrt{1045.25 \cdot 147.5}$$

$$= 306.267 \text{ mm.}$$

Circumferential Stress in shell, no rings (4.15.21) [σ_6]:

$$= -K5 \cdot Q \cdot k / (t(b + X1 + X2))$$

$$= -0.7603 \cdot 108616 \cdot 0.1 / (147.5(240.0 + 306.27 + 306.27))$$

$$= -0.64 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8Rm$ (4.15.22) [σ_7]:

$$= -Q / (4 \cdot t(b + X1 + X2)) - 3 \cdot K7 \cdot Q / (2 \cdot t^2)$$

$$= -108616 / (4 \cdot 147.5(240.0 + 306.267 + 306.267)) - 3 \cdot 0.0529 \cdot 108616 / (2 \cdot 147.5^2)$$

$$= -6.00 \text{ N./mm}^2$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{Rm \cdot t}, 2a)$$

$$= \min(240.0 + 1.56 \cdot \sqrt{1045.25 \cdot 147.5}, 2 \cdot 2200.0)$$

$$= 852.53 \text{ mm.}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	2040.0001	mm.
Baseplate Thickness	Bpthk	56.0000	mm.
Baseplate Width	Bpwid	280.0000	mm.
Number of Ribs (inc. outside ribs)	Nribs	4	
Rib Thickness	Ribtk	16.0000	mm.
Web Thickness	Webtk	16.0000	mm.
Web Location	Webloc	Side	
Saddle Yield Stress	Sy	205.0	N./mm ²
Height of Web at Center	Hw,c	306.0	mm.
Friction Coefficient	mu	0.100	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})^3) / 12$$

$$= ((1.56 \cdot \sqrt{(971.5 \cdot 147.5)} + 340.0)^3) / 12$$

$$= 24884.260 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}} (C1 - Y)^2$$

$$= 24884.26 + 1372.532 (148.692 - 73.75)^2$$

$$= 101970.734 \text{ cm}^4$$

	B mm.	D mm.	Y mm.	A cm ²	AY mm. ³	I + AD ² cm ⁴
Shell	930.5	147.5	73.8	1372.5	10122421.0	0.102E+06
Wearplate	340.0	16.0	155.5	54.4	845919.9	36.8
Web	16.0	484.0	405.5	77.4	3140192.2	0.662E+05
BasePlate	280.0	56.0	675.5	156.8	10591841.0	0.436E+06
Totals	...	...	...	1661.2	24700374.0	0.604E+06

Distance to Centroid [C1]:

$$= AY / A$$

$$= 24700374 / 1661.172$$

$$= 148.692 \text{ mm.}$$

Angle [β]:

$$= 180 - \text{Saddle Angle} / 2$$

$$= 180 - 120.0 / 2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120.0) - 0.5 \cdot \sin(120.0)^2) / (\pi - 2.094 + \sin(120.0) \cos(120.0))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.204 \cdot 108615.844$$

$$= 22105.6797 \text{ Kg}$$

Tension Stress, $St = (Fh / As) = 7.5106 \text{ N./mm}^2$
 Allowed Stress, $Sa = 0.6 \cdot \text{Yield Str} = 122.9978 \text{ N./mm}^2$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$

$$= 1675.0 - 971.5 \cdot \sin(120.0/2) / 1.0472$$

$$= 871.576 \text{ mm.}$$

Bending Moment, $M = Fh \cdot d = 19267.1602 \text{ Kg-m.}$

Bending Stress, $Sb = (M \cdot C1 / I) = 4.6533 \text{ N./mm}^2$
 Allowed Stress, $Sa = 2/3 \cdot \text{Yield Str} = 136.6642 \text{ N./mm}^2$

Minimum Thickness of Baseplate per Moss:

The maximum of $2/3 \cdot \text{Yield}$ and the Saddle Allowable Stress value was used to determine the baseplate thickness.

$$= \sqrt{3(Q + \text{SaddleWt}) \text{BasePlateWidth} / (2 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt{3(108616 + 680)280.0 / (2 \cdot 2040.0 \cdot 137.997)}$$

$$= 39.989 \text{ mm.}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle} / 2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120.0/2) (970.0 + 149.0 + 16.0)$$

$$= 1965.878 \text{ mm.}$$

Distance between Ribs [e]:
 = Web Length / (Nribs - 1)
 = 1965.8777/(4 - 1)
 = 655.293 mm.

Baseplate Pressure Area [Ap]:
 = e • Bpwid / 2
 = 655.2925 • 280.0/2
 = 917.410 cm²

Bearing Pressure [Bp]:
 = (Q + SaddleWt) / (BasePlateLength • BasePlateWidth)
 = 109295.68/(2040.0 • 280.0)
 = 19.134 Kg/cm²

Axial Load [P]:
 = Ap • Bp
 = 917.4 • 19.13
 = 17554.078 Kg

Area of the Rib and Web [Ar]:
 = Rib Area + Web Area
 = 35.84 + 52.423
 = 88.263 cm²

Compressive Stress [Sc]:
 = P/Ar
 = 17554.1/88.2634
 = 19.504 N./mm²

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib	16.0	224.0	128.0	35.8	458752.0	0.332E+04
Web	327.6	16.0	8.0	52.4	41938.7	0.126E+04
Totals	...	...	...	88.3	500690.7	0.458E+04

Rib dimension [D]:
 = Saddle Width - Web Thickness
 = 240.0 - 16.0
 = 224.000 mm.

Distance to Centroid from Datum [ytot]:
 = AY / A
 = 500691/88.263
 = 56.727 mm.

Distance to Centroid [C1]:
 = max(ytot, Saddle Width - ytot)
 = max(56.727, 240.0 - 56.727)
 = 183.273 mm.

Radius of Gyration [r]:
 = $\sqrt{\text{Total Inertia} / \text{Total Area}}$
 = $\sqrt{4575.1/88.263}$
 = 71.996 mm.

Length of Outer Rib [L]:
 = Saddle Height - cos(theta/2)(radius + shlthk + wpdthk) - bpthk
 = 1675.0 - cos(120.0/2)(970.0 + 149.0 + 16.0) - 56.0
 = 1051.500 mm.

Intermediate Term [Cc]:
 = $\sqrt{2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress}}$
 = $\sqrt{2 \cdot \pi^2 \cdot 0.18822\text{E}+09/205.0}$
 = 134.630

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 1051.5/71.996$$

$$= 14.605$$

Bending Moment [Rm]:

$$= FI / (2 \cdot Bplen) \cdot e \cdot L / 2$$

$$= 332.8 / (2 \cdot 2040.0) \cdot 655.293 \cdot 1051.5 / 2$$

$$= 28.103 \text{ Kg-m.}$$

Compressive Allowable, $KL/r < Cc$ (14.6049 < 134.6301) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$$

$$= (1 - (14.6)^2 / (2 \cdot 134.63^2)) 205 / (5/3 + 3 \cdot (14.6) / (8 \cdot 134.63) - (14.6^3) / (8 \cdot 134.63^3))$$

$$= 119.4 \text{ N./mm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= Sc/Sca + (Rm \cdot C1 / I) / Sba$$

$$= 19.5/119.37 + (28.1 \cdot 183.273/45750880) / 136.66$$

$$= 0.171$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	16.0	224.0	128.0	35.8	458752.0	0.436E+04
Web	655.3	16.0	8.0	104.8	83877.4	0.100E+04
Totals	...	...	...	140.7	542629.4	0.537E+04

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 542629/140.687$$

$$= 38.570 \text{ mm.}$$

Distance to Centroid [C1]:

$$= \max(ytot, \text{abs}(\text{Saddle Width} - ytot))$$

$$= \max(38.57, \text{abs}(240.0 - 38.57))$$

$$= 201.430 \text{ mm.}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \sqrt{ (Ro + Wpdthk)^2 - (Pitch/2)^2 } - Bpthk$$

$$= 1675.0 - \sqrt{ (1135.0 + 16.0)^2 - (655.293/2)^2 } - 56.0$$

$$= 532.320 \text{ mm.}$$

Radius of Gyration [r]:

$$= \sqrt{ (\text{Total Inertia} / \text{Total Area}) }$$

$$= \sqrt{ (5367.2/140.687) }$$

$$= 61.765 \text{ mm.}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 532.32/61.765$$

$$= 8.618$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 332.806 / (2 \cdot 2040.0)$$

$$= 0.082 \text{ Kg/mm.}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 0.082 \cdot 655.293 \cdot 532.32$$

$$= 28.454 \text{ Kg-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 28.454/266452.938$$

$$= 1.047 \text{ N./mm}^2$$

Compressive Allowable, $KL/r < C_c$ (8.6184 < 134.6301) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (8.62)^2 / (2 \cdot 134.63^2)) 205 / (5/3 + 3 \cdot (8.62) / (8 \cdot 134.63) - (8.62^3) / (8 \cdot 134.63^3))$$

$$= 121 \text{ N./mm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (M_{base,c} \cdot C_1 / I) / S_{ba}$$

$$= 24.07 / 121.01 + (28.45 \cdot 201.43 / 5367.156) / 136.66$$

$$= 0.207$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	3
Total Number of Bolts in Tension/Baseplate	Nbt	1
Bolt Material Specification		SA-36
Bolt Allowable Stress	Stba	114.00 N./mm ²
Bolt Corrosion Allowance	Bca	1.5 mm.
Distance from Bolts to Edge	Edgedis	85.0 mm.
Nominal Bolt Diameter	Bnd	42.0000 mm.
Thread Series	Series	TEMA Metric
BasePlate Allowable Stress	S	108.00 N./mm ²
Area Available in a Single Bolt	BltArea	8.5576 cm ²
Saddle Load QO (Weight)	QO	107340.1 Kg
Saddle Load QL (Wind/Seismic contribution)	QL	86.8 Kg
Maximum Transverse Force	Ft	765.1 Kg
Maximum Longitudinal Force	Fl	332.8 Kg
Saddle Bolted to Steel Foundation		No

Shear Stress in a Single Bolt, Longitudinal Direction [tb,l]:

$$= F_l / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 332.8 / (8.558 \cdot 3)$$

$$= 1.3 \text{ N./mm}^2. \text{ Must be less than } 68.4 \text{ N./mm}^2.$$

Shear Stress in a Single Bolt, Transverse Direction [tb,t]:

$$= F_t / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 765.1 / (8.558 \cdot 3)$$

$$= 2.9 \text{ N./mm}^2. \text{ Must be less than } 68.4 \text{ N./mm}^2.$$

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:

$$= 0.0 \text{ (} QO > QL \text{ --> No Uplift in Longitudinal direction)}$$

Bolt Area due to Shear Load [Bltarears]:

$$= F_l / (\text{BoltShearAllowable} \cdot \text{Nbolts})$$

$$= 332.81 / (68.4 \cdot 3.0)$$

$$= 0.1591 \text{ cm}^2$$

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:

$$= B \cdot F_t + \text{Sum of X Moments}$$

$$= 1675.0 \cdot 765.07 + 0.0$$

$$= 1281.52 \text{ Kg-m.}$$

Eccentricity (e):

$$= R_{mom} / QO$$

$$= 1281.52 / 107340.08$$

$$= 11.94 \text{ mm.} < B_{plen} / 6 \text{ --> No Uplift in Transverse direction}$$

Bolt Area due to Transverse Load [Bltareart]:

$$= 0 \text{ (No Uplift)}$$

Required Area of a Single Bolt [Bltarear]:

$$= \max[Bltarearl, Bltarears, Bltareart]$$

$$= \max[0.0, 0.1591, 0.0]$$

$$= 0.1591 \text{ cm}^2$$



Review notes about nozzle loadings and supports in the Warnings report.

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Input, Nozzle Desc: TS OUT NOZ. T2

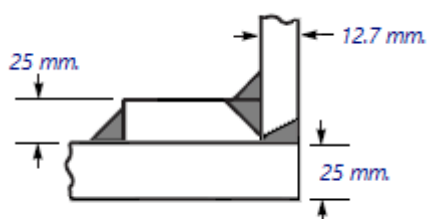
From: 30

Pressure for Reinforcement Calculations	P	10.000	bars
Temperature for Internal Pressure	Temp	85	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	1940.00	mm.
Shell Finished (Minimum) Thickness	t	25.0000	mm.
Shell Internal Corrosion Allowance	c	3.0000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		1060.00	mm.
User Entered Minimum Design Metal Temperature		0.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	N./mm ²
Allowable Stress At Ambient	Sna	118.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		90.00	deg
Diameter		30.0000	in.
Size and Thickness Basis		Nominal	
Nominal Thickness		20	
Flange Material		SA-105	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	3.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	350.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	16.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	0.0000	mm.
Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	138.00	N./mm ²
Pad Allowable Stress At Ambient	Spa	138.00	N./mm ²
Diameter of Pad along vessel surface	Dp	1001.9999	mm.
Thickness of Pad	te	25.0000	mm.
Weld leg size between Pad and Shell	Wp	20.0000	mm.
Groove weld depth between Pad and Nozzle	Wgpn	25.0000	mm.
Reinforcing Pad Width		120.0000	mm.
Flange Class		150	
Flange Grade		GR 1.1	
Flange Series		B	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: TS OUT NOZ. T2

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 29.000 in.
 Actual Thickness Used in Calculation 0.500 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 10.0 \cdot 973.0 / (138 \cdot 1.0 - 0.6 \cdot 10.0)$
 $= 7.0821$ mm.

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 10.0 \cdot 371.3 / (118 \cdot 1.0 - 0.6 \cdot 10.0)$
 $= 3.1629$ mm.

UG-40 Modified by Appendix 1-7, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit) D1 1485.2001 mm.
 Parallel to Vessel Wall, opening length d 742.6000 mm.
 Normal to Vessel Wall (Thickness Limit), pad side Tlwp 49.2500 mm.

Weld Strength Reduction Factor [fr1]:
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 118.0 / 138.0)$
 $= 0.855$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, S_p / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(0.855, 1.0)$
 $= 0.855$

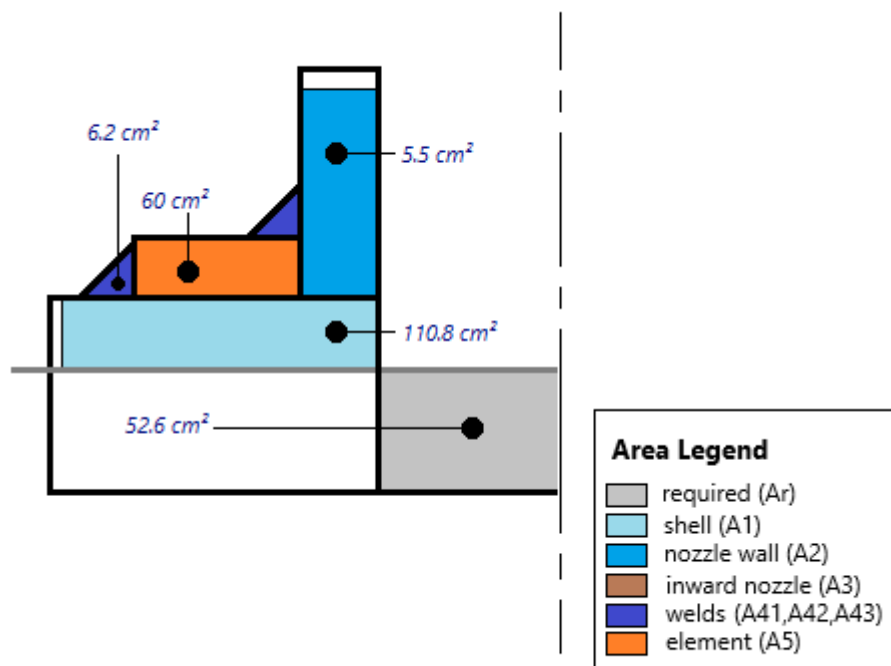


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5	Design	External	Mapnc
Area Required Ar	52.591	NA	NA
Area in Shell A1	110.781	NA	NA
Area in Nozzle Wall A2	5.506	NA	NA
Area in Inward Nozzle A3	0.000	NA	NA
Area in Welds A41+A42+A43	6.189	NA	NA
Area in Element A5	60.000	NA	NA
TOTAL AREA AVAILABLE Atot	182.475	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.
 The area available with the given pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1))$ UG-37(c)
 $= (742.6 \cdot 7.0821 \cdot 1.0 + 2 \cdot 9.7 \cdot 7.0821 \cdot 1.0 \cdot (1 - 1.0))$
 $= 52.591 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 742.6(1.0 \cdot 22.0 - 1.0 \cdot 7.082) - 2 \cdot 9.7$
 $(1.0 \cdot 22.0 - 1.0 \cdot 7.082) \cdot (1 - 1.0)$
 $= 110.781 \text{ cm}^2$

Area Available in Nozzle Wall Projecting Outward [A2]:
 $= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$
 $= (2 \cdot 49.25) \cdot (9.7 - 3.16) \cdot 0.8551$
 $= 5.506 \text{ cm}^2$

Area Available in Welds [A41 + A42 + A43]:
 $= Wo^2 \cdot fr3 + (Wi\text{-}can/0.707)^2 \cdot fr2 + Wp^2 \cdot fr4$
 $= 16.0^2 \cdot 0.86 + (0.0)^2 \cdot 0.86 + 20.0^2 \cdot 1.0$

= 6.189 cm²

Area Available in Element [A5]:

= (min(Dp,DL)-(Nozzle OD))*(min(tp,Tlwp,te)) • fr4
= (1002.0 - 762.0) • 25.0 • 1.0
= 60.000 cm²

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures ta = 6.1629 mm.
Wall Thickness per UG16(b), tr16b = 5.5000 mm.
Wall Thickness, shell/head, internal pressure trb1 = 10.0821 mm.
Wall Thickness tb1 = max(trb1, tr16b) = 10.0821 mm.
Wall Thickness tb2 = max(trb2, tr16b) = 5.5000 mm.
Wall Thickness per table UG-45 tb3 = 11.3312 mm.

Determine Nozzle Thickness candidate [tb]:

= min[tb3, max(tb1,tb2)]
= min[11.331, max(10.0821, 5.5)]
= 10.0821 mm.

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

= max(ta, tb)
= max(6.1629, 10.0821)
= 10.0821 mm.

Available Nozzle Neck Thickness = 0.875 * 12.700 = 11.113 mm. -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	40.6,	Allowable	:	118.0 N./mm ²	Passed
Expansion	:	0.0,	Allowable	:	254.4 N./mm ²	Passed
Occasional	:	18.9,	Allowable	:	156.9 N./mm ²	Passed
Shear	:	13.6,	Allowable	:	82.6 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Pad Weld for the Nozzle, Curve: B

Govrn. thk, tg = 11.113, tr = 3.163, c = 3.0 mm., E* = 1.0
Thickness Ratio = tr * E*/(tg - c) = 0.39, Temp. Reduction = 53 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-26 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-48 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

Govrn. thk, tg = 11.113, tr = 3.163, c = 3.0 mm., E* = 1.0
Thickness Ratio = tr * E*/(tg - c) = 0.39, Temp. Reduction = 53 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-26 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-48 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Shell to Pad Weld Junction at Pad OD, Curve: B

Govrn. thk, tg = 25.0, tr = 7.082, c = 3.0 mm., E* = 1.0
Thickness Ratio = tr * E*/(tg - c) = 0.322, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-1 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: B

Govrn. thk, tg = 11.113, tr = 3.163, c = 3.0 mm., E* = 1.0

Thickness Ratio = $tr * E / (tg - c) = 0.39$, Temp. Reduction = 53 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -26 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -48 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

Gov. MDMT of the Nozzle : -48 °C
 Gov. MDMT of the Reinforcement Pad : -48 °C
 Gov. MDMT of the nozzle to shell joint welded assembly : -48 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -48 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $10.00/19.60 = 0.510$

UG-40 Modified by Appendix 1-7, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit) D1 1113.9000 mm.
 Parallel to Vessel Wall, opening length d 556.9500 mm.
 Normal to Vessel Wall (Thickness Limit), pad side Tlwp 49.2500 mm.

Reinforcement Areas for Large Nozzle (Per App 1-7(a)): (cm²)

AREA AVAILABLE, A1 to A6		No Pad	With Pad
Area Required [Internal Pressure]	AR	35.063	35.063
Area Available in Shell	A1	55.390	55.390
Area Available in Nozzle Wall	A2	2.711	5.506
Area Available in Inward Nozzle	A3	0.000	0.000
Area Available in Welds	A4	2.189	6.189
Area Available in Pad	A5	0.000	60.000
Area Available in Hub	A6	0.000	0.000
TOTAL AREA AVAILABLE	Atot	60.290	127.085

The area available without a pad is Sufficient.
 The area available with the given pad is Sufficient.

The Nozzle does not fall within the scope of the diameter requirements of App. 1-7(b). Therefore, the Membrane and Bending stress checks are not required.

Weld Size Calculations, Description: TS OUT NOZ. T2

Intermediate Calc. for nozzle/shell Welds Tmin 9.7000 mm.
 Intermediate Calc. for pad/shell Welds TminPad 19.0000 mm.

Results Per UW-16.1:

Required Thickness Actual Thickness
 Nozzle Weld $6.0000 = \text{Min per Code}$ $11.3120 = 0.7 * Wo$ mm.
 Pad Weld $9.5000 = 0.5 * TminPad$ $14.1400 = 0.7 * Wp$ mm.

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= (Ar - A1) * Sv$
 $= (52.5913 - 110.7807) * 138$
 $= 0.00 \text{ Kgf}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2 + A5 + A4 - (Wi - Can / .707)^2 * fr2) * Sv$
 $= (5.5058 + 60.0 + 6.189 - 0.0 * 0.86) * 138$
 $= 100885.84 \text{ Kgf}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw$
 $= (3.1416/2.0) \cdot 762.0 \cdot 16.0 \cdot 0.49 \cdot 118$
 $= 112911. \text{ Kg}$

Shear, Pad Element Weld [Spew]:
 $= (\pi/2) \cdot DP \cdot WP \cdot 0.49 \cdot SEW$
 $= (3.1416/2.0) \cdot 1002.0 \cdot 20.0 \cdot 0.49 \cdot 138$
 $= 217048. \text{ Kg}$

Tension, Pad Groove Weld [Tpgw]:
 $= (\pi/2) \cdot Dlo \cdot Wgpn \cdot 0.74 \cdot Seg$
 $= (3.1416/2) \cdot 762.0 \cdot 25.0 \cdot 0.74 \cdot 138$
 $= 311600. \text{ Kg}$

Strength of Failure Paths:

Failure Path [Path 1-1]:
 $= (\text{SPEW} + \text{SNGW}) = (217048 + 0) = 217048 \text{ Kg}$

Summary of Failure Path Calculations:

Path 1-1 = 217048 Kg, must exceed W = 0 Kg or W1 = 100885 Kg

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 11 bars

The Drop for this Nozzle is : 77.9579 mm.
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 452.9579 mm.

Warning:
 Review Appendix 1-7 (b)(3) for notes regarding externally applied loads to large nozzles and U-2(g).

Input Echo, WRC297 Item 1, Description: TS OUT NOZ. T2:

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.0000	mm.
Shell External Corrosion Allowance	0.0000	mm.
Shell Diameter	1940.000	mm.
Shell Thickness	25.0000	mm.
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc 138.00	N./mm ²
Vessel Hot S.I. Allowable	Smh 138.00	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	3.0000	mm.
Nozzle Diameter	736.600	mm.
Nozzle Thickness	12.7000	mm.
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-106 B	
Nozzle Cold S.I. Allowable	SNmc 118.00	N./mm ²
Nozzle Hot S.I. Allowable	SNmh 118.00	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter of Reinforcing Pad	1002.000	mm.
Thickness of Reinforcing Pad	25.0000	mm.

Note:
 External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUSained loads.

Design Internal Pressure	Dp 10.00	bars
Radial Load	P -3671.23	Kgf



Circumferential Shear	Vc	2753.42	Kgf
Longitudinal Shear	Vl	3671.23	Kgf
Circumferential Moment	Mc	7156.13	Kg-m.
Longitudinal Moment	Ml	7156.13	Kg-m.
Torsional Moment	Mt	8257.07	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$
 $= 762.0 + 2 \cdot 1.65 \cdot \sqrt{(996.5(25.0 - 3.0))}$
 $= 1250.612 \text{ mm.}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	1993.000 mm.
Nozzle Outside Diameter	(d) =	762.000 mm.
Vessel Thickness used	(T) =	47.000 mm.
Nozzle Thickness used	(t) =	9.700 mm.
T / t	=	4.845
d / t	=	78.557

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 2.490$$

Nr/P	=	0.023
Mr/P	=	0.025
M0/P	=	0.050
N0/P	=	0.236
MrD/Mc	=	0.070
NrDL/Mc	=	0.057
M0d/Mc	=	0.177
N0DL/Mc	=	0.475
MrD/Ml	=	0.040
NrDL/Ml	=	0.034
M0D/Ml	=	0.075
N0DL/Ml	=	0.549

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	3	3	3	3
Outplane Bending (P)	4	-4	4	-4
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-22	-22	22	22
Outplane Bending (ML)	-18	18	18	-18
Normal Pressure Stress	20	21	20	21
Outplane Stress Summary	-12	16	70	24

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	0	0	0	0
Inplane Bending (P)	2	-2	2	-2
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-1	-1	1	1
Inplane Bending (ML)	-9	9	9	-9
Inplane Pressure Stress	10	10	10	10



Inplane Stress Summary 1 16 24 0

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE		Au	Al	Bu	B1
(Shear stress normal to longitudinal plane)		Top	Top	Bottom	Bottom
		Outside	Inside	Outside	Inside
Outplane Shear (Vc)		0	0	0	0
Outplane Shear (Vl)		0	0	0	0
Outplane Shear (Mt)		1	1	1	1

Shear Stress Summary 2 2 1 1

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE		Au	Al	Bu	B1
(Stress Intensities)		Top	Top	Bottom	Bottom
		Outside	Inside	Outside	Inside
Two * Max Shear Stress		15	18	70	24

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE		Cu	Cl	Du	D1
(Stresses Normal to circumferential plane)		Left	Left	Right	Right
		Outside	Inside	Outside	Inside
Outplane Membrane (P)		3	3	3	3
Outplane Bending (P)		4	-4	4	-4
Outplane Membrane (Mc)		-19	-19	19	19
Outplane Bending (Mc)		-44	44	44	-44
Outplane Membrane (ML)		0	0	0	0
Outplane Bending (ML)		0	0	0	0
Normal Pressure Stress		10	10	10	10

Outplane Stress Summary -45 33 82 -15

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE		Cu	Cl	Du	D1
(Stresses parallel to circumferential plane)		Left	Left	Right	Right
		Outside	Inside	Outside	Inside
Inplane Membrane (P)		0	0	0	0
Inplane Bending (P)		2	-2	2	-2
Inplane Membrane (Mc)		-2	-2	2	2
Inplane Bending (Mc)		-17	17	17	-17
Inplane Membrane (ML)		0	0	0	0
Inplane Bending (ML)		0	0	0	0
Inplane Pressure Stress		20	21	20	21

Inplane Stress Summary 3 34 42 4

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE		Cu	Cl	Du	D1
(Shear stress normal to circumferential plane)		Left	Left	Right	Right
		Outside	Inside	Outside	Inside
Outplane Shear (Vc)		0	0	0	0
Outplane Shear (Vl)		0	0	0	0
Torsional Shear (Mt)		1	1	1	1

Shear Stress Summary 1 1 2 2

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE		Cu	Cl	Du	D1
(Stress Intensities)		Left	Left	Right	Right
		Outside	Inside	Outside	Inside
Two * Max Shear Stress		48	35	83	20

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE		Au	Al	Bu	B1
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(Stresses in the hoop direction)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Hoop Membrane (P)	3	3	3	3
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-22	-22	22	22
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	45	46	45	46
Hoop Stress Summary	26	27	72	73

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Axial Membrane (P)	1	1	1	1
Axial Bending (P)	30	-30	30	-30
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-16	-16	16	16
Axial Bending (ML)	-132	132	132	-132
Axial Pressure Stress	22	22	22	22
Axial Stress Summary	-94	110	204	-122

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	2	2	-2	-2
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	9	9	9	9
Shear Stress Summary	11	11	6	6

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	123	111	204	196

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	3	3	3	3
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-19	-19	19	19
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	45	46	45	46
Hoop Stress Summary	29	30	68	70

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Axial Membrane (P)	1	1	1	1
Axial Bending (P)	30	-30	30	-30
Axial Membrane (Mc)	-16	-16	16	16
Axial Bending (Mc)	-244	244	244	-244
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0



Axial Pressure Stress	22	22	22	22
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Axial Stress Summary	-206	221	315	-234
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu	Cl	Du	Dl
	Outside	Inside	Outside	Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-3	-3	3	3
Shear due to Torsion	9	9	9	9
Shear Stress Summary	6	6	12	12

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu	Cl	Du	Dl
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	235	221	316	305

Stress Computations at the Edge of the Reinforcing Pad:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - ca)}$
 $= 762.0 + 2 \cdot 1.65 \cdot \sqrt{984.0(25.0 - 3.0)}$
 $= 1247.537 \text{ mm.}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter (D) = 1968.000 mm.
 Nozzle Outside Diameter (d) = 1002.000 mm.
 Vessel Thickness used (T) = 22.000 mm.
 Nozzle Thickness used (t) = 129.700 mm.
 $T / t = 0.170$
 $d / t = 10.000$

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 4.816$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

Nr/P = 0.043
 Mr/P = 0.069
 M0/P = 0.022
 N0/P = 0.042
 MrD/Mc = 0.246
 NrDL/Mc = 0.089
 M0d/Mc = 0.074
 N0DL/Mc = 0.074
 MrD/Ml = 0.057
 NrDL/Ml = 0.042
 M0D/Ml = 0.018
 N0DL/Ml = 0.039

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au	Al	Bu	Bl
	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Outplane Membrane (P)	3	3	3	3
Outplane Bending (P)	9	-9	9	-9
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-5	-5	5	5
Outplane Bending (ML)	-15	15	15	-15
Normal Pressure Stress	43	44	43	44
Outplane Stress Summary	35	47	77	28

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	3	3	3	3
Inplane Bending (P)	30	-30	30	-30
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-6	-6	6	6
Inplane Bending (ML)	-49	49	49	-49
Inplane Pressure Stress	21	21	21	21
Inplane Stress Summary	0	38	111	-49

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	2	2	2	2
Shear Stress Summary	3	3	1	1

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	36	48	111	77

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	3	3	3	3
Outplane Bending (P)	9	-9	9	-9
Outplane Membrane (Mc)	-10	-10	10	10
Outplane Bending (Mc)	-63	63	63	-63
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	21	21	21	21
Outplane Stress Summary	-39	68	109	-37

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Inplane Membrane (P)	3	3	3	3
Inplane Bending (P)	30	-30	30	-30
Inplane Membrane (Mc)	-12	-12	12	12
Inplane Bending (Mc)	-213	213	213	-213
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	43	44	43	44
Inplane Stress Summary	-148	217	304	-183

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-1	-1	1	1
Torsional Shear (Mt)	2	2	2	2

Shear Stress Summary 1 1 3 3

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress	148	217	304	183

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	20	21	20	21	20	21	20	21
Circ. Pl (SUS)	-19	-19	25	25	-2	-2	2	2
Circ. Q (SUS)	-14	14	22	-22	-15	15	19	-19
Long. Pm (SUS)	10	10	10	10	10	10	10	10
Long. Pl (SUS)	-1	-1	1	1	-16	-16	22	22
Long. Q (SUS)	-7	7	11	-11	-40	40	48	-48
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	1	1	1	1	1	1	1	1
Pm (SUS)	20.0	21.0	20.0	21.0	20.0	21.0	20.0	21.0
Pm+Pl (SUS)	9.0	9.0	45.0	46.0	24.0	25.0	32.0	32.0
Pm+Pl+Q (Total)	15.1	17.0	67.0	24.1	49.0	35.0	80.0	20.1

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	21.00	138.00	Passed
Pm+Pl (SUS)	46.00	207.00	Passed
Pm+Pl+Q (TOTAL)	80.03	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Reinforcing Pad Edge (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	43	44	43	44	43	44	43	44
Circ. Pl (SUS)	-2	-2	8	8	-9	-9	15	15
Circ. Q (SUS)	-6	6	24	-24	-183	183	243	-243
Long. Pm (SUS)	21	21	21	21	21	21	21	21
Long. Pl (SUS)	-3	-3	9	9	-7	-7	13	13
Long. Q (SUS)	-19	19	79	-79	-54	54	72	-72
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	-1	-1	1	1
Shear Q (SUS)	2	2	2	2	2	2	2	2
Pm (SUS)	43.0	44.0	43.0	44.0	43.0	44.0	43.0	44.0
Pm+Pl (SUS)	41.0	42.0	51.0	52.0	34.0	35.0	58.0	59.0



Pm+Pl+Q (Total)	36.2	48.4	109.1	77.1	149.0	218.0	301.0	184.1
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Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	44.00	138.00	Passed
Pm+Pl (SUS)	59.04	207.00	Passed
Pm+Pl+Q (TOTAL)	301.05	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	45	46	45	46	45	46	45	46
Circ. Pl (SUS)	-19	-19	25	25	-16	-16	22	22
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	22	22	22	22	22	22	22	22
Long. Pl (SUS)	-15	-15	17	17	-15	-15	17	17
Long. Q (SUS)	-102	102	162	-162	-214	214	274	-274
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	2	2	-2	-2	-3	-3	3	3
Shear Q (SUS)	9	9	9	9	9	9	9	9
Pm (SUS)	45.0	46.0	45.0	46.0	45.0	46.0	45.0	46.0
Pm+Pl (SUS)	26.2	27.2	70.1	71.1	29.4	30.4	67.3	68.3
Pm+Pl+Q (Total)	123.0	110.4	201.4	194.5	236.3	221.2	313.6	303.9

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	46.00	118.00	Passed
Pm+Pl (SUS)	71.12	177.00	Passed
Pm+Pl+Q (TOTAL)	313.58	353.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.



Input, Nozzle Desc: DRAIN NOZ. D1

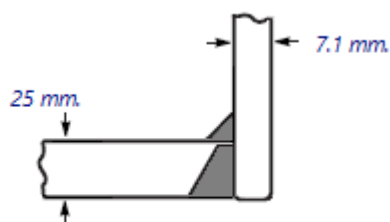
From: 30

Pressure for Reinforcement Calculations	P	10.093	bars
Temperature for Internal Pressure	Temp	85	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	1940.00	mm.
Shell Finished (Minimum) Thickness	t	25.0000	mm.
Shell Internal Corrosion Allowance	c	3.0000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		1060.00	mm.
User Entered Minimum Design Metal Temperature		0.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	N./mm ²
Allowable Stress At Ambient	Sna	118.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		-90.00	deg
Diameter		1.5000	in.
Size and Thickness Basis		Nominal	
Nominal Thickness		160	
Flange Material		SA-105	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	3.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	200.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	25.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		150	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: DRAIN NOZ. D1

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	1.338	in.
Actual Thickness Used in Calculation	0.281	in.



Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $10.09 \cdot 973.0 / (138 \cdot 1.0 - 0.6 \cdot 10.09)$
= 7.1483 mm.

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $10.09 \cdot 19.99 / (118 \cdot 1.0 - 0.6 \cdot 10.09)$
= 0.1719 mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	92.2600	mm.
Parallel to Vessel Wall	Rn+tn+tt	46.1300	mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	10.3435	mm.

*Taking a UG-36(c)(3)(a) exemption for nozzle: DRAIN NOZ. D1.
This calculation is valid for nozzles that meet all the requirements of
paragraph UG-36. Please check the Code carefully, especially for nozzles
that are not isolated or do not meet Code spacing requirements. To force
the computation of areas for small nozzles go to Tools->Configuration
and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 3.1719	mm.
Wall Thickness per UG16(b),	tr16b	= 5.5000	mm.
Wall Thickness, shell/head, internal pressure	trb1	= 10.1483	mm.
Wall Thickness	tb1 = max(trb1, tr16b)	= 10.1483	mm.
Wall Thickness	tb2 = max(trb2, tr16b)	= 5.5000	mm.
Wall Thickness per table UG-45	tb3	= 6.2200	mm.

Determine Nozzle Thickness candidate [tb]:
= min[tb3, max(tb1, tb2)]
= min[6.22, max(10.1483, 5.5)]
= 6.2200 mm.

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
= max(ta, tb)
= max(3.1719, 6.22)
= 6.2200 mm.

Available Nozzle Neck Thickness = $0.875 \cdot 7.137 = 6.245$ mm. -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), Curve: B

Govrn. thk, tg = 6.245, tr = 0.172, c = 3.0 mm., $E^* = 1.0$
Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.053$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-48 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = $10.09 / 19.60 = 0.515$

Weld Size Calculations, Description: DRAIN NOZ. D1

Intermediate Calc. for nozzle/shell Welds	Tmin	4.1374	mm.
---	------	--------	-----

Results Per UW-16.1:



	Required Thickness	Actual Thickness
Nozzle Weld	$2.8962 = 0.7 * t_{min.}$	$7.0700 = 0.7 * W_o \text{ mm.}$

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.3002 mm.
The Cut Length for this Nozzle is, Drop + Ho + H + T : 225.3002 mm.

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Input, Nozzle Desc: SH. IN A2

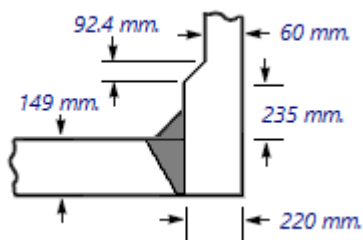
From: 40

Pressure for Reinforcement Calculations	P	191.000	bars
Temperature for Internal Pressure	Temp	260	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	1940.00	mm.
Shell Finished (Minimum) Thickness	t	149.0000	mm.
Shell Internal Corrosion Allowance	c	1.5000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		3479.00	mm.
User Entered Minimum Design Metal Temperature		0.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-266 2	
Material UNS Number		K03506	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	134.60	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		90.00	deg
Diameter		24.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	60.0000	mm.
Hub Height of Integral Nozzle	h	235.0000	mm.
Height of Beveled Transition	L'	92.3760	mm.
Hub Thickness of Integral Nozzle (tn or x+tp)		220.0000	mm.
Corrosion Allowance	can	1.5000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	455.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	80.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: SH. IN A2

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	24.000	in.
Actual Thickness Used in Calculation	2.362	in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $191.0 \cdot 971.5 / (138 \cdot 1.0 - 0.6 \cdot 191.0)$
= 146.6508 mm.

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $191.0 \cdot 306.3 / (135 \cdot 1.0 - 0.6 \cdot 191.0)$
= 47.5123 mm.

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
= (Hub Thickness - Neck Thickness) / $\tan(30)$
= $160.0 / 0.5773$
= 277.1282 mm.

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

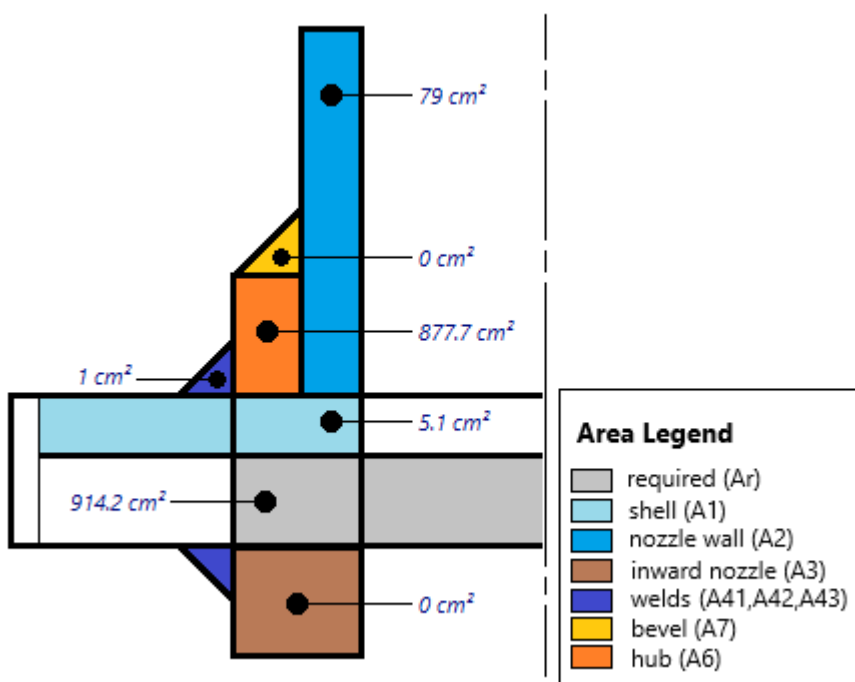
UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	1225.2000	mm.
Parallel to Vessel Wall, opening length	d	612.6000	mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	368.7500	mm.

Weld Strength Reduction Factor [fr1]:
= $\min(1, S_n / S_v)$
= $\min(1, 134.6 / 138.0)$
= 0.975

Weld Strength Reduction Factor [fr2]:
= $\min(1, S_n / S_v)$
= $\min(1, 134.6 / 138.0)$
= 0.975

Weld Strength Reduction Factor [fr3]:
= $\min(fr2, fr4)$
= $\min(0.975, 1.0)$
= 0.975





Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	914.158	NA	NA
Area in Shell	A1	5.111	NA	NA
Area in Nozzle Wall	A2	79.039	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	0.975	NA	NA
Area in Element	A5	0.000	NA	NA
Area in Hub	A6	877.653	NA	NA
TOTAL AREA AVAILABLE	Atot	962.779	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (612.6 \cdot 146.6508 \cdot 1.0 + 2 \cdot 218.5 \cdot 146.6508 \cdot 1.0 \cdot (1 - 0.98))$$

$$= 914.158 \text{ cm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 612.6(1.0 \cdot 147.5 - 1.0 \cdot 146.651) - 2 \cdot 218.5$$

$$(1.0 \cdot 147.5 - 1.0 \cdot 146.6508) \cdot (1 - 0.975)$$

$$= 5.111 \text{ cm}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot tnp)(tn - trn)fr2$$

$$= (2 \cdot 368.75)(58.5 - 47.51) \cdot 0.9754$$

$$= 79.039 \text{ cm}^2$$

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$$

$$= 10.0^2 \cdot 0.9754 + (0.0)^2 \cdot 0.9754$$

$$= 0.975 \text{ cm}^2$$

Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(Tnp, ho, Hubht)) \cdot (Hubtk - tn) \cdot fr2$$

$$= (2 \cdot \min(368.8, 455.0, 235.0)) \cdot (220.0 - 60.0) \cdot 0.975$$

$$= 877.653 \text{ cm}^2 \text{ Includes Bevel Area of } 144.163 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures $ta = 49.0123 \text{ mm.}$
 Wall Thickness per UG16(b), $tr16b = 3.0000 \text{ mm.}$
 Wall Thickness, shell/head, internal pressure $trb1 = 148.1508 \text{ mm.}$
 Wall Thickness $tb1 = \max(trb1, tr16b) = 148.1508 \text{ mm.}$
 Wall Thickness $tb2 = \max(trb2, tr16b) = 3.0000 \text{ mm.}$
 Wall Thickness per table UG-45 $tb3 = 9.8312 \text{ mm.}$

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$

$$= \min[9.831, \max(148.1508, 3.0)]$$

$$= 9.8312 \text{ mm.}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(ta, tb)$$

$$= \max(49.0123, 9.8312)$$

$$= 49.0123 \text{ mm.}$$

Available Nozzle Neck Thickness = 60.0000 mm. -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained :	61.0,	Allowable :	134.6 N./mm ²	Passed
Expansion :	0.0,	Allowable :	279.7 N./mm ²	Passed
Occasional :	45.6,	Allowable :	179.0 N./mm ²	Passed
Shear :	10.7,	Allowable :	94.2 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Warning:

*The minimum metal temperature could not be computed because the governing thickness for this component was greater than 100.000 mm..
This component must be impact tested.*

Weld Size Calculations, Description: SH. IN A2

Intermediate Calc. for nozzle/shell Welds T_{min} 19.0000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * W _o mm.

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$\begin{aligned} &= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - t_r)) \cdot S_v) \\ &= \max(0, (914.1577 - 5.1109 + 2 \cdot 218.5 \cdot 0.9754 \cdot (1.0 \cdot 147.5 - 146.6508)) \cdot 138) \\ &= 1284265.00 \text{ Kgf} \end{aligned}$$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$\begin{aligned} &= (A2+A4-(W_i-\text{Can}/.707)^2 \cdot f_{r2}) \cdot S_v \\ &= (956.6923 + 0.9754 - 0.0 \cdot 0.98) \cdot 138 \\ &= 1347588.75 \text{ Kgf} \end{aligned}$$

Weld Load [W2]:

$$\begin{aligned} &= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v \\ &= (956.6923 + 0.0 + 0.9754 + (628.7087)) \cdot 138 \\ &= 2232280.25 \text{ Kgf} \end{aligned}$$

Weld Load [W3]:

$$\begin{aligned} &= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S \\ &= (956.6923 + 0.0 + 0.9754 + 0.0 + (628.7087)) \cdot 138 \\ &= 2232280.25 \text{ Kgf} \end{aligned}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$\begin{aligned} &= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw} \\ &= (3.1416/2.0) \cdot 1049.6 \cdot 10.0 \cdot 0.49 \cdot 135 \\ &= 110881. \text{ Kgf} \end{aligned}$$

Shear, Nozzle Wall [Snw]:

$$\begin{aligned} &= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n \\ &= (3.1416 \cdot 415.55) \cdot (220.0 - 1.5) \cdot 0.7 \cdot 135 \\ &= 2740572. \text{ Kgf} \end{aligned}$$

Tension, Shell Groove Weld [Tngw]:

$$\begin{aligned} &= (\pi/2) \cdot D_{lo} \cdot W_{gnvi} \cdot 0.74 \cdot S_{ng} \\ &= (3.1416/2.0) \cdot 1049.6 \cdot (80.0 - 0.0) \cdot 0.74 \cdot 138 \\ &= 1373434. \text{ Kgf} \end{aligned}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
 $= (SONW + SNW) = (110881 + 2740572) = 2851453 \text{ Kg}$

Failure Path [Path 2-2]:
 $= (Sonw + Tpgw + Tngw + Sinw)$
 $= (110881 + 0 + 1373435 + 0) = 1484316 \text{ Kg}$

Failure Path [Path 3-3]:
 $= (Sonw + Tngw + Sinw)$
 $= (110881 + 1373435 + 0) = 1484316 \text{ Kg}$

Summary of Failure Path Calculations:

Path 1-1 = 2851453 Kg, must exceed W = 1284265 Kg or W1 = 1347588 Kg
 Path 2-2 = 1484315 Kg, must exceed W = 1284265 Kg or W2 = 2232280 Kg
 Path 3-3 = 1484315 Kg, must exceed W = 1284265 Kg or W3 = 2232280 Kg

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 192 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 154.2274 mm.
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 758.2274 mm.

Input Echo, WRC297 Item 1, Description: SH. IN A2 :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	1.5000	mm.
Shell External Corrosion Allowance	0.0000	mm.
Shell Diameter	1940.000	mm.
Shell Thickness	149.0000	mm.
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc 138.00	N./mm ²
Vessel Hot S.I. Allowable	Smh 138.00	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	1.5000	mm.
Nozzle Diameter	609.600	mm.
Nozzle Thickness	220.0000	mm.
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-266 2	
Nozzle Cold S.I. Allowable	SNmc 138.00	N./mm ²
Nozzle Hot S.I. Allowable	SNmh 134.60	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Note:

External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUstained loads.

Design Internal Pressure	Dp	191.00	bars
Radial Load	P	-14684.89	Kgf
Circumferential Shear	Vc	11013.68	Kgf
Longitudinal Shear	Vl	14684.89	Kgf
Circumferential Moment	Mc	22899.59	Kg-m.
Longitudinal Moment	Ml	22899.59	Kg-m.
Torsional Moment	Mt	26422.59	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No



Local Loads applied at end of Nozzle/Attachment

No

*Warning - The ratio, Dv/Tv (15.020) is < 20
 check the Limitations of WRC bulletin 297.*
*Warning - The ratio, Dn/Tn (4.771) is < 10
 check the Limitations of WRC bulletin 297.*

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$
 $= 1049.6 + 2 \cdot 1.65 \cdot \sqrt{(1045.25(149.0 - 1.5))}$
 $= 2345.346 \text{ mm.}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter (D) = 2090.500 mm.
 Nozzle Outside Diameter (d) = 1049.600 mm.
 Vessel Thickness used (T) = 147.500 mm.
 Nozzle Thickness used (t) = 218.500 mm.
 T / t = 0.675
 d / t = 10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.890$$

Nr/P = 0.077
 Mr/P = 0.138
 M0/P = 0.042
 N0/P = 0.102
 MrD/Mc = 0.274
 NrDL/Mc = 0.115
 M0d/Mc = 0.089
 N0DL/Mc = 0.063
 MrD/Ml = 0.118
 NrDL/Ml = 0.070
 M0D/Ml = 0.040
 N0DL/Ml = 0.128

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	0	0	0	0
Outplane Bending (P)	1	-1	1	-1
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-1	-1	1	1
Outplane Bending (ML)	-2	2	2	-2
Normal Pressure Stress	116	136	116	136
Outplane Stress Summary	115	136	122	133

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	0	0	0	0
Inplane Bending (P)	5	-5	5	-5
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	-6	6	6	-6
Inplane Pressure Stress	58	58	58	58
Inplane Stress Summary	56	59	72	47

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
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(Shear stress normal to longitudinal plane)			Top Outside	Top Inside	Bottom Outside	Bottom Inside
Outplane Shear (Vc)			0	0	0	0
Outplane Shear (Vl)			0	0	0	0
Outplane Shear (Mt)			1	1	1	1
Shear Stress Summary			1	1	0	0

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)		Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress		115	136	122	133

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)		Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)		0	0	0	0
Outplane Bending (P)		1	-1	1	-1
Outplane Membrane (Mc)		0	0	0	0
Outplane Bending (Mc)		-5	5	5	-5
Outplane Membrane (ML)		0	0	0	0
Outplane Bending (ML)		0	0	0	0
Normal Pressure Stress		58	58	58	58
Outplane Stress Summary		54	62	66	52

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)		Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Inplane Membrane (P)		0	0	0	0
Inplane Bending (P)		5	-5	5	-5
Inplane Membrane (Mc)		-1	-1	1	1
Inplane Bending (Mc)		-16	16	16	-16
Inplane Membrane (ML)		0	0	0	0
Inplane Bending (ML)		0	0	0	0
Inplane Pressure Stress		116	136	116	136
Inplane Stress Summary		105	146	140	116

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)		Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Shear (Vc)		0	0	0	0
Outplane Shear (Vl)		0	0	0	0
Torsional Shear (Mt)		1	1	1	1
Shear Stress Summary		0	0	1	1

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)		Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress		105	146	140	116

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)		Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Hoop Membrane (P)		0	0	0	0



Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-1	-1	1	1
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	23	46	23	46

Hoop Stress Summary	23	46	25	48
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses Normal to	Top	Top	Bottom	Bottom
pipe cross-section)	Outside	Inside	Outside	Inside

Axial Membrane (P)	0	0	0	0
Axial Bending (P)	1	-1	1	-1
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-2	-2	2	2
Axial Bending (ML)	-2	2	2	-2
Axial Pressure Stress	11	11	11	11

Axial Stress Summary	9	10	18	10
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress)	Outside	Inside	Outside	Inside

Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	0	0	0	0

Shear Stress Summary	0	0	0	0
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities	Outside	Inside	Outside	Inside

Two * Max Shear Stress	23	46	25	48
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses in the	Left	Left	Right	Right
hoop direction)	Outside	Inside	Outside	Inside

Hoop Membrane (P)	0	0	0	0
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	23	46	23	46

Hoop Stress Summary	23	46	24	47
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to	Left	Left	Right	Right
pipe cross-section)	Outside	Inside	Outside	Inside

Axial Membrane (P)	0	0	0	0
Axial Bending (P)	1	-1	1	-1
Axial Membrane (Mc)	-2	-2	2	2
Axial Bending (Mc)	-5	5	5	-5
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	11	11	11	11

Axial Stress Summary	5	13	21	6
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu	Cl	Du	Dl
	Outside	Inside	Outside	Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	0	0	0	0
Shear Stress Summary	0	0	1	1

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu	Cl	Du	Dl
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	23	46	25	47

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	116	136	116	136	116	136	116	136
Circ. Pl (SUS)	-1	-1	1	1	-1	-1	1	1
Circ. Q (SUS)	-1	1	3	-3	-11	11	21	-21
Long. Pm (SUS)	58	58	58	58	58	58	58	58
Long. Pl (SUS)	0	0	0	0	0	0	0	0
Long. Q (SUS)	-1	1	11	-11	-4	4	6	-6
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	1	1	1	1	1	1	1	1
Pm (SUS)	116.0	136.0	116.0	136.0	116.0	136.0	116.0	136.0
Pm+Pl (SUS)	115.0	135.0	117.0	137.0	115.0	135.0	117.0	137.0
Pm+Pl+Q (Total)	114.0	136.0	120.0	134.0	104.0	146.0	138.0	116.0

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	136.00	138.00	Passed
Pm+Pl (SUS)	137.00	207.00	Passed
Pm+Pl+Q (TOTAL)	146.01	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	23	46	23	46	23	46	23	46
Circ. Pl (SUS)	-1	-1	1	1	0	0	0	0
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	11	11	11	11	11	11	11	11
Long. Pl (SUS)	-2	-2	2	2	-2	-2	2	2
Long. Q (SUS)	-1	1	3	-3	-4	4	6	-6

Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	23.0	46.0	23.0	46.0	23.0	46.0	23.0	46.0
Pm+Pl (SUS)	22.0	45.0	24.0	47.0	23.0	46.0	23.0	46.0
Pm+Pl+Q (Total)	22.0	45.0	24.0	47.0	23.0	46.0	23.0	46.0

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	46.00	134.60	Passed
Pm+Pl (SUS)	47.00	201.90	Passed
Pm+Pl+Q (TOTAL)	47.00	403.80	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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Input, Nozzle Desc: SH. OUT NOZ B2

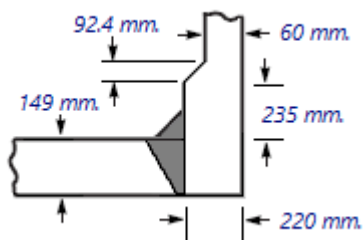
From: 40

Pressure for Reinforcement Calculations	P	191.112	bars
Temperature for Internal Pressure	Temp	260	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	1940.00	mm.
Shell Finished (Minimum) Thickness	t	149.0000	mm.
Shell Internal Corrosion Allowance	c	1.5000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		11759.00	mm.
User Entered Minimum Design Metal Temperature		0.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-266 2	
Material UNS Number		K03506	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	134.60	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		-90.00	deg
Diameter		24.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	60.0000	mm.
Hub Height of Integral Nozzle	h	235.0000	mm.
Height of Beveled Transition	L'	92.3760	mm.
Hub Thickness of Integral Nozzle (tn or x+tp)		220.0000	mm.
Corrosion Allowance	can	1.5000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	455.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	80.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: SH. OUT NOZ B2

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	24.000	in.
Actual Thickness Used in Calculation	2.362	in.



Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 191.11 \cdot 971.5 / (138 \cdot 1.0 - 0.6 \cdot 191.11)$
 $= 146.7448 \text{ mm.}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 191.11 \cdot 306.3 / (135 \cdot 1.0 - 0.6 \cdot 191.11)$
 $= 47.5428 \text{ mm.}$

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
 $= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$
 $= 160.0 / 0.5773$
 $= 277.1282 \text{ mm.}$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

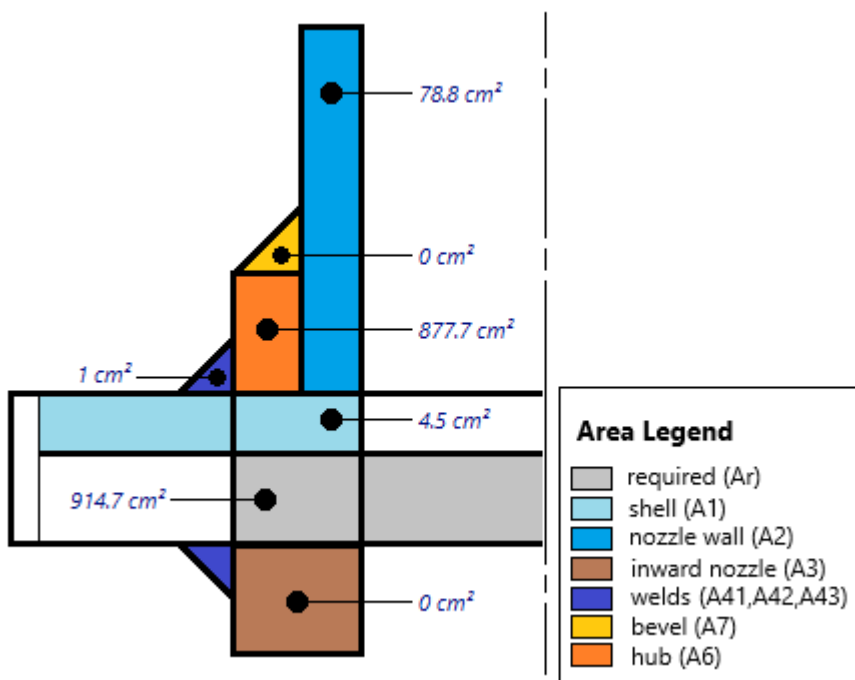
UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	1225.2000	mm.
Parallel to Vessel Wall, opening length	d	612.6000	mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	368.7500	mm.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 134.6 / 138.0)$
 $= 0.975$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 134.6 / 138.0)$
 $= 0.975$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(0.975, 1.0)$
 $= 0.975$



Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	914.743	NA	NA
Area in Shell	A1	4.545	NA	NA
Area in Nozzle Wall	A2	78.820	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	0.975	NA	NA
Area in Element	A5	0.000	NA	NA
Area in Hub	A6	877.653	NA	NA
TOTAL AREA AVAILABLE	Atot	961.994	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$
 $= (612.6 \cdot 146.7448 \cdot 1.0 + 2 \cdot 218.5 \cdot 146.7448 \cdot 1.0 \cdot (1 - 0.98))$
 $= 914.743 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 612.6(1.0 \cdot 147.5 - 1.0 \cdot 146.745) - 2 \cdot 218.5$
 $(1.0 \cdot 147.5 - 1.0 \cdot 146.7448) \cdot (1 - 0.975)$
 $= 4.545 \text{ cm}^2$

Area Available in Nozzle Projecting Outward [A2]:
 $= (2 \cdot tnp)(tn - trn)fr2$
 $= (2 \cdot 368.75)(58.5 - 47.54) \cdot 0.9754$
 $= 78.820 \text{ cm}^2$

Area Available in Inward Weld + Outward Weld [A41 + A43]:
 $= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$
 $= 10.0^2 \cdot 0.9754 + (0.0)^2 \cdot 0.9754$
 $= 0.975 \text{ cm}^2$

Area Available in the Hub Section [A6]:
 $= (2 \cdot \min(Tnp, ho, Hubht)) \cdot (Hubtk - tn) \cdot fr2$
 $= (2 \cdot \min(368.8, 455.0, 235.0)) \cdot (220.0 - 60.0) \cdot 0.975$
 $= 877.653 \text{ cm}^2$ Includes Bevel Area of 144.163 cm²

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures $ta = 49.0428 \text{ mm.}$
 Wall Thickness per UG16(b), $tr16b = 3.0000 \text{ mm.}$
 Wall Thickness, shell/head, internal pressure $trb1 = 148.2448 \text{ mm.}$
 Wall Thickness $tb1 = \max(trb1, tr16b) = 148.2448 \text{ mm.}$
 Wall Thickness $tb2 = \max(trb2, tr16b) = 3.0000 \text{ mm.}$
 Wall Thickness per table UG-45 $tb3 = 9.8312 \text{ mm.}$

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[9.831, \max(148.2448, 3.0)]$
 $= 9.8312 \text{ mm.}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(49.0428, 9.8312)$
 $= 49.0428 \text{ mm.}$

Available Nozzle Neck Thickness = 60.0000 mm. -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained :	61.1,	Allowable :	134.6 N./mm ²	Passed
Expansion :	0.0,	Allowable :	279.7 N./mm ²	Passed
Occasional :	45.7,	Allowable :	179.0 N./mm ²	Passed
Shear :	10.7,	Allowable :	94.2 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Warning:

*The minimum metal temperature could not be computed because the governing thickness for this component was greater than 100.000 mm..
This component must be impact tested.*

Weld Size Calculations, Description: SH. OUT NOZ B2

Intermediate Calc. for nozzle/shell Welds T_{min} 19.0000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * W _o mm.

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$\begin{aligned} &= \max(0, (A - A1 + 2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - t_r)) \cdot S_v) \\ &= \max(0, (914.7434 - 4.5454 + 2 \cdot 218.5 \cdot 0.9754 \cdot (1.0 \cdot 147.5 - 146.7448)) \cdot 138) \\ &= 1285321.25 \text{ Kgf} \end{aligned}$$

For hub type nozzles, A₂ includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$\begin{aligned} &= (A2 + A4 - (W_i - \text{Can} / .707)^2 \cdot f_{r2}) \cdot S_v \\ &= (956.4727 + 0.9754 - 0.0 \cdot 0.98) \cdot 138 \\ &= 1347280.00 \text{ Kgf} \end{aligned}$$

Weld Load [W2]:

$$\begin{aligned} &= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v \\ &= (956.4727 + 0.0 + 0.9754 + (628.7087)) \cdot 138 \\ &= 2231971.75 \text{ Kgf} \end{aligned}$$

Weld Load [W3]:

$$\begin{aligned} &= (A2 + A3 + A4 + A5 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S \\ &= (956.4727 + 0.0 + 0.9754 + 0.0 + (628.7087)) \cdot 138 \\ &= 2231971.75 \text{ Kgf} \end{aligned}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$\begin{aligned} &= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw} \\ &= (3.1416/2.0) \cdot 1049.6 \cdot 10.0 \cdot 0.49 \cdot 135 \\ &= 110881. \text{ Kgf} \end{aligned}$$

Shear, Nozzle Wall [Snw]:

$$\begin{aligned} &= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n \\ &= (3.1416 \cdot 415.55) \cdot (220.0 - 1.5) \cdot 0.7 \cdot 135 \\ &= 2740572. \text{ Kgf} \end{aligned}$$

Tension, Shell Groove Weld [Tngw]:

$$\begin{aligned} &= (\pi/2) \cdot D_{lo} \cdot W_{gnvi} \cdot 0.74 \cdot S_{ng} \\ &= (3.1416/2.0) \cdot 1049.6 \cdot (80.0 - 0.0) \cdot 0.74 \cdot 138 \\ &= 1373434. \text{ Kgf} \end{aligned}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
 $= (SONW + SNW) = (110881 + 2740572) = 2851453 \text{ Kg}$

Failure Path [Path 2-2]:
 $= (Sonw + Tpgw + Tngw + Sinw)$
 $= (110881 + 0 + 1373435 + 0) = 1484316 \text{ Kg}$

Failure Path [Path 3-3]:
 $= (Sonw + Tngw + Sinw)$
 $= (110881 + 1373435 + 0) = 1484316 \text{ Kg}$

Summary of Failure Path Calculations:

Path 1-1 = 2851453 Kg, must exceed W = 1285321 Kg or W1 = 1347280 Kg
 Path 2-2 = 1484315 Kg, must exceed W = 1285321 Kg or W2 = 2231971 Kg
 Path 3-3 = 1484315 Kg, must exceed W = 1285321 Kg or W3 = 2231971 Kg

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 192 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 154.2274 mm.
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 758.2274 mm.

Input Echo, WRC297 Item 1, Description: SH. OUT NOZ B2:

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	1.5000	mm.
Shell External Corrosion Allowance	0.0000	mm.
Shell Diameter	1940.000	mm.
Shell Thickness	149.0000	mm.
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc 138.00	N./mm ²
Vessel Hot S.I. Allowable	Smh 138.00	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	1.5000	mm.
Nozzle Diameter	609.600	mm.
Nozzle Thickness	220.0000	mm.
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-266 2	
Nozzle Cold S.I. Allowable	SNmc 138.00	N./mm ²
Nozzle Hot S.I. Allowable	SNmh 134.60	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Note:

External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUstained loads.

Design Internal Pressure	Dp	191.11	bars
Radial Load	P	-14684.89	Kgf
Circumferential Shear	Vc	11013.68	Kgf
Longitudinal Shear	Vl	14684.89	Kgf
Circumferential Moment	Mc	22899.59	Kg-m.
Longitudinal Moment	Ml	22899.59	Kg-m.
Torsional Moment	Mt	26422.59	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No



Local Loads applied at end of Nozzle/Attachment

No

*Warning - The ratio, D_v/T_v (15.020) is < 20
 check the Limitations of WRC bulletin 297.*
*Warning - The ratio, D_n/T_n (4.771) is < 10
 check the Limitations of WRC bulletin 297.*

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$
 $= 1049.6 + 2 \cdot 1.65 \cdot \sqrt{(1045.25(149.0 - 1.5))}$
 $= 2345.346 \text{ mm.}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter (D) = 2090.500 mm.
 Nozzle Outside Diameter (d) = 1049.600 mm.
 Vessel Thickness used (T) = 147.500 mm.
 Nozzle Thickness used (t) = 218.500 mm.
 $T / t = 0.675$
 $d / t = 10.000$

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.890$$

$N_r/P = 0.077$
 $M_r/P = 0.138$
 $M_0/P = 0.042$
 $N_0/P = 0.102$
 $M_rD/M_c = 0.274$
 $N_rDL/M_c = 0.115$
 $M_0d/M_c = 0.089$
 $N_0DL/M_c = 0.063$
 $M_rD/M_l = 0.118$
 $N_rDL/M_l = 0.070$
 $M_0D/M_l = 0.040$
 $N_0DL/M_l = 0.128$

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	0	0	0	0
Outplane Bending (P)	1	-1	1	-1
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-1	-1	1	1
Outplane Bending (ML)	-2	2	2	-2
Normal Pressure Stress	116	136	116	136
Outplane Stress Summary	115	136	122	134

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	0	0	0	0
Inplane Bending (P)	5	-5	5	-5
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	-6	6	6	-6
Inplane Pressure Stress	58	58	58	58
Inplane Stress Summary	56	59	72	47

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
--------------------	----	----	----	----

(Shear stress normal to longitudinal plane)			Top Outside	Top Inside	Bottom Outside	Bottom Inside
Outplane Shear (Vc)			0	0	0	0
Outplane Shear (Vl)			0	0	0	0
Outplane Shear (Mt)			1	1	1	1
Shear Stress Summary			1	1	0	0

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)			Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress			115	136	122	134

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)			Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)			0	0	0	0
Outplane Bending (P)			1	-1	1	-1
Outplane Membrane (Mc)			0	0	0	0
Outplane Bending (Mc)			-5	5	5	-5
Outplane Membrane (ML)			0	0	0	0
Outplane Bending (ML)			0	0	0	0
Normal Pressure Stress			58	58	58	58
Outplane Stress Summary			54	62	66	52

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)			Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Inplane Membrane (P)			0	0	0	0
Inplane Bending (P)			5	-5	5	-5
Inplane Membrane (Mc)			-1	-1	1	1
Inplane Bending (Mc)			-16	16	16	-16
Inplane Membrane (ML)			0	0	0	0
Inplane Bending (ML)			0	0	0	0
Inplane Pressure Stress			116	136	116	136
Inplane Stress Summary			105	146	140	116

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)			Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Shear (Vc)			0	0	0	0
Outplane Shear (Vl)			0	0	0	0
Torsional Shear (Mt)			1	1	1	1
Shear Stress Summary			0	0	1	1

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)			Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress			105	146	140	116

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)			Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Hoop Membrane (P)			0	0	0	0



Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-1	-1	1	1
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	23	46	23	46

Hoop Stress Summary	23	46	25	48
---------------------	----	----	----	----

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses Normal to	Top	Top	Bottom	Bottom
pipe cross-section)	Outside	Inside	Outside	Inside

Axial Membrane (P)	0	0	0	0
Axial Bending (P)	1	-1	1	-1
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-2	-2	2	2
Axial Bending (ML)	-2	2	2	-2
Axial Pressure Stress	11	11	11	11

Axial Stress Summary	9	10	18	10
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress)	Outside	Inside	Outside	Inside

Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	0	0	0	0

Shear Stress Summary	0	0	0	0
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities	Outside	Inside	Outside	Inside

Two * Max Shear Stress	23	46	25	48
------------------------	----	----	----	----

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses in the	Left	Left	Right	Right
hoop direction)	Outside	Inside	Outside	Inside

Hoop Membrane (P)	0	0	0	0
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	23	46	23	46

Hoop Stress Summary	23	46	24	47
---------------------	----	----	----	----

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to	Left	Left	Right	Right
pipe cross-section)	Outside	Inside	Outside	Inside

Axial Membrane (P)	0	0	0	0
Axial Bending (P)	1	-1	1	-1
Axial Membrane (Mc)	-2	-2	2	2
Axial Bending (Mc)	-5	5	5	-5
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	11	11	11	11

Axial Stress Summary	5	13	21	6
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu	Cl	Du	Dl
	Outside	Inside	Outside	Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	0	0	0	0
Shear Stress Summary	0	0	1	1

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu	Cl	Du	Dl
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	23	46	25	47

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	116	136	116	136	116	136	116	136
Circ. Pl (SUS)	-1	-1	1	1	-1	-1	1	1
Circ. Q (SUS)	-1	1	3	-3	-11	11	21	-21
Long. Pm (SUS)	58	58	58	58	58	58	58	58
Long. Pl (SUS)	0	0	0	0	0	0	0	0
Long. Q (SUS)	-1	1	11	-11	-4	4	6	-6
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	1	1	1	1	1	1	1	1
Pm (SUS)	116.0	136.0	116.0	136.0	116.0	136.0	116.0	136.0
Pm+Pl (SUS)	115.0	135.0	117.0	137.0	115.0	135.0	117.0	137.0
Pm+Pl+Q (Total)	114.0	136.0	120.0	134.0	104.0	146.0	138.0	116.0

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	136.00	138.00	Passed
Pm+Pl (SUS)	137.00	207.00	Passed
Pm+Pl+Q (TOTAL)	146.01	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	23	46	23	46	23	46	23	46
Circ. Pl (SUS)	-1	-1	1	1	0	0	0	0
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	11	11	11	11	11	11	11	11
Long. Pl (SUS)	-2	-2	2	2	-2	-2	2	2
Long. Q (SUS)	-1	1	3	-3	-4	4	6	-6



Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	23.0	46.0	23.0	46.0	23.0	46.0	23.0	46.0
Pm+Pl (SUS)	22.0	45.0	24.0	47.0	23.0	46.0	23.0	46.0
Pm+Pl+Q (Total)	22.0	45.0	24.0	47.0	23.0	46.0	23.0	46.0

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	46.00	134.60	Passed
Pm+Pl (SUS)	47.00	201.90	Passed
Pm+Pl+Q (TOTAL)	47.00	403.80	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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Input, Nozzle Desc: CH. IN NOZ. A1

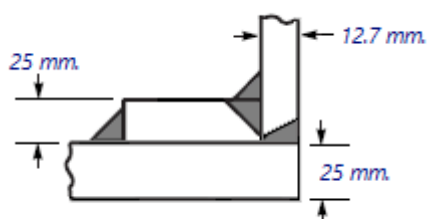
From: 50

Pressure for Reinforcement Calculations	P	10.095	bars
Temperature for Internal Pressure	Temp	85	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	1940.00	mm.
Shell Finished (Minimum) Thickness	t	25.0000	mm.
Shell Internal Corrosion Allowance	c	3.0000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		14178.00	mm.
User Entered Minimum Design Metal Temperature		0.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	N./mm ²
Allowable Stress At Ambient	Sna	118.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		30.0000	in.
Size and Thickness Basis		Nominal	
Nominal Thickness		20	
Flange Material		SA-350 LF2	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	3.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	350.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	16.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	0.0000	mm.
Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	138.00	N./mm ²
Pad Allowable Stress At Ambient	Spa	138.00	N./mm ²
Diameter of Pad along vessel surface	Dp	1001.9999	mm.
Thickness of Pad	te	25.0000	mm.
Weld leg size between Pad and Shell	Wp	20.0000	mm.
Groove weld depth between Pad and Nozzle	Wgpn	25.0000	mm.
Reinforcing Pad Width		120.0000	mm.
Flange Class		150	
Flange Grade		GR 1.1	
Flange Series		B	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: CH. IN NOZ. A1

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	29.000 in.
Actual Thickness Used in Calculation	0.500 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 10.09 \cdot 973.0 / (138 \cdot 1.0 - 0.6 \cdot 10.09)$
 $= 7.1494$ mm.

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 10.09 \cdot 371.3 / (118 \cdot 1.0 - 0.6 \cdot 10.09)$
 $= 3.1930$ mm.

UG-40 Modified by Appendix 1-7, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	1485.2001 mm.
Parallel to Vessel Wall, opening length	d	742.6000 mm.
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		49.2500 mm.

Weld Strength Reduction Factor [fr1]:
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 118.0 / 138.0)$
 $= 0.855$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, S_p / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(0.855, 1.0)$
 $= 0.855$

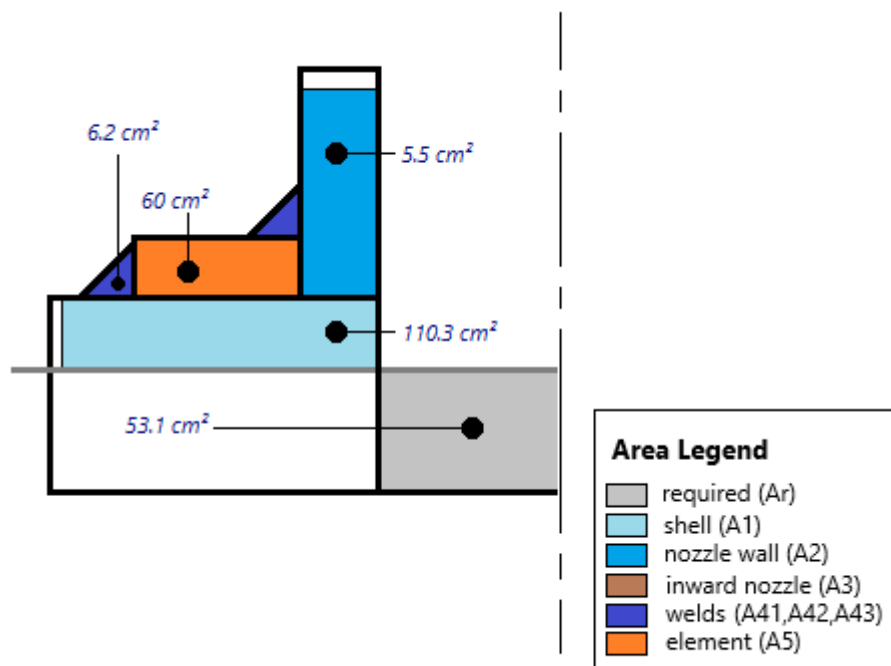


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	53.091	NA	NA
Area in Shell	A1	110.281	NA	NA
Area in Nozzle Wall	A2	5.480	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	6.189	NA	NA
Area in Element	A5	60.000	NA	NA
TOTAL AREA AVAILABLE	Atot	181.950	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

The area available with the given pad is Sufficient.

Area Required [A]:

$$\begin{aligned}
 &= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)} \\
 &= (742.6 \cdot 7.1494 \cdot 1.0 + 2 \cdot 9.7 \cdot 7.1494 \cdot 1.0 \cdot (1 - 1.0)) \\
 &= 53.091 \text{ cm}^2
 \end{aligned}$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$\begin{aligned}
 &= d(E1 \cdot tr - F \cdot tr) - 2 \cdot tn(E1 \cdot tr - F \cdot tr) \cdot (1 - fr1) \\
 &= 742.6(1.0 \cdot 22.0 - 1.0 \cdot 7.149) - 2 \cdot 9.7 \\
 &\quad (1.0 \cdot 22.0 - 1.0 \cdot 7.1494) \cdot (1 - 1.0) \\
 &= 110.281 \text{ cm}^2
 \end{aligned}$$

Area Available in Nozzle Wall Projecting Outward [A2]:

$$\begin{aligned}
 &= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2 \\
 &= (2 \cdot 49.25) \cdot (9.7 - 3.19) \cdot 0.8551 \\
 &= 5.480 \text{ cm}^2
 \end{aligned}$$

Area Available in Welds [A41 + A42 + A43]:

$$\begin{aligned}
 &= Wo^2 \cdot fr3 + (Wi\text{-}can/0.707)^2 \cdot fr2 + Wp^2 \cdot fr4 \\
 &= 16.0^2 \cdot 0.86 + (0.0)^2 \cdot 0.86 + 20.0^2 \cdot 1.0
 \end{aligned}$$



= 6.189 cm²

Area Available in Element [A5]:

= (min(Dp,DL)-(Nozzle OD))*(min(tp,Tlwp,te)) * fr4
 = (1002.0 - 762.0) * 25.0 * 1.0
 = 60.000 cm²

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures ta = 6.1930 mm.
 Wall Thickness per UG16(b), tr16b = 5.5000 mm.
 Wall Thickness, shell/head, internal pressure trb1 = 10.1494 mm.
 Wall Thickness tb1 = max(trb1, tr16b) = 10.1494 mm.
 Wall Thickness tb2 = max(trb2, tr16b) = 5.5000 mm.
 Wall Thickness per table UG-45 tb3 = 11.3312 mm.

Determine Nozzle Thickness candidate [tb]:

= min[tb3, max(tb1,tb2)]
 = min[11.331, max(10.1494, 5.5)]
 = 10.1494 mm.

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

= max(ta, tb)
 = max(6.193, 10.1494)
 = 10.1494 mm.

Available Nozzle Neck Thickness = 0.875 * 12.700 = 11.113 mm. -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	40.8,	Allowable	:	118.0 N./mm ²	Passed
Expansion	:	0.0,	Allowable	:	254.2 N./mm ²	Passed
Occasional	:	19.1,	Allowable	:	156.9 N./mm ²	Passed
Shear	:	13.6,	Allowable	:	82.6 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Pad Weld for the Nozzle, Curve: B

Govern. thk, tg = 11.113, tr = 3.193, c = 3.0 mm., E* = 1.0
 Thickness Ratio = tr * E*/(tg - c) = 0.394, Temp. Reduction = 52 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-26 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-48 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

Govern. thk, tg = 11.113, tr = 3.193, c = 3.0 mm., E* = 1.0
 Thickness Ratio = tr * E*/(tg - c) = 0.394, Temp. Reduction = 52 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-26 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-48 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Shell to Pad Weld Junction at Pad OD, Curve: B

Govern. thk, tg = 25.0, tr = 7.149, c = 3.0 mm., E* = 1.0
 Thickness Ratio = tr * E*/(tg - c) = 0.325, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-1 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: B

Govern. thk, tg = 11.113, tr = 3.193, c = 3.0 mm., E* = 1.0

Thickness Ratio = $tr * E_s / (tg - c) = 0.394$, Temp. Reduction = 52 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -26 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -48 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

Gov. MDMT of the Nozzle : -48 °C
 Gov. MDMT of the Reinforcement Pad : -48 °C
 Gov. MDMT of the nozzle to shell joint welded assembly : -48 °C

Flange MDMT including Temperature reduction per UCS-66.1:

MDMT of ASME B16.5/47 flange per Matl. Specification -46 °C
 Flange MDMT with Temp reduction per UCS-66(i)(2) -48 °C

Where the Stress Reduction Ratio per UCS-66(i)(2) is :
 Design Pressure/Ambient Rating = $10.09/19.60 = 0.515$

UG-40 Modified by Appendix 1-7, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit) D1 1113.9000 mm.
 Parallel to Vessel Wall, opening length d 556.9500 mm.
 Normal to Vessel Wall (Thickness Limit), pad side Tlwp 49.2500 mm.

Reinforcement Areas for Large Nozzle (Per App 1-7(a)): (cm²)

AREA AVAILABLE, A1 to A6		No Pad	With Pad
Area Required [Internal Pressure]	AR	35.396	35.396
Area Available in Shell	A1	55.140	55.140
Area Available in Nozzle Wall	A2	2.699	5.480
Area Available in Inward Nozzle	A3	0.000	0.000
Area Available in Welds	A4	2.189	6.189
Area Available in Pad	A5	0.000	60.000
Area Available in Hub	A6	0.000	0.000
TOTAL AREA AVAILABLE	Atot	60.028	126.810

The area available without a pad is Sufficient.
 The area available with the given pad is Sufficient.

The Nozzle does not fall within the scope of the diameter requirements of App. 1-7(b). Therefore, the Membrane and Bending stress checks are not required.

Weld Size Calculations, Description: CH. IN NOZ. A1

Intermediate Calc. for nozzle/shell Welds Tmin 9.7000 mm.
 Intermediate Calc. for pad/shell Welds TminPad 19.0000 mm.

Results Per UW-16.1:

Required Thickness Actual Thickness
 Nozzle Weld $6.0000 = \text{Min per Code}$ $11.3120 = 0.7 * W_o$ mm.
 Pad Weld $9.5000 = 0.5 * \text{TminPad}$ $14.1400 = 0.7 * W_p$ mm.

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= (Ar - A1) * Sv$
 $= (53.0911 - 110.2809) * 138$
 $= 0.00 \text{ Kg}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2 + A5 + A4 - (W_i - \text{Can} / .707)^2 * fr2) * Sv$
 $= (5.4805 + 60.0 + 6.189 - 0.0 * 0.86) * 138$
 $= 100850.20 \text{ Kg}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$
 $= (3.1416/2.0) \cdot 762.0 \cdot 16.0 \cdot 0.49 \cdot 118$
 $= 112911. \text{ Kg}$

Shear, Pad Element Weld [Spew]:
 $= (\pi/2) \cdot D_P \cdot W_P \cdot 0.49 \cdot S_{EW}$
 $= (3.1416/2.0) \cdot 1002.0 \cdot 20.0 \cdot 0.49 \cdot 138$
 $= 217048. \text{ Kg}$

Tension, Pad Groove Weld [Tpgw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_{gpn} \cdot 0.74 \cdot S_{eg}$
 $= (3.1416/2) \cdot 762.0 \cdot 25.0 \cdot 0.74 \cdot 138$
 $= 311600. \text{ Kg}$

Strength of Failure Paths:

Failure Path [Path 1-1]:
 $= (S_{PEW} + S_{NGW}) = (217048 + 0) = 217048 \text{ Kg}$

Summary of Failure Path Calculations:

Path 1-1 = 217048 Kg, must exceed W = 0 Kg or W1 = 100850 Kg

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 11 bars

The Drop for this Nozzle is : 77.9579 mm.
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 452.9579 mm.

Warning:
 Review Appendix 1-7 (b)(3) for notes regarding externally applied loads to large nozzles and U-2(g).

Input Echo, WRC297 Item 1, Description: CH. IN NOZ. A1:

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.0000	mm.
Shell External Corrosion Allowance	0.0000	mm.
Shell Diameter	1940.000	mm.
Shell Thickness	25.0000	mm.
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc 138.00	N./mm ²
Vessel Hot S.I. Allowable	Smh 138.00	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	3.0000	mm.
Nozzle Diameter	736.600	mm.
Nozzle Thickness	12.7000	mm.
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-106 B	
Nozzle Cold S.I. Allowable	SNmc 118.00	N./mm ²
Nozzle Hot S.I. Allowable	SNmh 118.00	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter of Reinforcing Pad	1002.000	mm.
Thickness of Reinforcing Pad	25.0000	mm.

Note:
 External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUSTAINED loads.

Design Internal Pressure	Dp 10.09	bars
Radial Load	P -3671.23	Kgf



Circumferential Shear	Vc	2753.42	Kgf
Longitudinal Shear	Vl	3671.23	Kgf
Circumferential Moment	Mc	7156.13	Kg-m.
Longitudinal Moment	Ml	7156.13	Kg-m.
Torsional Moment	Mt	8257.07	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$
 $= 762.0 + 2 \cdot 1.65 \cdot \sqrt{(996.5(25.0 - 3.0))}$
 $= 1250.612 \text{ mm.}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	1993.000 mm.
Nozzle Outside Diameter	(d) =	762.000 mm.
Vessel Thickness used	(T) =	47.000 mm.
Nozzle Thickness used	(t) =	9.700 mm.
T / t	=	4.845
d / t	=	78.557

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 2.490$$

Nr/P	=	0.023
Mr/P	=	0.025
M0/P	=	0.050
N0/P	=	0.236
MrD/Mc	=	0.070
NrDL/Mc	=	0.057
M0d/Mc	=	0.177
N0DL/Mc	=	0.475
MrD/Ml	=	0.040
NrDL/Ml	=	0.034
M0D/Ml	=	0.075
N0DL/Ml	=	0.549

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	3	3	3	3
Outplane Bending (P)	4	-4	4	-4
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-22	-22	22	22
Outplane Bending (ML)	-18	18	18	-18
Normal Pressure Stress	20	21	20	21
Outplane Stress Summary	-12	16	70	24

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	0	0	0	0
Inplane Bending (P)	2	-2	2	-2
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-1	-1	1	1
Inplane Bending (ML)	-9	9	9	-9
Inplane Pressure Stress	10	10	10	10



Inplane Stress Summary 1 16 24 0

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE		Au	Al	Bu	Bl
(Shear stress normal to longitudinal plane)		Top Outside	Top Inside	Bottom Outside	Bottom Inside
Outplane Shear (Vc)		0	0	0	0
Outplane Shear (Vl)		0	0	0	0
Outplane Shear (Mt)		1	1	1	1

Shear Stress Summary 2 2 1 1

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE		Au	Al	Bu	Bl
(Stress Intensities)		Top Outside	Top Inside	Bottom Outside	Bottom Inside
Two * Max Shear Stress					
		15	18	70	25

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE		Cu	Cl	Du	Dl
(Stresses Normal to circumferential plane)		Left Outside	Left Inside	Right Outside	Right Inside
Outplane Membrane (P)		3	3	3	3
Outplane Bending (P)		4	-4	4	-4
Outplane Membrane (Mc)		-19	-19	19	19
Outplane Bending (Mc)		-44	44	44	-44
Outplane Membrane (ML)		0	0	0	0
Outplane Bending (ML)		0	0	0	0
Normal Pressure Stress		10	10	10	10

Outplane Stress Summary -45 33 83 -15

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE		Cu	Cl	Du	Dl
(Stresses parallel to circumferential plane)		Left Outside	Left Inside	Right Outside	Right Inside
Inplane Membrane (P)		0	0	0	0
Inplane Bending (P)		2	-2	2	-2
Inplane Membrane (Mc)		-2	-2	2	2
Inplane Bending (Mc)		-17	17	17	-17
Inplane Membrane (ML)		0	0	0	0
Inplane Bending (ML)		0	0	0	0
Inplane Pressure Stress		20	21	20	21

Inplane Stress Summary 3 34 43 4

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE		Cu	Cl	Du	Dl
(Shear stress normal to circumferential plane)		Left Outside	Left Inside	Right Outside	Right Inside
Outplane Shear (Vc)		0	0	0	0
Outplane Shear (Vl)		0	0	0	0
Torsional Shear (Mt)		1	1	1	1

Shear Stress Summary 1 1 2 2

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE		Cu	Cl	Du	Dl
(Stress Intensities)		Left Outside	Left Inside	Right Outside	Right Inside

Two * Max Shear Stress 48 35 83 20

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE		Au	Al	Bu	Bl
--------------------	--	----	----	----	----



(Stresses in the hoop direction)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Hoop Membrane (P)	3	3	3	3
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-22	-22	22	22
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	45	46	45	46
Hoop Stress Summary	26	27	72	73

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Axial Membrane (P)	1	1	1	1
Axial Bending (P)	30	-30	30	-30
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-16	-16	16	16
Axial Bending (ML)	-132	132	132	-132
Axial Pressure Stress	22	22	22	22
Axial Stress Summary	-94	110	204	-122

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	2	2	-2	-2
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	9	9	9	9
Shear Stress Summary	11	11	6	6

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	123	111	204	196

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	3	3	3	3
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-19	-19	19	19
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	45	46	45	46
Hoop Stress Summary	29	31	69	70

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Axial Membrane (P)	1	1	1	1
Axial Bending (P)	30	-30	30	-30
Axial Membrane (Mc)	-16	-16	16	16
Axial Bending (Mc)	-244	244	244	-244
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0



Axial Pressure Stress	22	22	22	22
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Axial Stress Summary	-205	221	315	-234
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu	Cl	Du	Dl
	Outside	Inside	Outside	Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-3	-3	3	3
Shear due to Torsion	9	9	9	9
Shear Stress Summary	6	6	12	12

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu	Cl	Du	Dl
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	236	222	316	305

Stress Computations at the Edge of the Reinforcing Pad:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - ca)}$
 $= 762.0 + 2 \cdot 1.65 \cdot \sqrt{984.0(25.0 - 3.0)}$
 $= 1247.537 \text{ mm.}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	1968.000 mm.
Nozzle Outside Diameter	(d) =	1002.000 mm.
Vessel Thickness used	(T) =	22.000 mm.
Nozzle Thickness used	(t) =	129.700 mm.
T / t	=	0.170
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 4.816$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

Nr/P	=	0.043
Mr/P	=	0.069
M0/P	=	0.022
N0/P	=	0.042
MrD/Mc	=	0.246
NrDL/Mc	=	0.089
M0d/Mc	=	0.074
N0DL/Mc	=	0.074
MrD/Ml	=	0.057
NrDL/Ml	=	0.042
M0D/Ml	=	0.018
N0DL/Ml	=	0.039

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au	Al	Bu	Bl
	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Outplane Membrane (P)	3	3	3	3
Outplane Bending (P)	9	-9	9	-9
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-5	-5	5	5
Outplane Bending (ML)	-15	15	15	-15
Normal Pressure Stress	44	45	44	45
Outplane Stress Summary	35	48	78	28

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	3	3	3	3
Inplane Bending (P)	30	-30	30	-30
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-6	-6	6	6
Inplane Bending (ML)	-49	49	49	-49
Inplane Pressure Stress	22	22	22	22
Inplane Stress Summary	0	38	111	-49

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	2	2	2	2
Shear Stress Summary	3	3	1	1

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	36	49	111	77

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	3	3	3	3
Outplane Bending (P)	9	-9	9	-9
Outplane Membrane (Mc)	-10	-10	10	10
Outplane Bending (Mc)	-63	63	63	-63
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	22	22	22	22
Outplane Stress Summary	-39	68	109	-37

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Inplane Membrane (P)	3	3	3	3
Inplane Bending (P)	30	-30	30	-30
Inplane Membrane (Mc)	-12	-12	12	12
Inplane Bending (Mc)	-213	213	213	-213
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	44	45	44	45
Inplane Stress Summary	-148	218	304	-182

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-1	-1	1	1
Torsional Shear (Mt)	2	2	2	2

Shear Stress Summary 1 1 3 3

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress	148	218	304	183

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	20	21	20	21	20	21	20	21
Circ. Pl (SUS)	-19	-19	25	25	-2	-2	2	2
Circ. Q (SUS)	-14	14	22	-22	-15	15	19	-19
Long. Pm (SUS)	10	10	10	10	10	10	10	10
Long. Pl (SUS)	-1	-1	1	1	-16	-16	22	22
Long. Q (SUS)	-7	7	11	-11	-40	40	48	-48
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	1	1	1	1	1	1	1	1
Pm (SUS)	20.0	21.0	20.0	21.0	20.0	21.0	20.0	21.0
Pm+Pl (SUS)	9.0	9.0	45.0	46.0	24.0	25.0	32.0	32.0
Pm+Pl+Q (Total)	15.1	17.0	67.0	24.1	49.0	35.0	80.0	20.1

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	21.00	138.00	Passed
Pm+Pl (SUS)	46.00	207.00	Passed
Pm+Pl+Q (TOTAL)	80.03	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Reinforcing Pad Edge (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	44	45	44	45	44	45	44	45
Circ. Pl (SUS)	-2	-2	8	8	-9	-9	15	15
Circ. Q (SUS)	-6	6	24	-24	-183	183	243	-243
Long. Pm (SUS)	22	22	22	22	22	22	22	22
Long. Pl (SUS)	-3	-3	9	9	-7	-7	13	13
Long. Q (SUS)	-19	19	79	-79	-54	54	72	-72
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	-1	-1	1	1
Shear Q (SUS)	2	2	2	2	2	2	2	2
Pm (SUS)	44.0	45.0	44.0	45.0	44.0	45.0	44.0	45.0
Pm+Pl (SUS)	42.0	43.0	52.0	53.0	35.0	36.0	59.0	60.0



Pm+Pl+Q (Total)	36.2	49.4	110.1	77.1	148.0	219.0	302.0	183.1
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Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	45.00	138.00	Passed
Pm+Pl (SUS)	60.04	207.00	Passed
Pm+Pl+Q (TOTAL)	302.05	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	45	46	45	46	45	46	45	46
Circ. Pl (SUS)	-19	-19	25	25	-16	-16	22	22
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	22	22	22	22	22	22	22	22
Long. Pl (SUS)	-15	-15	17	17	-15	-15	17	17
Long. Q (SUS)	-102	102	162	-162	-214	214	274	-274
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	2	2	-2	-2	-3	-3	3	3
Shear Q (SUS)	9	9	9	9	9	9	9	9
Pm (SUS)	45.0	46.0	45.0	46.0	45.0	46.0	45.0	46.0
Pm+Pl (SUS)	26.2	27.2	70.1	71.1	29.4	30.4	67.3	68.3
Pm+Pl+Q (Total)	123.0	110.4	201.4	194.5	236.3	221.2	313.6	303.9

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	46.00	118.00	Passed
Pm+Pl (SUS)	71.12	177.00	Passed
Pm+Pl+Q (TOTAL)	313.58	353.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.



Input, Nozzle Desc: SAFETY VALVE S1

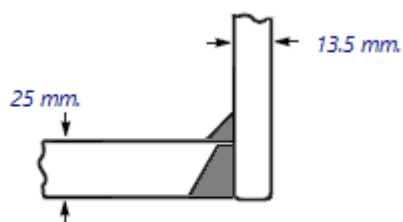
From: 50

Pressure for Reinforcement Calculations	P	10.000	bars
Temperature for Internal Pressure	Temp	85	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	1940.00	mm.
Shell Finished (Minimum) Thickness	t	25.0000	mm.
Shell Internal Corrosion Allowance	c	3.0000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		14178.00	mm.
User Entered Minimum Design Metal Temperature		0.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	N./mm ²
Allowable Stress At Ambient	Sna	118.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		90.00	deg
Diameter		4.0000	in.
Size and Thickness Basis		Nominal	
Nominal Thickness		160	
Flange Material		SA-105	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	3.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	300.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	8.7376	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		150	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: SAFETY VALVE S1

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	3.438	in.
Actual Thickness Used in Calculation	0.531	in.



Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 10.0 \cdot 973.0 / (138 \cdot 1.0 - 0.6 \cdot 10.0)$
 $= 7.0821 \text{ mm.}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 10.0 \cdot 46.66 / (118 \cdot 1.0 - 0.6 \cdot 10.0)$
 $= 0.3975 \text{ mm.}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit) D1 186.6504 mm.
 Parallel to Vessel Wall, opening length d 93.3252 mm.
 Normal to Vessel Wall (Thickness Limit), no pad Tlnp 26.2185 mm.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 118.0 / 138.0)$
 $= 0.855$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 118.0 / 138.0)$
 $= 0.855$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(0.855, 1.0)$
 $= 0.855$

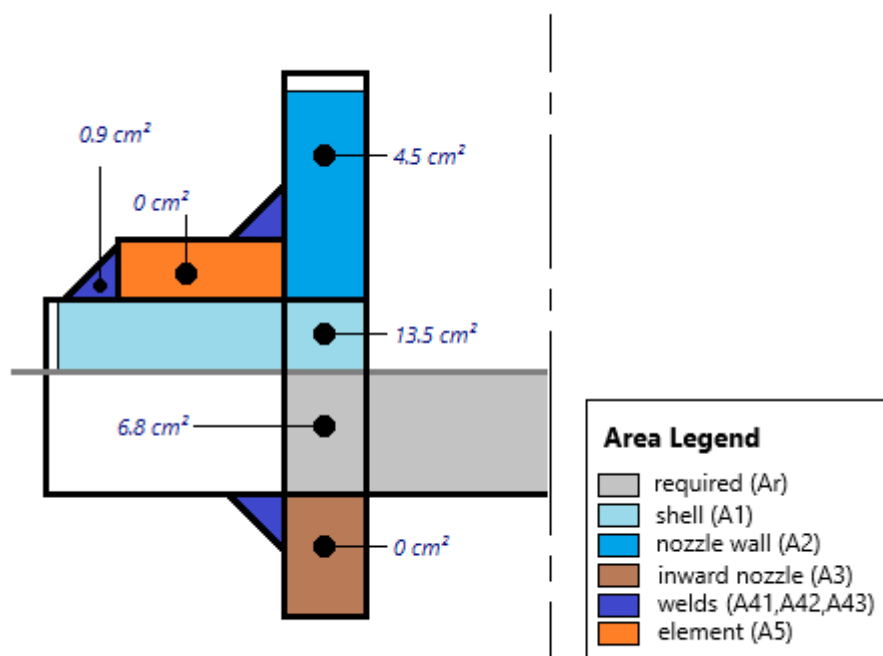


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	6.825	NA	NA
Area in Shell	A1	13.469	NA	NA
Area in Nozzle Wall	A2	4.524	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA



Area in Welds	A41+A42+A43	0.855	NA	NA
Area in Element	A5	0.000	NA	NA
TOTAL AREA AVAILABLE	Atot	18.848	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$
 $= (93.3252 \cdot 7.0821 \cdot 1.0 + 2 \cdot 10.4874 \cdot 7.0821 \cdot 1.0 \cdot (1 - 0.86))$
 $= 6.825 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 93.325(1.0 \cdot 22.0 - 1.0 \cdot 7.082) - 2 \cdot 10.487$
 $(1.0 \cdot 22.0 - 1.0 \cdot 7.0821) \cdot (1 - 0.855)$
 $= 13.469 \text{ cm}^2$

Area Available in Nozzle Projecting Outward [A2]:
 $= (2 \cdot tnp)(tn - trn)fr2$
 $= (2 \cdot 26.22)(10.49 - 0.4) \cdot 0.8551$
 $= 4.524 \text{ cm}^2$

Area Available in Inward Weld + Outward Weld [A41 + A43]:
 $= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$
 $= 10.0^2 \cdot 0.8551 + (0.0)^2 \cdot 0.8551$
 $= 0.855 \text{ cm}^2$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures $ta = 3.3975 \text{ mm.}$
 Wall Thickness per UG16(b), $tr16b = 5.5000 \text{ mm.}$
 Wall Thickness, shell/head, internal pressure $trb1 = 10.0821 \text{ mm.}$
 Wall Thickness $tb1 = \max(trb1, tr16b) = 10.0821 \text{ mm.}$
 Wall Thickness $tb2 = \max(trb2, tr16b) = 5.5000 \text{ mm.}$
 Wall Thickness per table UG-45 $tb3 = 8.2578 \text{ mm.}$

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[8.258, \max(10.0821, 5.5)]$
 $= 8.2578 \text{ mm.}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(3.3975, 8.2578)$
 $= 8.2578 \text{ mm.}$

Available Nozzle Neck Thickness = $0.875 \cdot 13.487 = 11.801 \text{ mm.} \rightarrow \text{OK}$

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: B

Govrn. thk, $t_g = 11.801$, $tr = 0.397$, $c = 3.0 \text{ mm.}$, $E* = 1.0$
 Thickness Ratio = $tr \cdot E* / (t_g - c) = 0.045$, Temp. Reduction = $78 \text{ }^\circ\text{C}$

Min Metal Temp. w/o impact per UCS-66, Curve B $-24 \text{ }^\circ\text{C}$
 Min Metal Temp. at Required thickness (UCS 66.1) $-104 \text{ }^\circ\text{C}$
 Min Metal Temp. w/o impact per UG-20(f) $-29 \text{ }^\circ\text{C}$

Gov. MDMT of the nozzle to shell joint welded assembly : $-104 \text{ }^\circ\text{C}$

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) $-18 \text{ }^\circ\text{C}$



Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -48 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = 10.00/19.60 = 0.510

Weld Size Calculations, Description: SAFETY VALVE S1

Intermediate Calc. for nozzle/shell welds Tmin 10.4874 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm.

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - t_r)) \cdot S_v)$
 $= \max(0, (6.8246 - 13.4687 + 2 \cdot 10.4874 \cdot 0.8551 \cdot (1.0 \cdot 22.0 - 7.0821)) \cdot 138)$
 $= \max(0, -5584.40) \text{ Kgf}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A5+A4-(W_i-\text{Can}/.707)^2 \cdot f_{r2}) \cdot S_v$
 $= (4.5241 + 0.0 + 0.8551 - 0.0 \cdot 0.86) \cdot 138$
 $= 7569.28 \text{ Kgf}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$
 $= (4.5241 + 0.0 + 0.8551 + (3.9457)) \cdot 138$
 $= 13121.49 \text{ Kgf}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$
 $= (4.5241 + 0.0 + 0.8551 + 0.0 + (3.9457)) \cdot 138$
 $= 13121.49 \text{ Kgf}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$
 $= (3.1416/2.0) \cdot 114.3 \cdot 10.0 \cdot 0.49 \cdot 118$
 $= 10585. \text{ Kgf}$

Shear, Nozzle Wall [Snw]:
 $= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n$
 $= (3.1416 \cdot 51.9063) \cdot (13.4874 - 3.0) \cdot 0.7 \cdot 118$
 $= 14404. \text{ Kgf}$

Tension, Shell Groove Weld [Tngw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_{gnv} \cdot 0.74 \cdot S_{ng}$
 $= (3.1416/2.0) \cdot 114.3 \cdot (8.7376 - 0.0) \cdot 0.74 \cdot 138$
 $= 16336. \text{ Kgf}$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
 $= (\text{SONW} + \text{SNW}) = (10585 + 14404) = 24989 \text{ Kgf}$

Failure Path [Path 2-2]:
 $= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw})$
 $= (10585 + 0 + 16336 + 0) = 26921 \text{ Kgf}$

Failure Path [Path 3-3]:
 $= (\text{Sonw} + \text{Tngw} + \text{Sinw})$
 $= (10585 + 16336 + 0) = 26921 \text{ Kgf}$



Summary of Failure Path Calculations:

Path 1-1 = 24989	Kgf, must exceed W = 0 Kgf	or W1 = 7569 Kgf
Path 2-2 = 26920	Kgf, must exceed W = 0 Kgf	or W2 = 13121 Kgf
Path 3-3 = 26920	Kgf, must exceed W = 0 Kgf	or W3 = 13121 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 19 bars

The Drop for this Nozzle is : 1.6850 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 326.6850 mm.

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Input, Nozzle Desc: DRAIN NOZ. D2

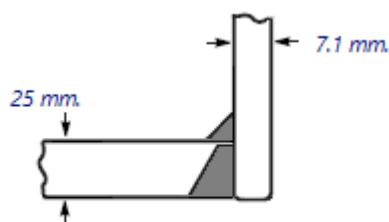
From: 50

Pressure for Reinforcement Calculations	P	10.189	bars
Temperature for Internal Pressure	Temp	85	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	1940.00	mm.
Shell Finished (Minimum) Thickness	t	25.0000	mm.
Shell Internal Corrosion Allowance	c	3.0000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		14128.00	mm.
User Entered Minimum Design Metal Temperature		0.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	N./mm ²
Allowable Stress At Ambient	Sna	118.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		-90.00	deg
Diameter		1.5000	in.
Size and Thickness Basis		Nominal	
Nominal Thickness		160	
Flange Material		SA-105	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	3.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	200.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	25.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		150	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: DRAIN NOZ. D2

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	1.338	in.
Actual Thickness Used in Calculation	0.281	in.



Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $10.19 \cdot 973.0 / (138 \cdot 1.0 - 0.6 \cdot 10.19)$
= 7.2167 mm.

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $10.19 \cdot 19.99 / (118 \cdot 1.0 - 0.6 \cdot 10.19)$
= 0.1735 mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	92.2600	mm.
Parallel to Vessel Wall	Rn+tn+t	46.1300	mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	10.3435	mm.

*Taking a UG-36(c)(3)(a) exemption for nozzle: DRAIN NOZ. D2.
This calculation is valid for nozzles that meet all the requirements of
paragraph UG-36. Please check the Code carefully, especially for nozzles
that are not isolated or do not meet Code spacing requirements. To force
the computation of areas for small nozzles go to Tools->Configuration
and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 3.1735	mm.
Wall Thickness per UG16(b),	tr16b	= 5.5000	mm.
Wall Thickness, shell/head, internal pressure	trb1	= 10.2167	mm.
Wall Thickness	tb1 = max(trb1, tr16b)	= 10.2167	mm.
Wall Thickness	tb2 = max(trb2, tr16b)	= 5.5000	mm.
Wall Thickness per table UG-45	tb3	= 6.2200	mm.

Determine Nozzle Thickness candidate [tb]:
= min[tb3, max(tb1, tb2)]
= min[6.22, max(10.2167, 5.5)]
= 6.2200 mm.

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
= max(ta, tb)
= max(3.1735, 6.22)
= 6.2200 mm.

Available Nozzle Neck Thickness = $0.875 \cdot 7.137 = 6.245$ mm. -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), Curve: B

Govrn. thk, tg = 6.245, tr = 0.174, c = 3.0 mm., E* = 1.0
Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.053$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

MDMT of ASME B16.5/47 flange per Matl. Specification	-18 °C
Flange MDMT with Temp reduction per UCS-66(i)(2)	-48 °C

Where the Stress Reduction Ratio per UCS-66(i)(2) is :
Design Pressure/Ambient Rating = $10.19 / 19.60 = 0.520$

Weld Size Calculations, Description: DRAIN NOZ. D2

Intermediate Calc. for nozzle/shell Welds	Tmin	4.1374	mm.
---	------	--------	-----

Results Per UW-16.1:



	Required Thickness	Actual Thickness
Nozzle Weld	$2.8962 = 0.7 * t_{min.}$	$7.0700 = 0.7 * W_o \text{ mm.}$

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.3002 mm.
The Cut Length for this Nozzle is, Drop + Ho + H + T : 225.3002 mm.

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Input Echo, Tubesheet Number 1, Description: FIXED TS

Shell Data:

Main Shell Description: MAIN SHELL

Shell Maximum Design Pressure	Psd,max	191.00 bars
Shell Maximum Operating Pressure	Psox,max	191.00 bars
Shell Minimum Operating Pressure	Psox,min	0.00 bars
Shell Thickness	ts	149.0000 mm.
Shell Internal Corrosion Allowance	cas	1.5000 mm.
Shell External Corrosion Allowance	casext	0.0000 mm.
Inside Diameter of Shell	Ds	1940.000 mm.
Shell Circumferential Joint Efficiency	Esw	1.000
Shell Temperature for Internal Pressure	Ts	260.00 °C
Shell Material		SA-516 70

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Shell Material UNS Number		K02700
Shell Allowable Stress at Temperature	Ss	137.60 N./mm ²
Shell Allowable Stress at Ambient		138.00 N./mm ²

Channel Data:

Channel Description: CH. SHELL-1

Channel Type:		Cylinder
Channel Maximum Design Pressure	Ptd,max	10.00 bars
Channel Maximum Operating Pressure	Ptox,max	10.00 bars
Channel Minimum Operating Pressure	Ptox,min	0.00 bars
Channel Thickness	tc	25.0000 mm.
Channel Internal Corrosion Allowance	cac	3.0000 mm.
Channel External Corrosion Allowance	cacext	0.0000 mm.
Inside Diameter of Channel	Dc	1940.000 mm.
Channel Design Temperature	TEMPC	85.00 °C
Channel Flange Matl		SA-516 70

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Channel Material UNS Number		K02700
Channel Allowable Stress at Temperature	Sc	138.00 N./mm ²
Channel Allowable Stress at Ambient		138.00 N./mm ²

Tube Data:

Number of Tube Holes	Nt	3297
Tube Wall Thickness	et	2.7700 mm.
Tube Outside Diameter	D	19.0500 mm.
Straight Tube Length (bet. inner tubsht faces) L		10820.00 mm.
Design Temperature of the Tubes		260.00 °C
Tube Material		SA-210 A-1
Tube Material UNS Number		K02707
Is this a Welded Tube		No
Tube Material Specification used		Smls. tube
Tube Allowable Stress at Temperature		118.00 N./mm ²
Tube Allowable Stress At Ambient		118.00 N./mm ²
Tube Yield Stress At design Temperature	Syt	208.00 N./mm ²
Tube Pitch (Center to Center Spacing)	P	30.0000 mm.
Tube Layout Pattern		Triangular
Fillet Weld Leg	af	2.7000 mm.
Groove Weld Leg	ag	2.7000 mm.
Tube-Tubesheet Joint Weld Type		Full Strength
Method for Tube-Tubesheet Jt. Allowable		UW-20.3
Is the Tube Inset		No
Radius to Outermost Tube Hole Center	ro	939.980 mm.
Largest Center-to-Center Tube Distance	Ul	0.0000 mm.
Length of Expanded Portion of Tube	ltx	0.0000 mm.
Tube-side pass partition groove depth	hg	0.0000 mm.
Area of Tube Layout	Ap	24988.3 cm ²
Perimeter of Tube Layout	Cp	5991.4 mm.



Tubesheet Data:

Tubesheet TYPE: Fixed Tubesheet Exchanger, Conf A

Tubesheet Design Metal Temperature T 260.00 °C
 Tubesheet Material SA-266 2

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Tubesheet Material UNS Number K03506
 Tubesheet Allowable Stress at Temperature S 134.60 N./mm²
 Tubesheet Allowable Stress at Ambient Tt 138.00 N./mm²
 Thickness of Tubesheet h 349.0000 mm.
 Tubesheet Corr. Allowance (Shell side) Cats 1.5000 mm.
 Tubesheet Corr. Allowance (Channel side) Catc 3.0000 mm.
 Tubesheet Outside Diameter A 2240.000 mm.

Area of the Untubed Lanes AL 0.0 cm²

Additional Data for Fixed/Floating Tubesheet Exchangers:

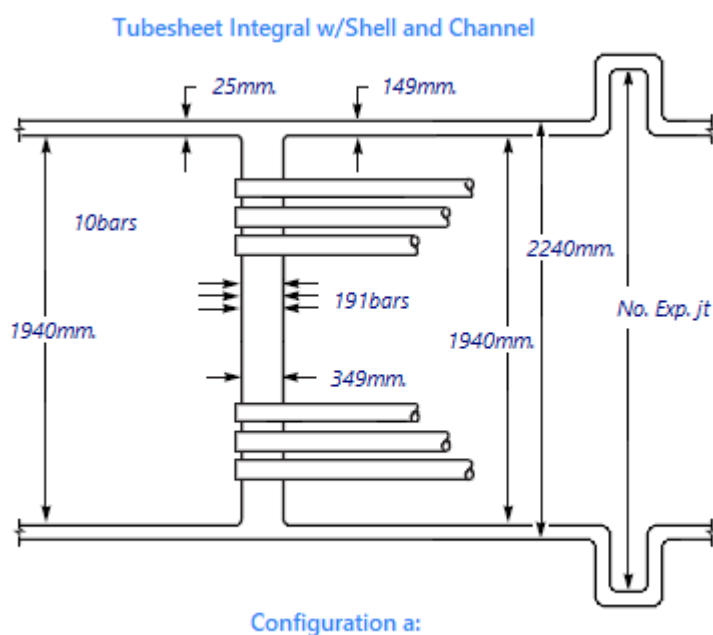
Unsupported Tube Span under consideration l 1510.000 mm.
 Tube End condition corresponding to Span (l) k 0.80

Ignore Radial Thermal Exp. effects (4.18.8/4.18.9) Yes
 Perform Differential Pressure Design No
 Run Multiple Load Cases Yes

Mean Shell Metal Temp. along Shell len. Tsm 45.00 °C
 Mean Tube Metal Temp. along Tube length Ttm 40.00 °C
 Junction Stress Reduction option Perform Plastic Calculation

Tubesheet Integral with Both
 Tubesheet Extended as Flange No

Is Exchanger in Creep range (skip EP, Use 3S for Sps) No



ASME TubeSheet Results per Part UHX/4.18, 2025

Elasticity/Expansion Material Properties:

Shell - TE-1 Carbon & Low Alloy Steels, Group 1
 Shell - TM-1 Carbon Steels with C<= 0.3%



Th. Exp. Coeff. Metal Temp. along Len	45.0 °C	0.0000117458 /°C
Elastic Mod. at Design Temperature	260.0 °C	0.18822E+09 KPa.
Th. Exp. Coeff. Metal Temp. at Tubsht	21.1 °C	0.0000115190 /°C
Elastic Mod. at Metal Temp. along Len	45.0 °C	0.20133E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

Channel - TE-1 Carbon & Low Alloy Steels, Group 1
 Channel - TM-1 Carbon Steels with C<= 0.3%

Th. Exp. Coeff. Metal Temp. at Tubsht	21.1 °C	0.0000115190 /°C
Elastic Mod. at Design Temperature	85.0 °C	0.19904E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

Tubes - TE-1 Carbon & Low Alloy Steels, Group 1
 Tubes - TM-1 Carbon Steels with C<= 0.3%

Th. Exp. Coeff. Metal Temp. along Len	40.0 °C	0.0000117134 /°C
Elastic Mod. at Design Temperature	260.0 °C	0.18822E+09 KPa.
Elastic Mod. at Metal Temp. along Len	40.0 °C	0.20162E+09 KPa.
Elastic Mod. at Tubsht. Design Temp.	260.0 °C	0.18822E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

TubeSheet - TE-1 Carbon & Low Alloy Steels, Group 1
 TubeSheet - TM-1 Carbon Steels with C> 0.3%

Elastic Mod. at Design Temperature	260.0 °C	0.18685E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20132E+09 KPa.

Note:

The Elasticity and Alpha values are taken from Tables in ASME II D.
 Please insure these properties are consistent with the
 type of Material for the tubes, shell, channel etc.

Tube Required Thickness under Internal Pressure (Tubeside pressure):

Thickness Due to Internal Pressure:
 $= (P \cdot (D/2 - CAE)) / (S \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= (10.0 \cdot (19.05/2 - 0)) / (118.0 \cdot 1.0 + 0.4 \cdot 10.0)$
 $= 0.0805 + 0.0000 = 0.0805 \text{ mm.}$

Tube Required Thickness under External Pressure (Shellside pressure) :

External Pressure Chart CS-2 at 260.00 °C
 Elastic Modulus for Material 186156960.00 KPa.

Results for Max. Allowable External Pressure (Emawp):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
2.7700	19.05	10820.00	6.88	50.0000	0.0232575	114.08
EMAWP = $(2.167 / (D/T) - 0.0833) \cdot B = 264.4294 \text{ bars}$						

Results for Req'd Thickness for Ext. Pressure (Tca):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
2.2777	19.05	10820.00	8.36	50.0000	0.0157258	108.66
EMAWP = $(2.167 / (D/T) - 0.0833) \cdot B = 191.0133 \text{ bars}$						

Summary of Tube Required Thickness Results:

Total Required Thickness including Corrosion all.	2.2777 mm.
Allowable Internal Pressure at Corroded thickness	388.30 bars
Required Internal Design Pressure	10.00 bars
Allowable External Pressure at Corroded thickness	264.43 bars
Required External Design Pressure	191.00 bars
Required Thickness due to Shell Side pressure	2.2777 mm.

Detailed Results for load Case D3 un-corr. (Psd,max + Ptd,max)

Results for ASME Fixed Tubesheet Calculations for Configuration a:



Results for Tubesheet Calculations Original Thickness:

ASME Section VIII-2 Part 4.18.8.4 Step 1:

Tube expansion depth ratio [p]:

$$\begin{aligned} &= l_{tx} / h \text{ (modified for corrosion if present)} \\ &= 0/349.0 \\ &= 0.0000 \text{ (must be } 0 \leq \rho \leq 1 \text{)} \end{aligned}$$

Effective tube hole diameter [d*]:

$$\begin{aligned} &= \max(dt - 2 \cdot tt(Et/E) \cdot (St/S) \cdot (\rho); dt - 2 \cdot tt) \\ &= \max(19.05 - 2 \cdot 2.77 (0.18822E+09/.18685E+09) \cdot \\ &\quad (117/134) \cdot (0); 19.05 - 2 \cdot 2.77) \\ &= 19.0500 \text{ mm.} \end{aligned}$$

Equivalent outer tube limit circle diameter [Do]:

$$\begin{aligned} &= 2 \cdot r_o + dt \\ &= 2 \cdot 939.98 + 19.05 \\ &= 1899.010 \text{ mm.} \end{aligned}$$

Determine the basic ligament efficiency for shear [μ]:

$$\begin{aligned} &= (p - dt) / p \\ &= (30.0 - 19.05) / 30.0 \\ &= 0.365 \end{aligned}$$

Equivalent outer tube limit radius [ao]:

$$\begin{aligned} &= Do / 2 \\ &= 1899.01/2 \\ &= 949.5050 \text{ mm.} \end{aligned}$$

Effective tube pitch [p*]:

$$\begin{aligned} &= p / \sqrt{ (1 - 4 \cdot \min(AL \cdot CNVfactor; 4 \cdot Do \cdot p) / (\pi \cdot Do^2)) } \\ &= 30.0 / \sqrt{ (1 - 4 \cdot \min(0 \cdot 100.0 ; 4 \cdot 1899.01 \cdot 30.0) / (\pi \cdot 1899.01^2)) } \\ &= 30.0000 \text{ mm.} \end{aligned}$$

Effective ligament efficiency for bending [μ^*]:

$$\begin{aligned} &= (p^* - d^*) / p^* \\ &= (30.0 - 19.05) / 30.0 \\ &= 0.3650 \end{aligned}$$

Ratio [ps]:

$$\begin{aligned} &= a_s / a_o \\ &= 970.0/949.505 \\ &= 1.0216 \end{aligned}$$

Ratio [pc]:

$$\begin{aligned} &= a_c / a_o \\ &= 970.0/949.505 \\ &= 1.0216 \end{aligned}$$

Compute parameter [xt]:

$$\begin{aligned} &= 1 - Nt((dt - 2 \cdot tt) / (2 \cdot a_o))^2 \\ &= 1 - 3297 ((19.05 - 2 \cdot 2.77) / (2 \cdot 949.505))^2 \\ &= 0.8331 \end{aligned}$$

Determine parameter [xs]:

$$\begin{aligned} &= 1 - Nt(dt / (2 \cdot a_o))^2 \\ &= 1 - 3297 (19.05 / (2 \cdot 949.505))^2 \\ &= 0.6682 \end{aligned}$$

Determine the value [h'g]:

$$\begin{aligned} &= \max((h_g - CATC); 0) \text{ (For pressure only cases)} \\ &= \max((0 - 0); 0) \\ &= 0.000 \text{ mm.} \end{aligned}$$

ASME Section VIII-2 Part 4.18.8.4 Step 2:

Axial Shell Stiffness [Ks]:

$$= \pi \cdot t_s(D_s + t_s) E_s / L$$



$$= \pi \cdot 149.0 (1940.0 + 149.0) \cdot 1.8822E+09 / 10820.0$$

$$= 1734680.0000 \text{ Kgf/mm.}$$

Axial Tube Stiffness [Kt]:

$$= \pi \cdot t_t (D_t - t_t) E_t / L$$

$$= \pi \cdot 2.77 (19.05 - 2.77) \cdot 1.8822E+09 / 10820.0$$

$$= 251.3210 \text{ Kgf/mm.}$$

Stiffness Factor [Ks,t]:

$$= K_s / (N_t \cdot K_t)$$

$$= 1734680 / (3297 \cdot 251.321)$$

$$= 2.09349$$

Rigidty Ratio [J]:

$$= 1 / (1 + K_s / K_j)$$

$$= 1 / (1 + 1734680 / 0)$$

$$= 1.0000 (= 1 \text{ if No Exp. Jt.})$$

Shell coefficient [β_s]:

$$= ((12(1 - \nu_s^2))^{1/4}) / ((D_s + t_s)t_s)$$

$$= ((12(1 - 0.3^2))^{1/4}) / ((1940.0 + 149.0)149.0)$$

$$= 0.0033 \text{ 1/mm.}$$

Shell coefficient [ks]:

$$= \beta_s \cdot E_s \cdot t_s^3 / (6(1 - \nu_s^2))$$

$$= 0.003 \cdot 0.18822E+09 \cdot 149.0^3 / (6(1 - 0.3^2))$$

$$= 3715651328.0000 \text{ bars} \cdot \text{mm.}^2$$

Shell coefficient [λ_s]:

$$= 6(D_s + t_s) \cdot k_s / h^3 \cdot (1 + h \cdot \beta_s + (h \cdot \beta_s)^2 / 2)$$

$$= 6(1940.0 + 149.0) \cdot 0.37157E+1 / 349.0^3 \cdot (1 + 349.0 \cdot 0.003 + (349.0 \cdot 0.003)^2 / 2)$$

$$= 3049807.7500 \text{ bars}$$

Shell coefficient [δ_s]:

$$= D_s^2 / (4 \cdot E_s \cdot t_s) \cdot (1 - D_s \cdot \nu_s / (2(D_s + t_s)))$$

$$= 1940.0^2 / (4 \cdot 0.18822E+09 \cdot 149.0) \cdot (1 - 1940.0 \cdot 0.3 / (2(1940.0 + 149.0)))$$

$$= 0.28874E-01 \text{ mm./N./mm}^2$$

Channel coefficient [β_c]:

$$= ((12(1 - \nu_c^2))^{1/4}) / ((D_c + t_c)t_c)$$

$$= ((12(1 - 0.3^2))^{1/4}) / ((1940.0 + 25.0)25.0)$$

$$= 0.0082 \text{ 1/mm.}$$

Channel coefficient [kc]:

$$= \beta_c \cdot E_c \cdot t_c^3 / (6(1 - \nu_c^2))$$

$$= 0.008 \cdot 0.19904E+09 \cdot 25.0^3 / (6(1 - 0.3^2))$$

$$= 46717224.0000 \text{ bars} \cdot \text{mm.}^2$$

Channel coefficients [CC, CqDelta, CqTheta, CmTheta, CmDelta]:

$$CqDelta = CqTheta = CmTheta = CmDelta = 1$$

Channel coefficient [λ_c]:

$$= 6(D_c + t_c / \cos(\theta)) \cdot k_c / h^3 \cdot (CmTheta + h \cdot \beta_c \cdot (Cm\Delta + CqTheta) / 2 + (h \cdot \beta_c)^2 / 2 \cdot Cq\Delta)$$

$$= 6(1940.0 + 25.0 / \cos(0)) \cdot 46717224 / 349.0^3 \cdot (1.0 + 349.0 \cdot 0.008 \cdot (1.0 + 1.0) / 2 + (349.0 \cdot 0.008)^2 / 2 \cdot 1.0)$$

$$= 103127.8 \text{ bars}$$

Channel coefficient [δ_c]:

$$= D_c^2 / (4 \cdot E_c \cdot t_c) \cdot (1 - D_c \cdot \nu_c / (2(D_c + t_c)))$$

$$= 1940.0^2 / (4 \cdot 0.19904E+09 \cdot 25.0) \cdot (1 - 1940.0 \cdot 0.3 / (2(1940.0 + 25.0)))$$

$$= 0.16107E+00 \text{ mm./N./mm}^2$$

Channel coefficient [zetac]:

$$zetac = 0$$

ASME Section VIII-2 Part 4.18.8.4 Step 3:

$$h/p = 11.633333; \mu^* = 0.365000; E^* = 68583928 \text{ KPa.}$$

$$E^*/E = 0.367063; \nu^* = 0.320752$$



Note: As h/p (11.633) is > 2 , data values for $h/p = 2$ were used.

Tube Bundle Stiffness Factor $[X_a]$:

$$= \left(\frac{24(1 - \nu^2)N_t \cdot E_t \cdot t_t (d_t - t_t) a_o^2}{E \cdot L \cdot H^3} \right)^{1/4}$$

$$= \left(\frac{24(1 - 0.321^2)3297 \cdot 0.18822E+09 \cdot 2.77}{(19.05 - 2.77)949.505^2 (68583928 \cdot 10820.0 \cdot 349.0^3)} \right)^{1/4}$$

$$= 2.0371$$

Values from Table 4.18.3

$$Z_d = 0.164435 ; Z_v = 0.160996 ; Z_m = 0.671948$$

$$Z_a = 0.826502E+00 ; Z_w = 0.160996$$

ASME Section VIII-2 Part 4.18.8.4 Step 4:

Diameter ratio $[K]$:

$$= A / D_o$$

$$= 2240.0/1899.01$$

$$= 1.1796$$

Compute coefficient $[F]$:

$$= (1 - \nu) / E \cdot (\lambda_s + \lambda_c + E \cdot \ln(K))$$

$$= (1 - 0.32) / 68583928 \cdot (3049807 + 103127.83 + 0.18685E+09 \cdot \ln(1.18))$$

$$= 3.4282$$

Compute parameter $[\phi]$:

$$= (1 + \nu) F$$

$$= (1 + 0.3208) 3.4282$$

$$= 4.5278$$

Compute parameter $[Q_1]$:

$$= (\rho_s - 1 - \phi \cdot Z_v) / (1 + \phi \cdot Z_m)$$

$$= (1.0216 - 1 - 4.5278 \cdot 0.161) / (1 + 4.5278 \cdot 0.6719)$$

$$= -0.174987$$

Compute Parameter $[Q_{z1}]$:

$$= (Z_d + Q_1 \cdot Z_w) / 2 \cdot X_a^4$$

$$= (0.16444 + -0.17499 \cdot 0.161) / 2 \cdot 2.03711^4$$

$$= 1.1733$$

Compute parameter $[Q_{z2}]$:

$$= (Z_v + Q_1 \cdot Z_m) / 2 \cdot X_a^4$$

$$= (0.161 + -0.17499 \cdot 0.67195) / 2 \cdot 2.03711^4$$

$$= 0.3738$$

Compute parameter $[U]$:

$$= (Z_w + (\rho_s - 1) Z_m) X_a^4 / (1 + \phi \cdot Z_m)$$

$$= (0.161 + (1.0216 - 1) 0.6719) 2.03711^4 / (1 + 4.5278 \cdot 0.6719)$$

$$= 0.7476$$

ASME Section VIII-2 Part 4.18.8.4 Step 5:

Determine factor $[y_b]$:

$$= 0 \text{ (config a)}$$

Compute Parameter $[y]$:

$$= 0.000 \text{ mm. (For Pressure only cases)}$$

Calculate parameter $[\omega_s]$:

$$= \rho_s \cdot k_s \cdot \beta_s \cdot \delta S (1 + h \cdot \beta_s)$$

$$= 1.0216 \cdot 0.37157E+1 \cdot 0.0033 \cdot 0.028874 (1 + 349.0 \cdot 0.0033)$$

$$= 76325.7656 \text{ mm.}^2$$

Calculate parameter $[\omega^*s]$:

$$= a_o^2 (\rho_s^2 - 1) \cdot (\rho_s - 1) / 4 - \omega_s$$

$$= 949.505^2 (1.022^2 - 1) \cdot (1.022 - 1) / 4 - 76325.766$$

$$= -76113.4766 \text{ mm.}^2$$

Parameter $[\omega_c]$:



$$= \rho_c \cdot k_c [\delta C (C_m \theta + \beta C + C_q \Delta \cdot h \cdot \beta C^2) - \zeta_c (C_m \theta + h \cdot C_q \theta \cdot h \cdot \beta C / 2)]$$

$$= 1.022 \cdot 46717224 [0.161074 (1.0 + 0.008 \cdot 1.0 \cdot 349.0) - 0 (1.0/349.0 + 1.0 \cdot 0.008/2)]$$

$$= 24353.6328 \text{ mm}^2$$

Parameter [$\omega \cdot c$]:

$$= a_o^2 [(\rho_c^2 + 1) \cdot (\rho_c - 1)/4 - (\rho_s - 1)/2] - \omega c$$

$$= 949.505^2 [(1.02158^2 + 1) \cdot (1.02158 - 1)/4 - (1.02158 - 1)/2] - 24353.63281$$

$$= -24141.3438 \text{ mm}^2$$

Pressure [$P \cdot S$]:

= 0 For Pressure only cases or Configurations d,e,f,A,B,C,D

Pressure [$P \cdot C$]:

= 0 For Pressure only cases or Configurations b,c,d,B,C,D

ASME Section VIII-2 Part 4.18.8.4 Step 6:

Pressure [$P's$]:

$$= P_s \{ x_s + 2(1 - x_s) v_t + [2/K_{st} \cdot (D_s/D_o)^2] v_s - [(\rho_s^2 - 1) / (J \cdot K_{st})] - [(1 - J) / (2J \cdot K_{st})] [(D_j^2 - (D_s)^2) / D_o^2] \}$$

$$= 191.0 \{ 0.668 + 2(1 - 0.668) 0.3 + [2/2.093 \cdot (1940.0/1899.01)^2] 0.3 - [(1.022^2 - 1) / (1.0 \cdot 2.093)] - [(1 - 1.0) / (2 \cdot 1.0 \cdot 2.093)] [(0^2 - (1940.0)^2) / 1899.01^2] \}$$

$$= 218.8004 \text{ bars}$$

Pressure [$P't$]:

$$= [x_t + 2(1 - x_t) v_t + 1/(J \cdot K_{st})] P_t$$

$$= [0.833 + 2(1 - 0.833) 0.3 + 1/(1 \cdot 2.093)] 10.0$$

$$= 14.1092 \text{ bars}$$

Pressure [P_y]:

$$= N_t \cdot K_t \cdot \gamma / (\pi \cdot a_o^2)$$

$$= 3297 \cdot 251.321 \cdot 0 / (3.142 \cdot 949.505^2)$$

$$= 0.0000 \text{ bars}$$

Pressure [P_w]:

$$= -\gamma_b \cdot U \cdot W \cdot / (2 \cdot \pi \cdot a_o^2)$$

$$= -0 \cdot 0.748 \cdot 0 / (2 \cdot \pi \cdot 949.505^2)$$

$$= -0.0000 \text{ bars}$$

Calculate the pressure [$Prim$]:

$$= -(U/a_o^2) \cdot (\omega \cdot s \cdot P_s - \omega \cdot c \cdot P_t)$$

$$= -(0.748/949.505^2) \cdot (-76113.477 \cdot 191.0 - -24141.344 \cdot 10.0)$$

$$= 11.8554 \text{ bars}$$

Calculate the pressure [P_w]:

$$= (U/a_o^2) \cdot (\omega \cdot s \cdot P_s - \omega \cdot c \cdot P_t)$$

$$= (0.748/949.505^2) \cdot (76325.7656 \cdot 0 - 24353.6328 \cdot 0)$$

$$= 0.0000 \text{ bars}$$

Determine the effective pressure [Pe]:

$$= J \cdot K_{st} / (1 + J \cdot K_{st} (Q_{z1} + (\rho_s - 1) Q_{z2})) \cdot (P's - P't + P_y + P_w + P_w + Prim)$$

$$= 0.1000E+01 \cdot 2.093 / (1 + 1.0 \cdot 2.093 (1.173 + (1.022 - 1) 0.374)) \cdot (218.8 - 14.109 + 0 + 0 + 0 + 11.855)$$

$$= 130.5257 \text{ bars}$$

ASME Section VIII-2 Part 4.18.8.4 Step 7:

Determine factor [$Q2$]:

$$= [((\omega \cdot s \cdot P_s - \omega \cdot c \cdot P_t) - (\omega \cdot s \cdot P_s - \omega \cdot c \cdot P_t)) + W \cdot \gamma_b / (2 \cdot \pi)] / (1 + \phi \cdot Z_m)$$

$$= [((-76113.477 \cdot 191.0 - -24141.344 \cdot 10.0) - (76325.766 \cdot 0 - 24353.633 \cdot 0)) \cdot 0.01 + 0 \cdot 0 / (2 \cdot \pi)] / (1 + 4.52784 \cdot 0.67195)$$

$$= -36063.8125 \text{ Kgf}$$

Calculate factor [$Q3$]:



$$\begin{aligned}
 &= Q1 + 2 \cdot Q2 / (Pe \cdot ao^2) \\
 &= -0.175 + 2 \cdot -36063.812 / (130.526 \cdot 949.505^2) \\
 &= -0.23509
 \end{aligned}$$

Fm Value from Table 4.18.3 = 0.117546

$$\begin{aligned}
 &\text{Tubesheet Bending Stress - original thickness } [\sigma]: \\
 &= (1.5 \cdot Fm / \mu \cdot) \cdot (2 \cdot ao / (H - h'g))^2 \cdot Pe \\
 &= (1.5 \cdot 0.1175 / 0.365) \cdot (2 \cdot 949.505 / (349.0 - 0))^2 \cdot 130.53 \\
 &= 186.6944 \text{ N./mm}^2
 \end{aligned}$$

$$\begin{aligned}
 &\text{Allowable Tubesheet Bending Stress } [\sigma \text{ allowed}]: \\
 &= 1.5 \cdot S = 1.5 \cdot 134.6 = 201.9 \text{ N./mm}^2
 \end{aligned}$$

$$\begin{aligned}
 &\text{Tubesheet Bending Stress - final thickness } [\sigma f]: \\
 &= (1.5 \cdot Fm / \mu \cdot) \cdot (2 \cdot ao / (h - h'g))^2 \cdot Pe \\
 &= (1.5 \cdot 0.1167 / 0.365) \cdot (2 \cdot 949.505 / (332.991 - 0))^2 \cdot 129.46 \\
 &= 201.8880 \text{ N./mm}^2
 \end{aligned}$$

$$\begin{aligned}
 &\text{Reqd Tubesheet Thickness, for Bending Stress (including CA) } [HReqB]: \\
 &= h + Cats + Catc \\
 &= 332.9908 + 0 + 0 \\
 &= 332.9908 \text{ mm.}
 \end{aligned}$$

ASME Section VIII-2 Part 4.18.8.4 Step 8:

$$\begin{aligned}
 &\text{Shear Stress check } [\tau \text{ limit}]: \\
 &= (2 \cdot \mu \cdot h / ao) \cdot \min(0.8S; 0.533Sy) \\
 &= (2 \cdot 0.365 \cdot 349.0 / 949.505) \cdot 107.666 \\
 &= 28.8889 \text{ N./mm}^2
 \end{aligned}$$

Shear Stress is not required to be computed; $[Pe] \leq \tau_{\text{limit}}$

$$\begin{aligned}
 &\text{Tubesheet Shear Stress - final thickness } [\tau f]: \\
 &= (Ap \cdot Pe) / (\mu \cdot h \cdot Cp) \\
 &= (24988.3 \cdot 73.08) / (0.365 \cdot 81.385 \cdot 5991.4) \\
 &= 102.6059 \text{ N./mm}^2
 \end{aligned}$$

$$\begin{aligned}
 &\text{Reqd Tubesheet Thickness, for Shear Stress (including CA) } [HreqS]: \\
 &= H + Cats + Catc \\
 &= 81.3846 + 0 + 0 \\
 &= 81.3846 \text{ mm.}
 \end{aligned}$$

$$\begin{aligned}
 &\text{Reqd Tubesheet Thickness for Given Loadings (including CA) } [Hreqd]: \\
 &= \max(HreqB; HreqS) \\
 &= \max(332.9908; 81.3846) \\
 &= 332.9908 \text{ mm.}
 \end{aligned}$$

ASME Section VIII-2 Part 4.18.8.4 Step 9:

$$\begin{aligned}
 &\text{The } F_{tmin} \text{ and } F_{tmax} \text{ Coefficients from Table 4.18.4:} \\
 &F_{tmin} = 0.8293, F_{tmax} = 1.09
 \end{aligned}$$

$$\begin{aligned}
 &\text{First Extreme Tube Axial Stress from among all the tubes } [\sigma t1]: \\
 &= ((Ps \cdot xs - Pt \cdot xt) - Pe \cdot F_{tmin}) / (Xt - Xs) \\
 &= ((191.0 \cdot 0.6682 - 10.0 \cdot 0.8331) - (130.526) \cdot 0.829) / \\
 &\quad (0.8331 - 0.6682) \\
 &= 6.7060 \text{ N./mm}^2
 \end{aligned}$$

$$\begin{aligned}
 &\text{Second Extreme value of Tube Axial Stress from among all the tubes } [\sigma t2]: \\
 &= ((Ps \cdot xs - Pt \cdot xt) - Pe \cdot F_{tmax}) / (Xt - Xs) \\
 &= ((191.0 \cdot 0.6682 - 10.0 \cdot 0.8331) - (130.526) \cdot 1.09) / \\
 &\quad (0.8331 - 0.6682) \\
 &= -13.9301 \text{ N./mm}^2
 \end{aligned}$$

$$\begin{aligned}
 &\text{Maximum Tube Axial Stress } [\sigma t, \max]: \\
 &= \max(|\sigma t,1|; |\sigma t,2|) = 13.93 \text{ N./mm}^2
 \end{aligned}$$

$$\begin{aligned}
 &\text{Allowable Tube Stress } [\sigma t, A]: \\
 &= Sot = 117.9979 \text{ N./mm}^2
 \end{aligned}$$



Check for Buckling as some of the tubes are in compression.

Determine the Factor of Safety [Fs]:

$$\begin{aligned} &= \max((3.25 - 0.25(Z_d + Q_3 \cdot Z_w)X_a^4); 1.25) \\ &= \max((3.25 - 0.25(0.164 + -0.235 \cdot 0.161)2.037^4); 1.25) \\ &= 2.0000 \text{ (Should be } \leq 2 \text{)} \end{aligned}$$

Determine the factor [rt]:

$$\begin{aligned} &= (\sqrt{ (\frac{dt^2}{4} + (dt - 2 \cdot tt)^2) }) / 4 \\ &= (\sqrt{ (\frac{19.05^2}{4} + (19.05 - 2 \cdot 2.77)^2) }) / 4 \\ &= 5.8386 \text{ mm.} \end{aligned}$$

Determine the factor [Ct]:

$$\begin{aligned} &= \sqrt{ (2 \cdot \pi^2 \cdot Et / S_{yt}) } \\ &= \sqrt{ (2 \cdot \pi^2 \cdot 0.18822E+09 / 208) } \\ &= 133.6543 \end{aligned}$$

Determine the factor [Ft]:

$$\begin{aligned} &= k \cdot L / r \\ &= 0.8 \cdot 1510.0 / 5.839 \\ &= 206.8999 \end{aligned}$$

Buckling Allowable Stress [Stb]:

$$\begin{aligned} &= (1 / F_s) \cdot (\pi^2 \cdot Et / F_t^2) (C_t \leq F_t) \\ &= (1 / 2.0) \cdot (\pi^2 \cdot 0.18822E+09 / 206.8999^2) \\ &= 21.700 \text{ N./mm}^2 \text{ (Never greater than } S_{ot} \text{)} \end{aligned}$$

Note: The Axial Compressive stress in tubes is within limits.

Largest tube-to-tubesheet Joint Load [Wt]:

$$\begin{aligned} &= \sigma_{t,max} \cdot \text{Tube Area} \\ &= 13.93 \cdot 1.4167 \\ &= 201.24 \text{ Kgf} \end{aligned}$$

Tube Metal Area [At]:

$$\begin{aligned} &= \pi (d_o - t) t \\ &= \pi (19.05 - 2.77) 2.77 \\ &= 1.417 \text{ cm}^2 \end{aligned}$$

Tube Weld Size Results per UW-20.3:

Tube Strength [Ft]:

$$\begin{aligned} &= \pi \cdot t (d_o - t) S \\ &= \pi \cdot 2.77 (19.05 - 2.77) 117.998 \\ &= 1704.628 \text{ Kgf} \end{aligned}$$

Groove Weld Strength [Fg]:

$$\begin{aligned} &= 0.85 \cdot \pi \cdot a_g (d_o + 0.67 \cdot a_g) S_w \text{ (but not } > F_t \text{)} \\ &= 0.85 \cdot \pi \cdot 2.7 (19.05 + 0.67 \cdot 2.7) 118.0 \\ &= 1704.6277 \text{ Kgf} \end{aligned}$$

Weld Strength Factor [fw]:

$$\begin{aligned} &= S / S_w \text{ where } S_w = \min(S \text{ and } S_t) \\ &= 117.998 / 117.998 \\ &= 1.000 \end{aligned}$$

Design Strength Ratio [fd]:

$$= 1 \text{ for full strength welds}$$

Fillet Weld Strength [Ff]:

$$\begin{aligned} &= 0.55 \cdot \pi \cdot a_f (d_o + 0.67 \cdot a_f) S_w \text{ (but not } > F_t \text{)} \\ &= 0.55 \cdot \pi \cdot 2.7 (19.05 + 0.67 \cdot 2.7) 118.0 \\ &= 1170.8878 \text{ Kgf} \end{aligned}$$

Minimum Length of Weld [ar]:

$$\begin{aligned} &= 2 (\sqrt{ ((0.75 \cdot d_o)^2 + 1.07 \cdot t \cdot (d_o - t) f_w \cdot f_d) - 0.75 \cdot d_o }) \\ &= 2 (\sqrt{ ((0.75 \cdot 19.05)^2 + 1.07 \cdot 2.77 \cdot (19.05 - 2.77) 1.0 \cdot 1.0) - 0.75 \cdot 19.05 }) \\ &= 3.198 \text{ mm.} \end{aligned}$$



Minimum Required Fillet Weld Leg [afr]: 1.6620 mm.
 Minimum Required Groove Weld Leg [agr]: 1.6620 mm.

Tube-Tubesheet Jt allowable, 1704.63 is >= tube strength 1704.63 Kgf

Note: This tube-tubesheet joint is a Full Strength joint

ASME Section VIII-2 Part 4.18.8.4 Step 10:

Shell Axial Membrane Allowable Stress:
 $= S_s \cdot E_{sw} = 137.6 \cdot 1.0 = 137.6 \text{ N./mm}^2$

Axial Membrane Stress in Shell [σ_s, m]:
 $= a_o^2 / ((D_s + t_s) t_s) \cdot [P_e + (p_s^2 - 1)(P_s - P_t)] + a_s^2 \cdot P_t / ((D_s + t_s) t_s)$
 $= 949.505^2 / ((1940.0 + 149.0) 149.0) \cdot [130.53 + (1.022^2 - 1)$
 $(191.0 - 10.0)] + 970.0^2 \cdot 10.0 / ((1940.0 + 149.0) 149.0)$
 $= 43.1195 \text{ N./mm}^2$

ASME Section VIII-2 Part 4.18.8.4 Step 11:

Note:

For a given Shell thickness of 149.0 mm., the minimum Shell length adjacent to the tubesheet should be 967.757 mm.

Shell Membrane Stress due to Joint Interaction [σ_s, m]:
 $= a_o^2 / ((D_s + t_s) t_s) [P_e + (p_s^2 - 1)(P_s - P_t)] + a_s^2 \cdot P_t / ((D_s + t_s) t_s)$
 $= 949.505^2 / ((1940.0 + 149.0) 149.0) [130.53 + (1.022^2 - 1)$
 $(191.0 - 10.0)] + 970.0^2 \cdot 10.0 / ((1940.0 + 149.0) 149.0)$
 $= 43.1195 \text{ N./mm}^2$

Shell Bending Stress due to Joint Interaction [σ_s, b]:
 $= 6 \cdot k_s / t_s^2 \{ \beta_s [\delta_s \cdot P_s + a_s^2 \cdot P \cdot s / (E_s \cdot t_s)] +$
 $6(1 - \nu^2) / (E^*) (a_o / h)^3 (1 + h \cdot \beta_s / 2) [P_e (Z_v + Z_m \cdot Q_1) + 2 / a_o^2 \cdot Z_m \cdot Q_2] \}$
 $= 6 \cdot 0.37157 E + 1 / 149.0^2 \{ 0.003 [0.028874 \cdot 191.0 + 970.0^2$
 $\cdot 0 / (.28045 E + 11)] +$
 $6(1 - 0.32^2) / (68583928) (949.51 / 349.0)^3 (1 + 349.0 \cdot$
 $0. / 2) [130.5 (0.161 + 0.672 \cdot -0.175) +$
 $2 / 949.51^2 \cdot 0.672 \cdot -36063.812] \}$
 $= 190.3011 \text{ N./mm}^2$

Shell Stress Summation vs. Allowable

$\therefore | \sigma_{s, m} | + | \sigma_{s, b} | \leq 1.5 \cdot S_s$
 $\therefore | 43.12 | + | 190.3 | \leq 206.4 \text{ N./mm}^2$
 $\therefore 233.42, \text{ must be } \leq 206.4 \text{ N./mm}^2 \text{ (See Elastic-Plastic iteration below)}$

Note:

For a given Channel thickness of 25.0 mm., the minimum Channel length adjacent to the tubesheet should be 396.409 mm.

Channel Membrane Stress due to Joint Interaction [σ_c, m]:
 $= a_c^2 \cdot P_t / ((D_c + t_c) t_c \cdot \cos(\theta))$
 $= 970.0^2 \cdot 10.0 / ((1940.0 + 25.0) 25.0 \cdot \cos(0))$
 $= 19.154 \text{ N./mm}^2$

Channel Bending Stress due to Joint Interaction [σ_c, b]:
 $= 6 \cdot k_c / t_c^2 \cdot \{ (\delta_c \cdot C_m \Delta + \beta_c - \zeta_c \cdot C_m \theta) \cdot P_t$
 $- 6(1 - \nu^2) / (E^*) \cdot (a_o / h)^3 \cdot (C_m \theta +$
 $h \cdot \beta_c / 2 \cdot C_m \Delta) \cdot [P_e (Z_v + Z_m \cdot Q_1) + 2 / a_o^2 \cdot Z_m \cdot Q_2] \}$
 $= 6 \cdot 46717224 / 25.0^2 \cdot \{ (0.161074 \cdot 1.0 \cdot 0.008 - 0 \cdot 1.0) \cdot 10.0$
 $- 6(1 - 0.321^2) / 68583928 \cdot$
 $(949.505 / 349.0)^3 \cdot (1.0 + 349.0 \cdot 0.008 / 2 \cdot 1.0) \cdot [130.5 (0.161 + 0.67195$
 $\cdot -0.175) + 2 / 949.505^2 \cdot 0.67195 \cdot -36063.812] \}$
 $= 52.4483 \text{ N./mm}^2$

Channel Stress Summation vs. Allowable

$\therefore | \sigma_{c, m} | + | \sigma_{c, b} | \leq 1.5 \cdot S_c$
 $\therefore | 19.15 | + | 52.45 | \leq 207.0 \text{ N./mm}^2$
 $\therefore 71.6 \text{ must be } \leq 207.0 \text{ N./mm}^2$



Stress/Force Summary for Loadcase D3 un-corr. (Psd,max + Ptd,max):

Stress Description	Actual	Allowable	Pass/Fail
Tubesheet Bend. Stress	186.7 <=	201.9 N./mm ²	Pass
Tubesheet Shear Stress	42.7 <=	107.7 N./mm ²	Pass
Maximum Tube Stress	13.9 <=	118.0 N./mm ²	Pass
Minimum Tube Stress (Buckling)	-13.9 <=	-21.7 N./mm ²	Pass
Maximum Force on any one Tube	201.2 <=	1704.6 Kg _f	Pass
Axial Membrane Stress in Shell	43.1 <=	137.6 N./mm ²	Pass
Shell Stress (jt. inter.)	233.4 <=	206.4 N./mm ²	See below
Channel Stress (jt. inter.)	71.6 <=	207.0 N./mm ²	Pass

Additional Results for the 2nd Elastic-Plastic Iteration.



These calculations are not intended to be used in the creep range. Check Part 4.18 in Section VIII Division 2.

Compute Allowable [Sstars]:
 = min(Sys, SPS,s/2)
 = min(213, 427/2) = 213.61 N./mm²

Compute Allowable [S*c]:
 = min(Syc, SPS,c/2)
 = min(242, 484/2) = 242.01 N./mm²

Compute Factor [facts]:
 = min([1.4 - 0.4* abs(σs,b) / Sstars], 1)
 = min([1.4 - 0.4* abs(190.3)/213.61], 1) = 1.0

Compute Factor [factc]:
 = min([1.4 - 0.4* abs(σc,b) / Sstarc], 1)
 = min([1.4 - 0.4* abs(52.45)/242.01], 1) = 1.0

Note: Since facts & factc are both >= 1, the design Passes.

Stress/Force Summary for Loadcase D3 un-corr. (Psd,max + Ptd,max):

Stress Description	Actual	Allowable	Pass/Fail
Shell Stress (jt. inter.)	233.4 <=	427.2 N./mm ²	Pass
Channel Stress (jt. inter.)	71.6 <=	207.0 N./mm ²	Pass

Thickness Results for Loadcase D3 un-corr. (Psd,max + Ptd,max):

Thickness (mm.)	Required	Actual	P/F
Tubesheet Thickness :	332.991	349.000	Ok
Tube-Tubesheet Fillet Weld Leg :	1.662	2.700	Ok
Tube-Tubesheet Groove Weld Leg :	1.662	2.700	Ok

Fixed Tubesheet results per ASME Div. 2 Part 4.18.8 2025

Results for 14 Load Cases:

Case#	--Reqd. Thk. + CA Tbsht Extnsn	----	Tubesheet Bend	Stresses Allwd	Shear	Allwd	Case Type	Pass/ Fail
D1uc	14.392	...	10	202	3	108	Ps+Pt-Th	D1 Ok
D2uc	343.623	...	197	202	46	108	Ps+Pt-Th	D2 Ok
D3uc	332.991	...	187	202	43	108	Ps+Pt-Th	D3 Ok
O1uc	18.157	...	23	404	6	108	Ps+Pt+Th	O1 Ok
O2uc	207.843	...	185	404	42	108	Ps+Pt+Th	O2 Ok
O3uc	197.866	...	175	404	39	108	Ps+Pt+Th	O3 Ok



O4uc	6.923	...	13	404	4	108	Ps+Pt+Th	O4	Ok
D1c	18.916	...	10	202	3	108	Ps+Pt-Th-c	D1	Ok
D2c	347.154	...	200	202	46	108	Ps+Pt-Th-c	D2	Ok
D3c	336.671	...	190	202	43	108	Ps+Pt-Th-c	D3	Ok
O1c	22.644	...	23	404	7	108	Ps+Pt+Th-c	O1	Ok
O2c	211.828	...	188	404	42	108	Ps+Pt+Th-c	O2	Ok
O3c	201.964	...	178	404	39	108	Ps+Pt+Th-c	O3	Ok
O4c	11.414	...	14	404	4	108	Ps+Pt+Th-c	O4	Ok

Max: 347.1535 ... mm. 0.991 0.427 (Str. Ratio)

Load Case Definitions:

[Ps & Pt]:

Shell-side and Tube-side Design or Operating Pressures
 derived from Psd,min Ptd,max, Psox,min, Ptox,max etc. per the
 Load Case Tables

[(+)-Th]:

With or Without Thermal Expansion, Tt,mx & Ts,mx

[c]:

With or Without Corrosion Allowance

[D1, D2, D3]:

Design Load Cases using the Maximum and Minimum Design Pressures

[D4]:

Design Load Case using the Minimum (Vacuum) Pressures (if specified)

[O1, O2, O3, O4]:

Operating Load Cases using the Maximum and Minimum Operating Pressures and
 Operating Temperatures

Shell Axial Membrane Stress Summary:

Case#	---- Shell Stresses ----				:	---- Shell Band Stress ----				: Pass
	Ten	Allwd	Cmp	Allwd		Ten	Allwd	Cmp	Allwd	
D1uc	...	137	...	...	:	...	...	...	...	Ok
D2uc	43	137	...	...	:	...	...	...	...	Ok
D3uc	43	137	...	...	:	...	...	...	...	Ok
O1uc	3	427	-3	-109	:	...	...	...	...	Ok
O2uc	40	427	...	...	:	...	...	...	...	Ok
O3uc	40	427	...	...	:	...	...	...	...	Ok
O4uc	3	427	-3	-109	:	...	...	...	...	Ok
D1c	...	137	...	...	:	...	...	...	...	Ok
D2c	43	137	...	...	:	...	...	...	...	Ok
D3c	44	137	...	...	:	...	...	...	...	Ok
O1c	3	427	-3	-109	:	...	...	...	...	Ok
O2c	40	427	...	...	:	...	...	...	...	Ok
O3c	40	427	...	...	:	...	...	...	...	Ok
O4c	3	427	-3	-109	:	...	...	...	...	Ok
Max RATIO	0.317			0.029	:	...			...	

Tube, Shell and Channel Stress Summary:

Case#	---- Tube Stresses ----				Tube Loads		Shell Stress		Channel Stress		Pass
	Ten	Allwd	Cmp	Allwd	Ld	Allwd	Stress	Allwd	Stress	Allwd	
D1uc	1	117	-1	-21	15	1704	8	206	83	206	Ok
D2uc	15	117	-15	-21	215	1704	240	427	12	206	Ok
D3uc	14	117	-14	-21	201	1704	233	427	72	206	Ok
O1uc	9	235	...	...	128	3409	22	427	92	484	Ok
O2uc	13	235	-7	-21	185	3409	225	427	3	484	Ok
O3uc	12	235	-6	-21	170	3409	218	427	80	484	Ok
O4uc	8	235	...	...	114	3409	15	427	8	484	Ok
D1c	1	117	-1	-21	16	1704	8	206	95	206	Ok
D2c	15	117	-15	-21	215	1704	245	427	13	206	Ok



D3c	14	117	-14	-21	201	1704	238	427	82	206	Ok
01c	9	235	...	...	128	3409	23	427	103	484	Ok
02c	14	235	-7	-21	196	3409	229	427	5	484	Ok
03c	12	235	-6	-21	180	3409	222	427	90	484	Ok
04c	8	235	...	...	114	3409	16	427	8	484	Ok

Max RATIO 0.126 0.687 0.126 0.573 0.457

Summary of Thickness Comparisons for 14 Load Cases:

Thickness (mm.)	Required	Actual	P/F
Tubesheet thickness :	347.154	349.000	Ok
Tube thickness :	2.278	2.770	Ok
Tube-tubesheet fillet weld leg :	1.662	2.700	Ok
Tube-tubesheet groove weld leg :	1.662	2.700	Ok

Min Shell length of thk, (149.000) adj. to tubesheet: 967.757 mm.

Min Channel length of thk, (25.000) adj. to tubesheet: 396.409 mm.

Note: This is a full strength Tube to Tubesheet Joint.

Summary of Axial Differential Expansion between Shell and Tubes :

Due to Thermal Expansion Shell Compresses by : -0.642 mm.
 Due to Pressure Shell Compresses by : -1.077 mm.
 Due to Pressure + Thermal Shell Compresses by : -1.719 mm.

Tubesheet MAWP used to Compute Hydrotest Pressure:

Stress / Force Condition	Tubeside MAWP	Ø shellside Stress Rat.	Shellside MAWP	Ø tubeside Stress Rat.
Tubesheet Bending Stress	200.597	1.000	192.738	1.000
Tubesheet Shear Stress	365.453	1.000	447.341	1.000
Tube Tensile Stress	1074.86	1.000	1512.20	1.000
Tube Compressive Stress	197.663	1.000	278.099	1.000
Tube-Tubesheet Joint load	1074.86	1.000	1512.26	1.000
Shell Stress (Axial, Junction)	57.087	0.132	192.738	0.578
Tubesheet-Channel Junction	50.285	1.000	192.738	0.063
Tube Pressure Stress	388.303	1.000	264.428	1.000
Minimum MAWP	50.285		192.738	

Tubesheet MAPnc used to Compute Hydrotest Pressure:

Stress / Force Condition	Tubeside MAPnc	Ø shellside Stress Rat.	Shellside MAPnc	Ø tubeside Stress Rat.
Tubesheet Bending Stress	207.313	1.000	201.018	1.000
Tubesheet Shear Stress	390.919	1.000	462.790	1.000
Tube Tensile Stress	1156.04	1.000	1512.34	1.000
Tube Compressive Stress	228.944	1.000	299.504	1.000
Tube-Tubesheet Joint load	1156.04	1.000	1512.33	1.000
Shell Stress (Axial, Junction)	207.313	0.299	201.018	0.486
Tubesheet-Channel Junction	62.410	0.997	201.018	0.056
Tube Pressure Stress	388.303	1.000	284.469	1.000
Minimum MAPnc	62.410		201.018	

(*) Only the design load cases were analyzed to compute the MAWP for determining the test pressure.

Summary of Exchanger Pressure and Thermal Displacements: (mm.)

Load Case#	Due to Pressure UnCorroded	Due to Pressure Corroded	Due to Thermal Exp. UnCorroded	Due to Thermal Exp. Corroded	Computed UnCorroded	Delta Ms Corroded
1	0.0075	0.0075	...	...	...	...
2	-1.0763	-1.0766	...	...	...	...



3	-1.0688	-1.0691	...	...	...	...
4	...	...	...	...	...	...
5	0.0075	0.0075	-0.6421	-0.6422	...	...
6	-1.0763	-1.0766	-0.6421	-0.6422	...	...
7	-1.0688	-1.0691	-0.6421	-0.6422	...	...
8	...	...	-0.6421	-0.6422	...	...

Tubesheet MDMT Calculations:

Shell Side MDMT calculation:

Governing thickness on the shell side per figure UCS-66.3 (e):

$$= \max(\text{tubesheet thk}/4, \min(\text{tubesheet thk}, \text{shell thickness}))$$

$$= \max(349.0/4, \min(349.0, 149.0))$$

$$= 149.000 \text{ mm.}$$

MDMT reduction ratio per UCS 66 (b)(1)(b):

$$= \min(\text{ps}/\text{Shellside MAPnc}, 1)$$

$$= \min(191.0/201.02, 1)$$

$$= 0.950$$

Warning:

The minimum metal temperature could not be computed because the governing thickness for this component was greater than 100.000 mm.. This component must be impact tested.

Channel Side MDMT calculation:

Governing Channel side thickness per figure UCS-66.3 (e):

$$= \max(\text{tubesheet thk}/4, \min(\text{tubesheet thk}, \text{channel thickness}))$$

$$= \max(349.0/4, \min(349.0, 25.0))$$

$$= 87.250 \text{ mm.}$$

MDMT reduction ratio per UCS 66 (b)(1)(b):

$$= \min(\text{pt}/\text{Tubeside MAPnc}, 1)$$

$$= \min(10.0/62.41, 1)$$

$$= 0.160$$

Thickness Ratio = 0.16, Temperature Reduction per Fig. UCS 66.1 = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

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